NORAXON

MR4 Help Documentation

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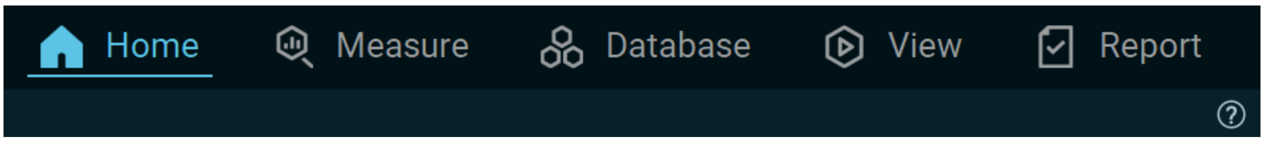
Home Screen

Software Navigation Overview

The Main Navigation Bar

On the top right of the software screen there is a software **navigation bar** that visualizes the workflow of a typical recording from the **Home** screen (start) to the **Report** (analysis):

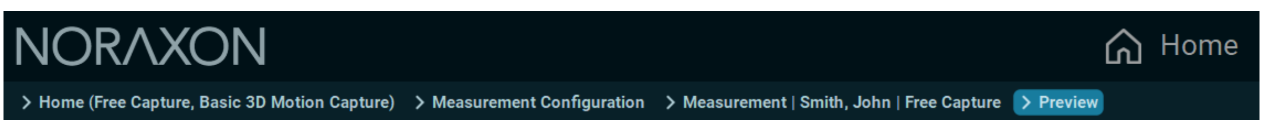


The currently open tab/menu is highlighted in blue. This navigation bar allows the user to jump directly into a certain menu if needed.

*Note: Some menus can only be accessed from the previous one in workflow. For example, a record can only be viewed (**View Tab**) if a record was selected in the previous **Database Tab**.*

Breadcrumb Bar

The **Breadcrumb Bar** is a navigational tool that sits below the **Main Navigation Bar** and indicates the user's current location within myoRESEARCH workflow.



The **Breadcrumb Bar** is structured as a pathway that follows the user's actions in the software. In MR4's workflow, all functions(Measure, Database, View, and Report) of the software are completed under a protocol created and selected on the home screen.

The first *breadcrumb* always contains the name of the current myoRESEARCH application and protocol in the format of **Home** (Application, Protocol), in the far left of the above screenshot. The Measurement *breadcrumb* will instead contain **Measurement** | Subject Name | Application.

Clicking any point along this path allows the user to return to a previous step. For example, clicking *Measurement Configuration* will bring the user back to the device configuration menu.

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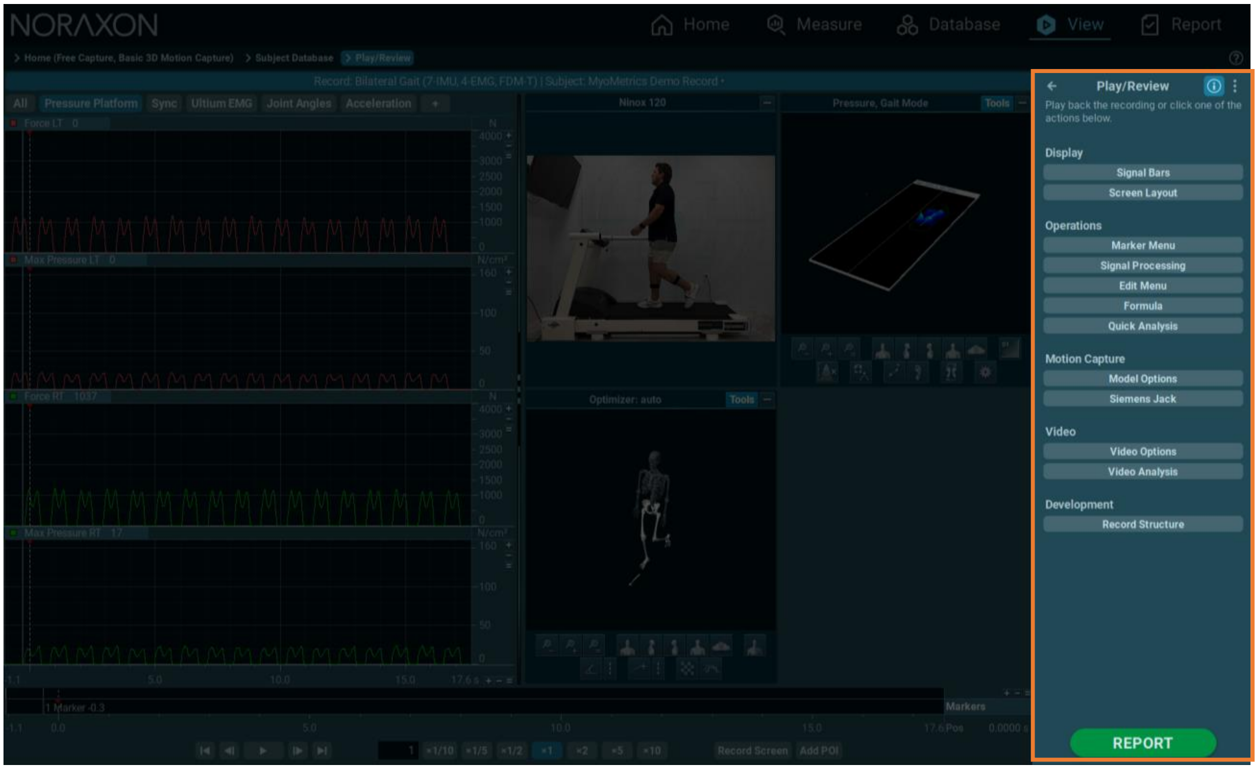
Operations Menu and Instruction Panel

The panel on the right of the software is known as the **operations menu**, which contains useful information and instructions about the current active function of myoRESEARCH, as well as various options and actions pertaining to the workspace.



In this screenshot, the **operations menu** contains various functions that can be used to modify the demo record currently being viewed. myoRESEARCH offers a set of optional tools to adjust or fine tune certain selections, operate menu specific options, or apply optional processing steps to the records.

The text at the top of the menu is the **instruction panel**, which contains contextual information that changes depending on what is currently being performed. In this instance, it provides simple instruction about the **operations menu**, but in more complex tasks such as myoMOTION calibration, the prompts can be very useful for a smooth setup.

Lastly, the large green button in the bottom right will lead to the next logical step in the workflow.

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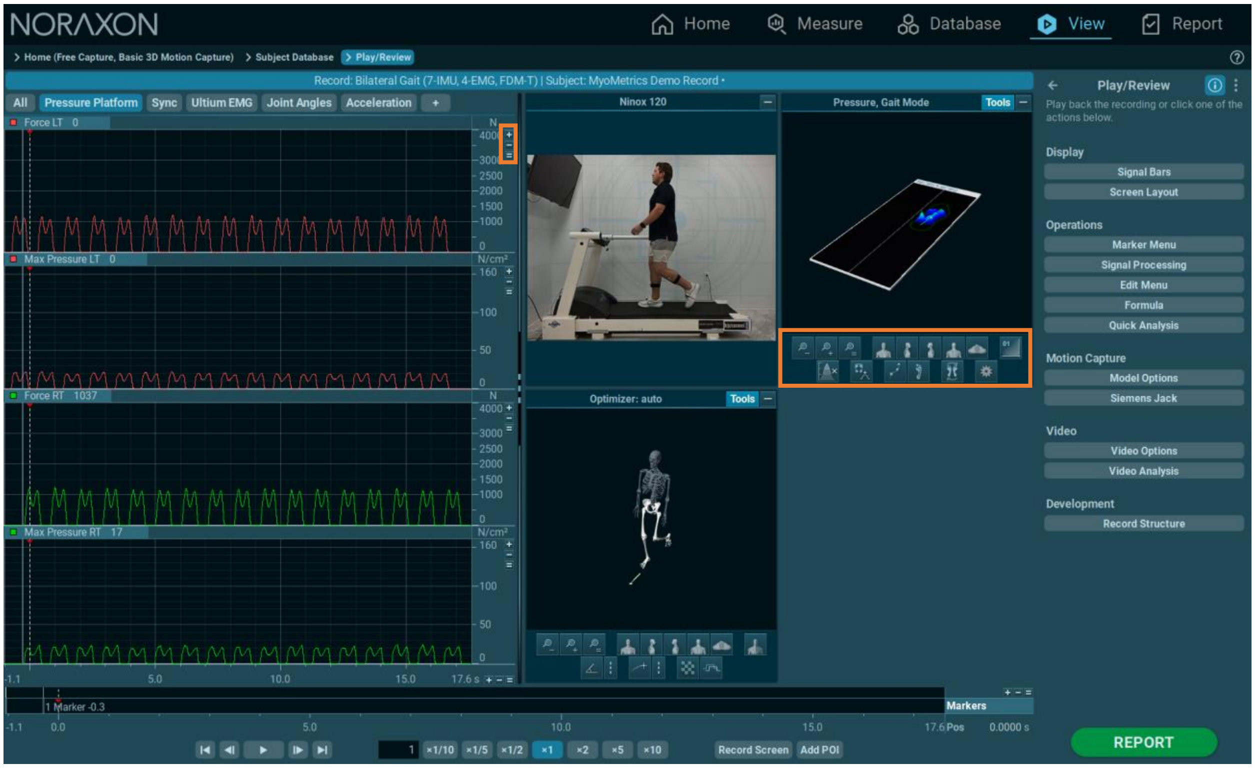
Local Menu Element Controls

Each menu consists of a set of menu elements, which are equipped with a set of local control tools. These local tools can be used to perform element specific selections and operations:



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Main Operations

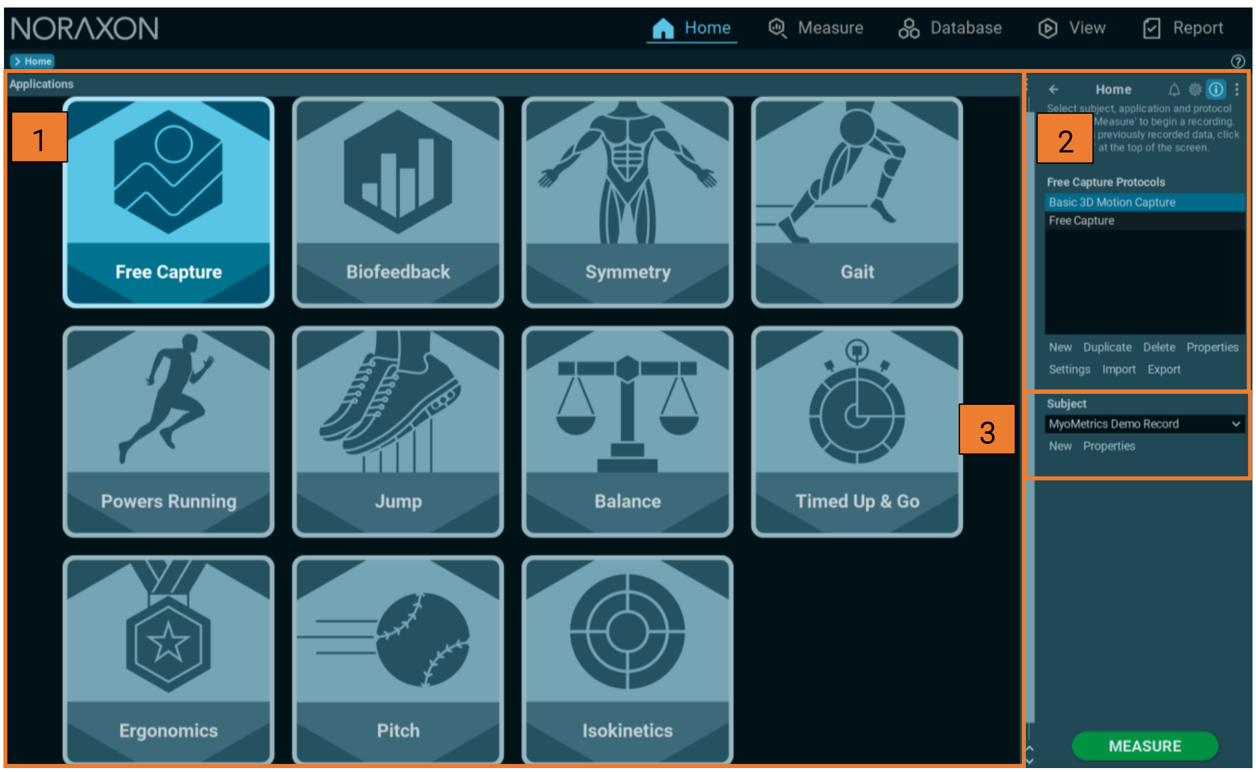
Introduction

The Home screen contains three major sections that can be used to configure a measurement.

The **Applications** menu can be used to select an **application** (such as Gait Analysis, Balance, Isokinetics, etc.) based off of an owned myoRESEARCH module(myoFORCE, myoMOTION, myoMUSCLE, etc.).

The Home screen also allows the user to create and modify a **protocol**, which is a preset that contains measurement settings, signal processing, report generation, and other modifications.

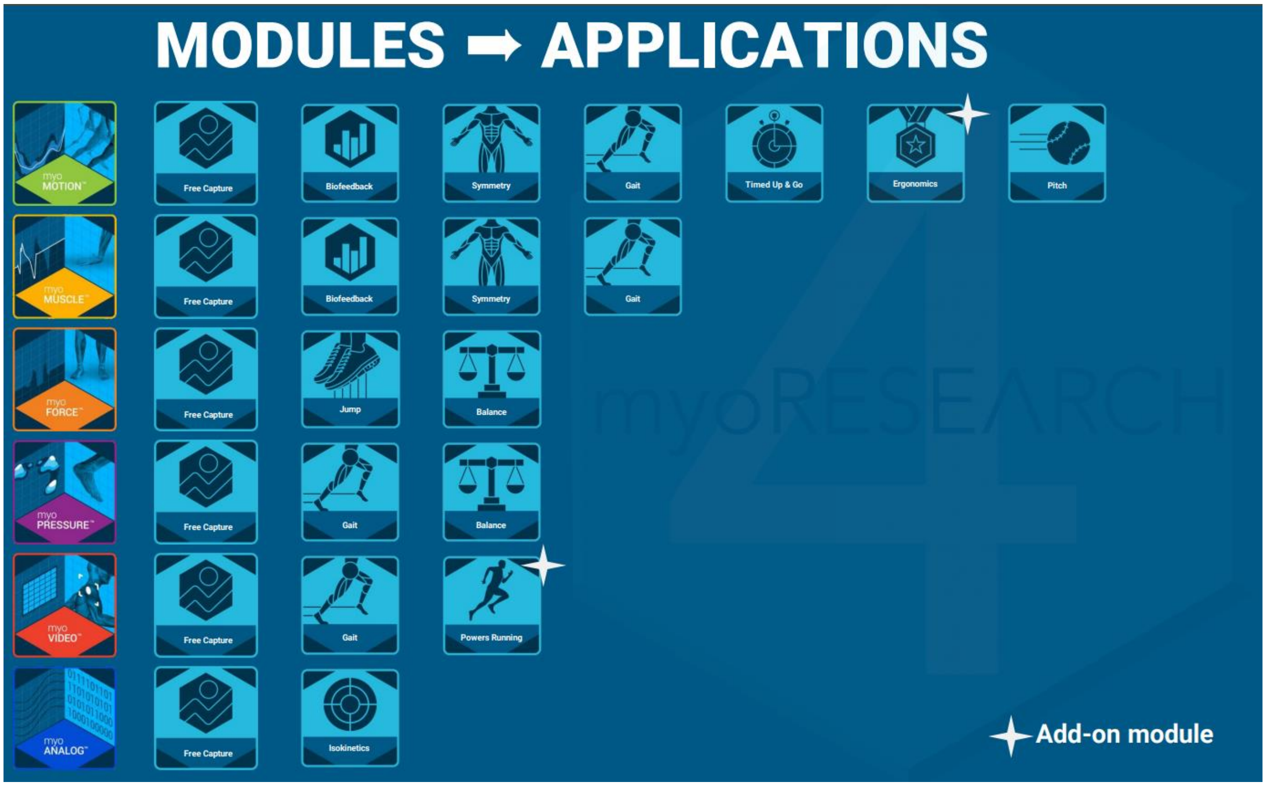
Lastly, the user may create or select a **subject** for the measurement from the Home screen.



The Home screen menu consists of 3 sub steps that can be used to select an **application** (in this case Free Capture) and select or create a measurement **protocol** and **subject**. To do this, simply perform steps 1 through 3 on the **Home** screen:

1. Select an Application

myoRESEARCH is a multi-functional software that can operate numerous sensors and devices. Each device can function stand-alone or be used in combination with each other. Applications are used to organize protocols under different applications. The applications visible on the home screen will depend on which myoRESEARCH modules the user owns.



Note: Applications do not possess any inherent differences in configuration. myoRESEARCH functionality is tied to the modules that are part of the license in use. Modifications to the protocol *determine how data is collected and analyzed.*

myoMUSCLE	EMG recording based on Noraxon EMG systems
myoMOTION	3D motion analysis based on Noraxon's inertial sensor system
myoPRESSURE	foot pressure analysis operated with pressure plates and insoles
myoVIDEO	video recording and marker-based 2D motion capture
myoFORCE	force plate integration with jump analysis
myoANALOG	16/32 channel AD board for the integration of third-party devices

Select an appropriate application for your use case.

2. Create a Protocol

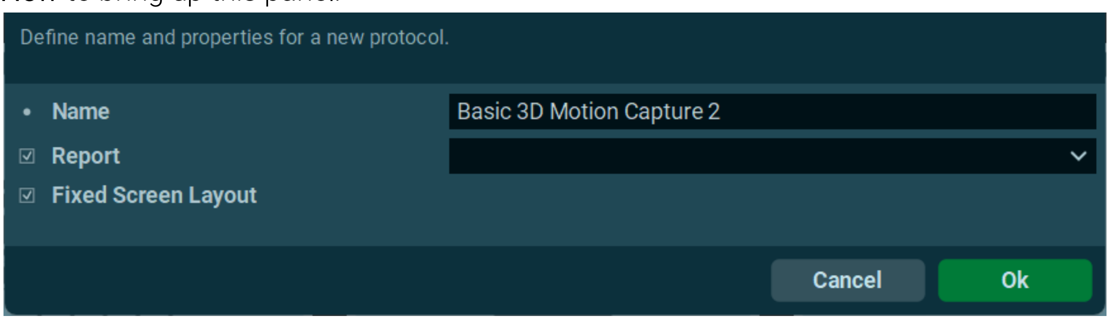
Any function performed in myoRESEARCH must be done under a **protocol** – a group of settings that controls how measurements are performed and analyzed. **Protocols** streamline the workflow of data collection by allowing users to create measurement presets that are personalized to their use case and hardware configuration.

A **protocol** stores the measurement configuration (the hardware and its settings), screen layout, and report type/format. Each protocol is stored under an **application** (Free Capture, Gait, Symmetry, etc.), which act as a folder to store and organize different protocols.

Protocols can be created from the **Home** screen using the local menu element controls underneath the main panel.

A video tutorial explaining the new **Protocol** system is available on the Noraxon website.

Click **New** to bring up this panel:

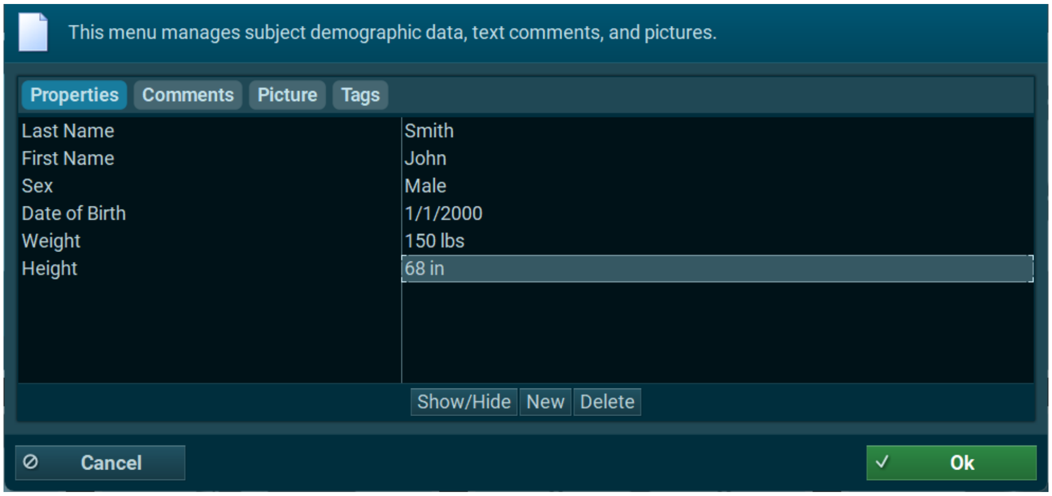


This screen will allow you to give a new protocol a unique name, as well as decide what report the protocol should run when the **Report** button is clicked when viewing a record. The checkbox, *Fixed Screen Layout* will set the screen layout (including the channel configuration, tabs, and video/animation tools) to be the same whenever the protocol is used. The *Report* checkbox can also be left unchecked, which will allow the user to select a report after collecting data – useful for situations where multiple reports could be applicable to the same record.

Next, create a new configuration or select/edit an existing configuration. The measurement setup menu is module and device specific. Please read the following sections for details on Measurement Configurations.

3. Select or Create a Subject

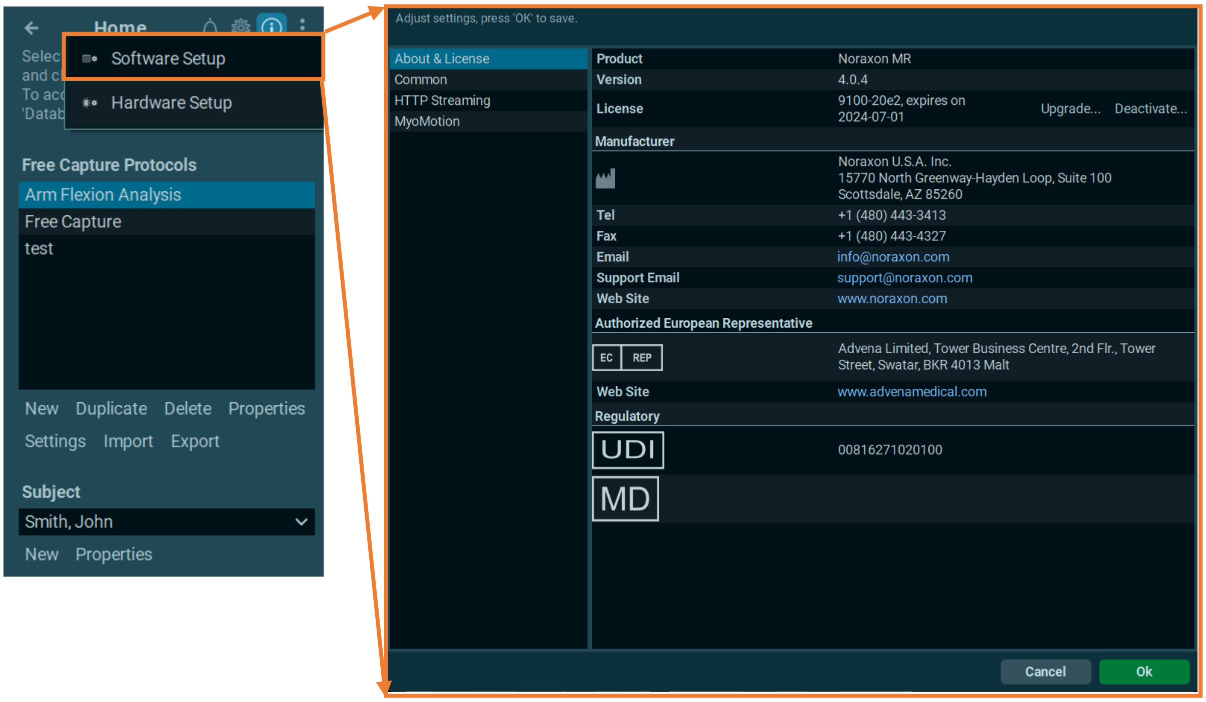
Measurements in myoRESEARCH are performed under a patient profile. Each profile stores useful information about the subject – for example, the patient's height is used to scale their bones in a 3D motion capture.



Either select a new subject from the subject list, or click **New/Properties** to create or modify an existing subject profile.

4. Hardware/Software Setup

Located in the top right of the **Operations Menu**, the gear icon allows users to configure the hardware and software setup.

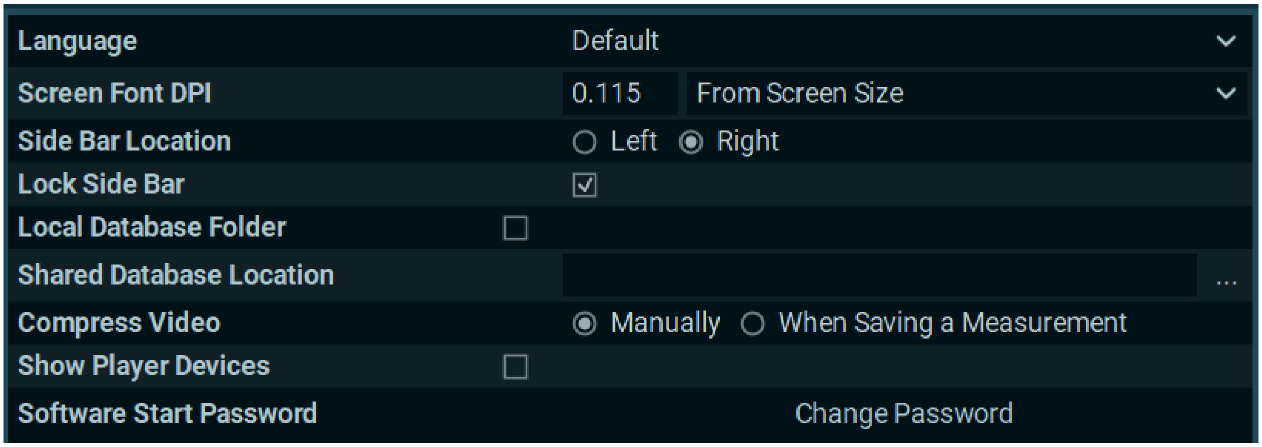


Software Setup

After selecting this menu item, some general software settings can be accessed alongside some module-specific functions. The software setup is sorted in 3 sections:

Common	section for general settings like language selection.
License	possibility to upgrade the license or deactivate a license.
MyoMOTION	specific software settings for the 3D Inertial sensor system.

Common



Language

This entry allows toggling between the default, the currently installed Windows Operating system, and the currently supported languages: English, German, Chinese, Russian and Japanese.

Screen font DPI

Three font size resolution options are available:

From screen size	is the recommended setting. Resolution is scaled in ratio to screen size and auto scales if myoRESEARCH is moved to a window, or the screen size temporarily changes, like when connected to a lower resolution LCD screen projector.
Fixed	manually selects a font resolution in DPI.
From system	checks the general font size of operating systems and will also be used for myoRESEARCH.

Side Bar Location

Change the location of the **Operations Menu/Instruction Panel** to either the right or left of the screen (right by default).

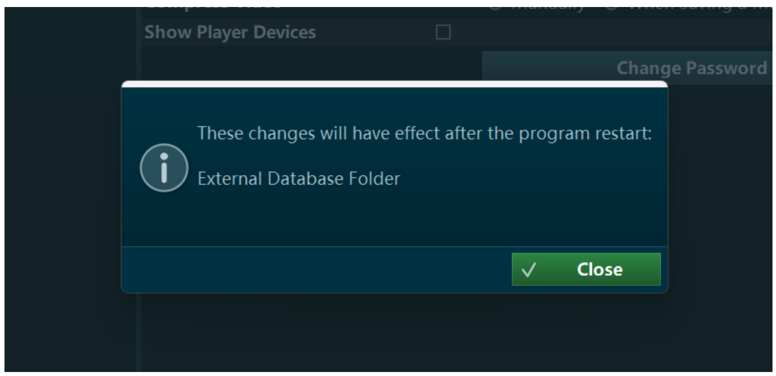
Lock Side Bar

Selected by default, while unchecked, the user may rescale the size of the **Operations Menu** by dragging the edge of the window.

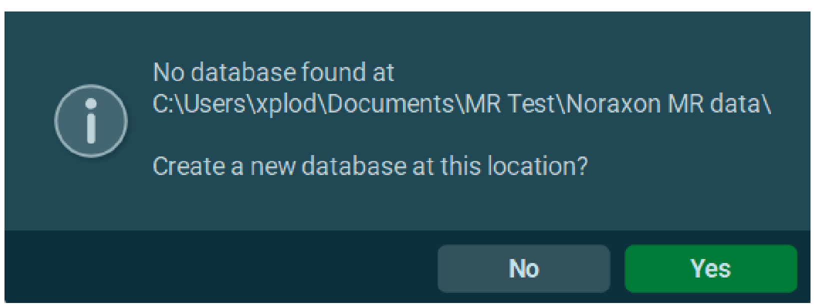
Local database folder

By default, all measurement data is stored inside the myoRESEARCH installation directory. This option allows the user to choose a specific location for the database, which is where myoRESEARCH will read and store data instead.

To create an external database, select the desired path. myoRESEARCH will then display a message and restart to implement the change to the database in use.



myoRESEARCH will look for already existing databases. In case this is the first time to define a new external database, it will offer to create a new one:



Confirm with **Yes**. myoRESEARCH will start and now use the new data directory path. To change back to the original internal database, re-enter the Software setup again and uncheck **External Database Folder**.

An alternative way of creating an external database is to export Subjects/Records in the **Database** menu and export dialog located at the right toolbar (see instructions for **Database**).

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Attention: Do not use Windows Explorer or comparable file/editor tools to copy data into or out of myoRESEARCH external databases. myoRESEARCH requires certain naming conventions and initialization files to run its complex database structure! Always use the myoRESEARCH export "to external location" function in the Operations Menu of the Database menu to export/import data.

Once created, myoRESEARCH can toggle between external and internal databases by changing the data directory path in Software setup.

***Attention:** If you plan to use networked storage, keep in mind that the loading records may take longer due to the slower data transmission speed of wireless storage. It is not recommended to use networked storage for recordings.*

To learn more about data backup, management, and sharing in MR4, please visit this guide on the Noraxon website.

Compress Video

This function compresses the data files captured from standard and high-speed cameras integrated in the setup to $^{1}/_{10}$ of its original size. This helps manage large databases and keeps them as small as possible. Because video compression requires time to process, it may not be possible to always compress videos in real time, especially if high speed video is involved. As a result, myoRESEARCH offers 2 different operation modes of video compression:

Manually	Default mode. Videos are not compressed in or after saving a record but can be compressed afterwards by using the **Database** menu function **Recompress Video.** This mode is recommended in multi-device setups that have several recording cameras and measurement devices.
When saving data	After each recording, the videos are compressed. ***Note:** This operation may double or triple the time needed to save data.*

Change Password

If needed, myoRESEARCH can be password protected. Enter the password dialog and type in a new password, confirm it again and press OK

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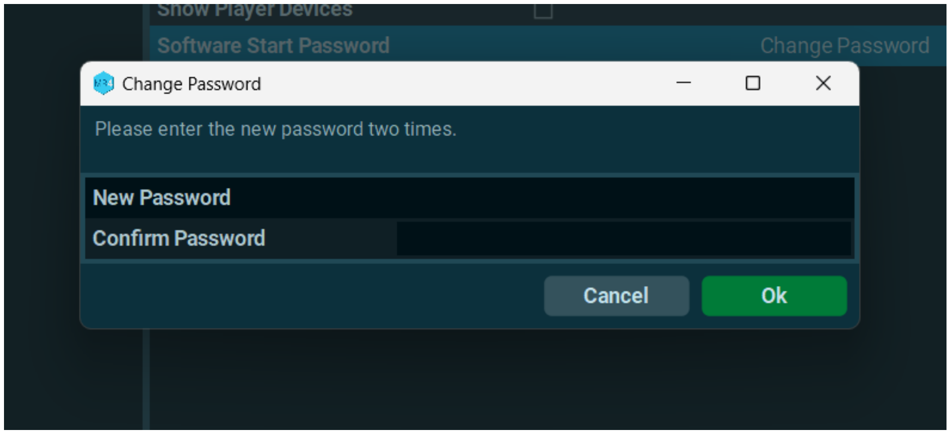


At the next program start the new password must be entered. To change a password, navigate to the Change Password menu again and enter the old password. The next two entry lines create a new password. If the boxes are blank, there is no password currently setup.

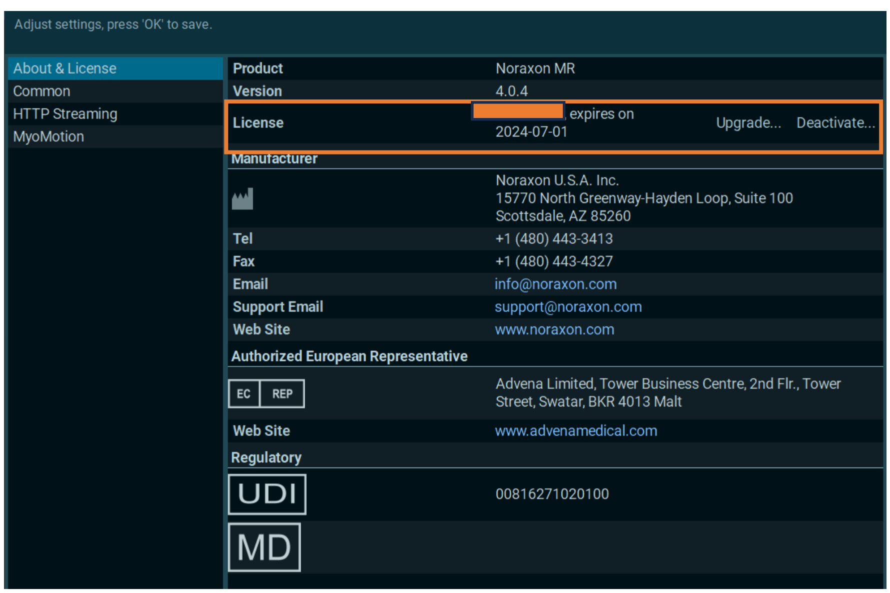
<u>HTTP Streaming</u>

If enabled, myoRESEARCH will stream data to the specified port address. HTTP stream runs in parallel to the regular recording and requires a measurement to be started.

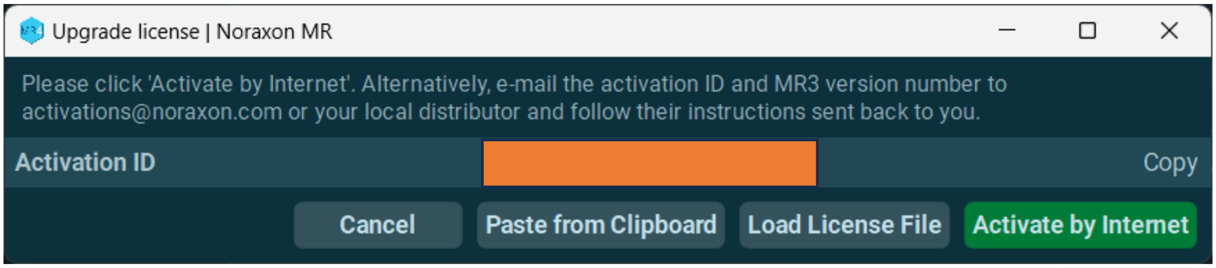
License

The software and each given module within are license protected. To upgrade an existing license with a new module click Upgrade and follow the on-screen instructions.

The license ID is already pre-filled from the initial activation routine when installing the software and may need to be changed if instructed by the Noraxon service engineer.



Click on **OK** to continue. A new dialog box will open, with several options available to receive the updated license file.



The recommended option to select is **Activate by the internet**, which requires that the PC is connected to the internet.

If internet access is not available, click **Copy** to copy it to clipboard and paste it into an email to support@noraxon.com. A support engineer will email an activation string, which should be copied to the clipboard. Select **Paste from clipboard** to load it for this activation routine.

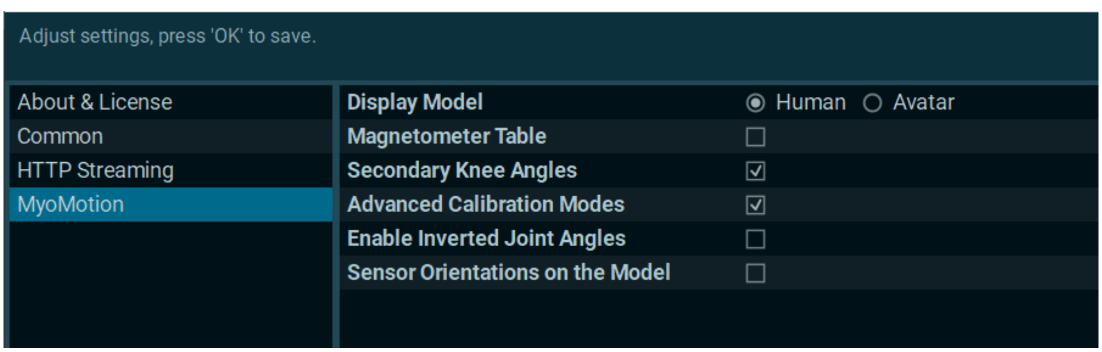
Deactivate License

This function is used to migrate the license to another PC. First deactivate the license on the old PC and install the software on the new PC. Proceed to use any of the previously described methods to reactivate the license. *Note: This process requires an internet connection.*

myoMOTION

This software setup section is meant to be used by myoMOTION (Inertial Sensor System) users.

Attention: It is recommended that functions be configured by an inertial sensor specialist. They require deeper knowledge of inertial sensor technology and kinematic application techniques.



Display Model

Select "Human" to display a human skeletal representation of the subject during 3D motion capture, or "Avatar" to use a 3D motion avatar.

Magnetometer table

If checked, a table will be shown in the right toolbar of the myoMOTION measurement screen. This data will show the magnetometer vector magnitude, dip angle, and the delta values related to the given overall weighted mean value. This is intended to assist with characterizing the magnetic environment of a test area.

Secondary knee angles

This option allows one to switch the knee joint from 1D (Flexion – Extension) to 3D (plus rotation and varus/valgus).

Note*: the latter two angles are heavily influenced by sensor fixation techniques and temporary soft tissue motions within a given activity, and should be evaluated carefully during dynamic activities.*

Advanced Calibration Modes

Enables the selection of three additional calibration modes for myoMOTION recordings, useful for subjects unable to perform Walking or Forward Lean calibration. Enables Multipose, Static, and Static with Ref Poses (see Motion Capture Systems setup for details)

Enable Inverted Joint Angles

Allows the calculation of negative joint angles to capture data such as elbow extension from motion data.

Sensor Orientations on the Model

Includes sensor orientations on the visual model – useful for troubleshooting or support.

Hardware Setup

The setup menu connects and assigns measurement hardware to myoRESEARCH. When opened for the first time, there are no devices shown in the Selected Devices section. Turn on and connect all measurement components purchased with the software and check if they appear as an icon on the right side of the hardware selection screen labeled **New Devices**:

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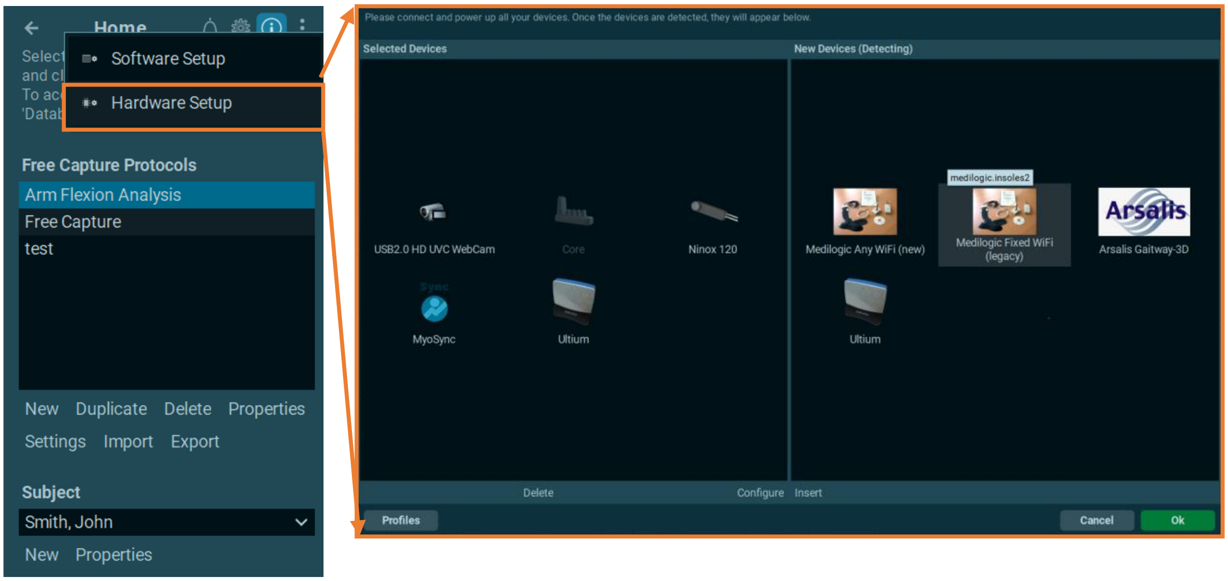
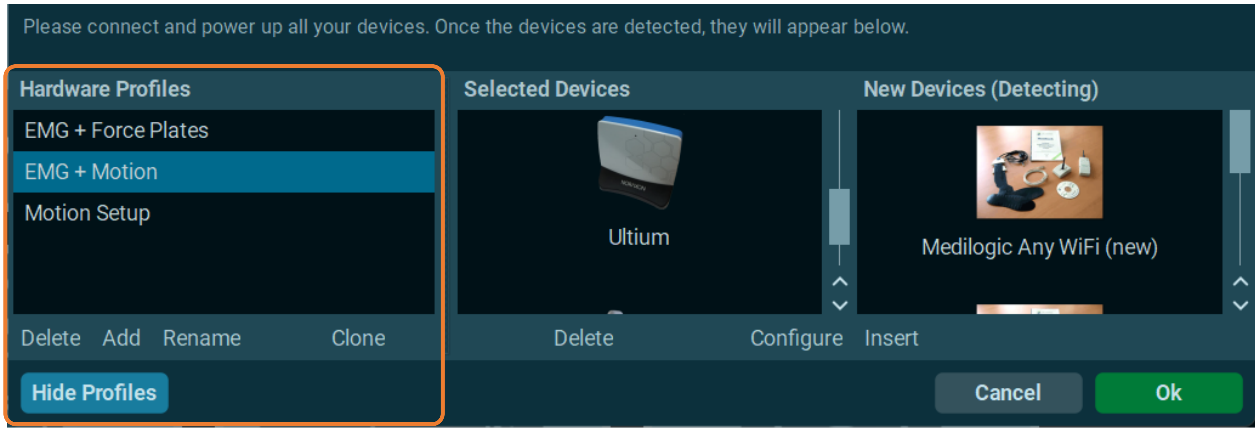


Note: Some devices such as Medilogic Insoles are already pre-filled even if they are not connected by USB cable. All devices labeled with "Player" are simulation devices to test/demo the software without having hardware at hand. The only way to access these "Player" devices is by enabling them in the Software Setup.

Each device requires additional hardware settings setup. This is specified in a device specific configuration menu. To enter these details, click the **Configure** button below the **Selected devices** list and follow the instructions within the menu. A more detailed description of the hardware configuration settings can be found in the hardware manual shipped with each device.

The lower left button, **Profiles,** creates different hardware profiles. Four different icons exist:

Add	creates a new hardware profile.
Clone	duplicates the selected profile.
Delete	deletes the selected profile.
Rename	renames the selected profile.



An unlimited number of hardware profiles can be created. The user can toggle between them at any time. The function to create several hardware profiles is especially helpful in instances where the user wants to change sensor assignments for different applications. For example, a smaller set of 7-8 Ultium Motion sensors can be assigned to unique upper body or lower body setups. In another scenario, the given sensor assignment for Ultium EMG and Ultium Motion can be assigned to certain channels within a 16-channel system.

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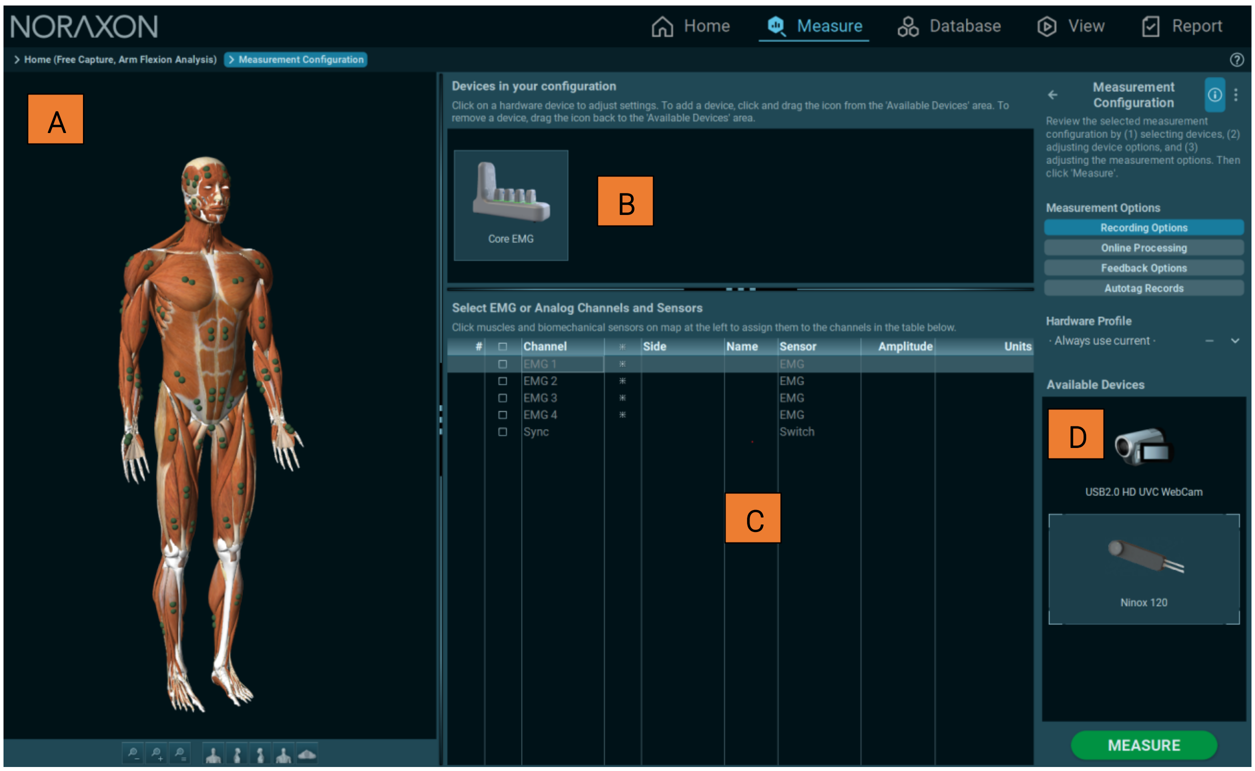
Measure

Measurement configuration

Introduction

In myoRESEARCH 4, clicking the **measure** button from the **Home** screen will send you to the active measurement screen. However, if the protocol's hardware configuration is empty (such is the case when a protocol has just been created), or if the **settings** button is clicked on the protocol list, myoRESEARCH will instead open the **measurement configuration** screen.



The **measurement configuration** screen consists of four major sections:

[A] Setup Visualizer — located on the left of the screen, displays relevant visualizations for the selected device (muscle map for EMG, skeletal model for myoMOTION, camera output for cameras, etc.)

[B] Configured Devices — displays a list of devices active in the current configuration.

[C] Device Options — located under configured devices, displays various device-specific options (sensor placement, camera gain/focus, etc.)

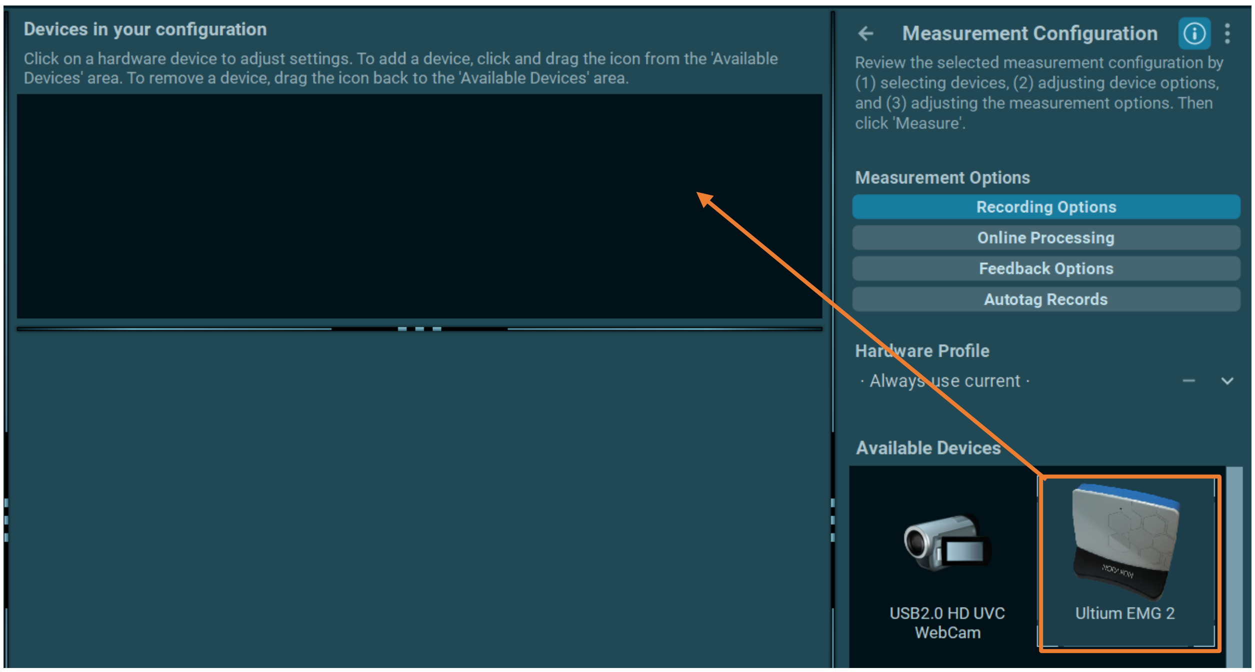
[D] Available Devices — displays devices active in **hardware setup** but not in the current **measurement configuration**.

In myoRESEARCH, devices fit into one of six categories/modules(myoMUSCLE, myoMOTION, myoFORCE, myoPRESSURE, myoANALOG, myoVIDEO), and all devices in a specific category will follow a similar setup process.

This document provides an overview of the workflow process and setup in myoRESEARCH – please see your device-specific manual for more detailed information on measurement configuration options.

EMG Systems

All measurement/recording devices inserted in the initial Hardware Setup dialog menu are listed on the right-hand side in Available Devices.



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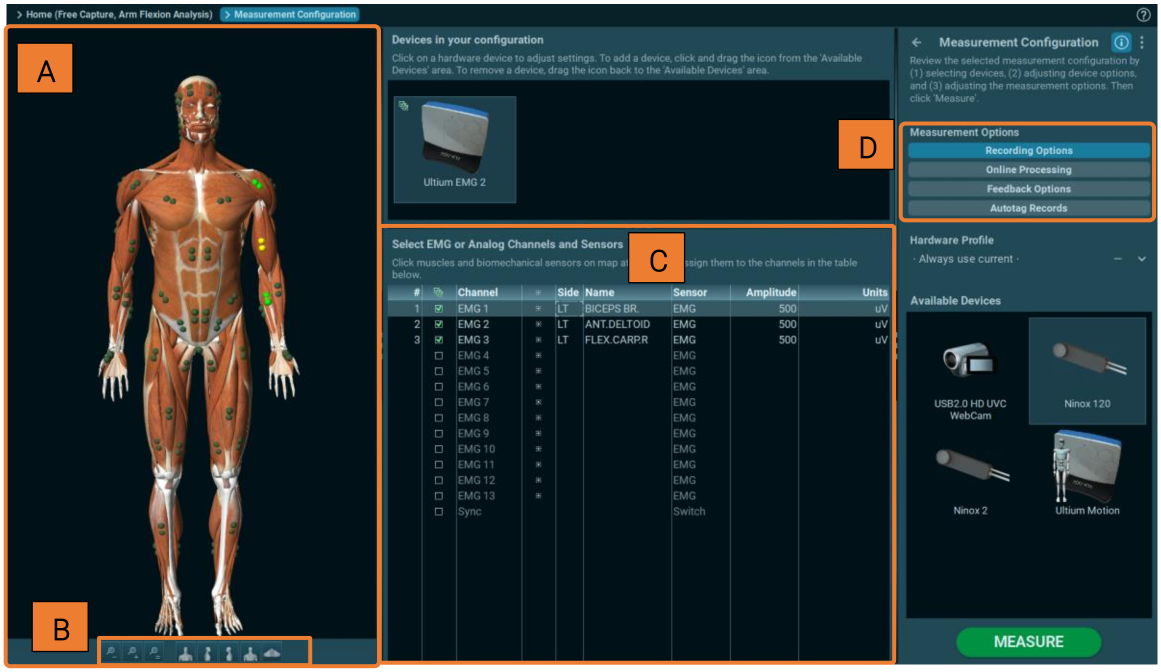
Click on the EMG system icon and use the mouse to drag it to **Section 1: Devices in your configuration.** When done, the myoMUSCLE sensor selection screen will appear:



[A] 3D-Muscle Map Click on the desired muscle sites in numerical order. For example, EMG 1, EMG 2, EMG 3, etc.

[B] Map Rotation Tools Use these tools to rotate, zoom and move the 3D animated muscle man.

[C] Channel Selection List Each selected sensor will automatically be inserted to the list of selected sensors. This list has more edit functions for individual naming (Column Sensors), side selection (Side), and amplitude scaling range (Amplitude).

[D] Measurement Options These options are related to real-time recording and will be explained in an extra chapter below.

Sensor Map

The sensor map manages all non-EMG sensors such as acceleration, goniometer, and heart rate. It appears in two modes:

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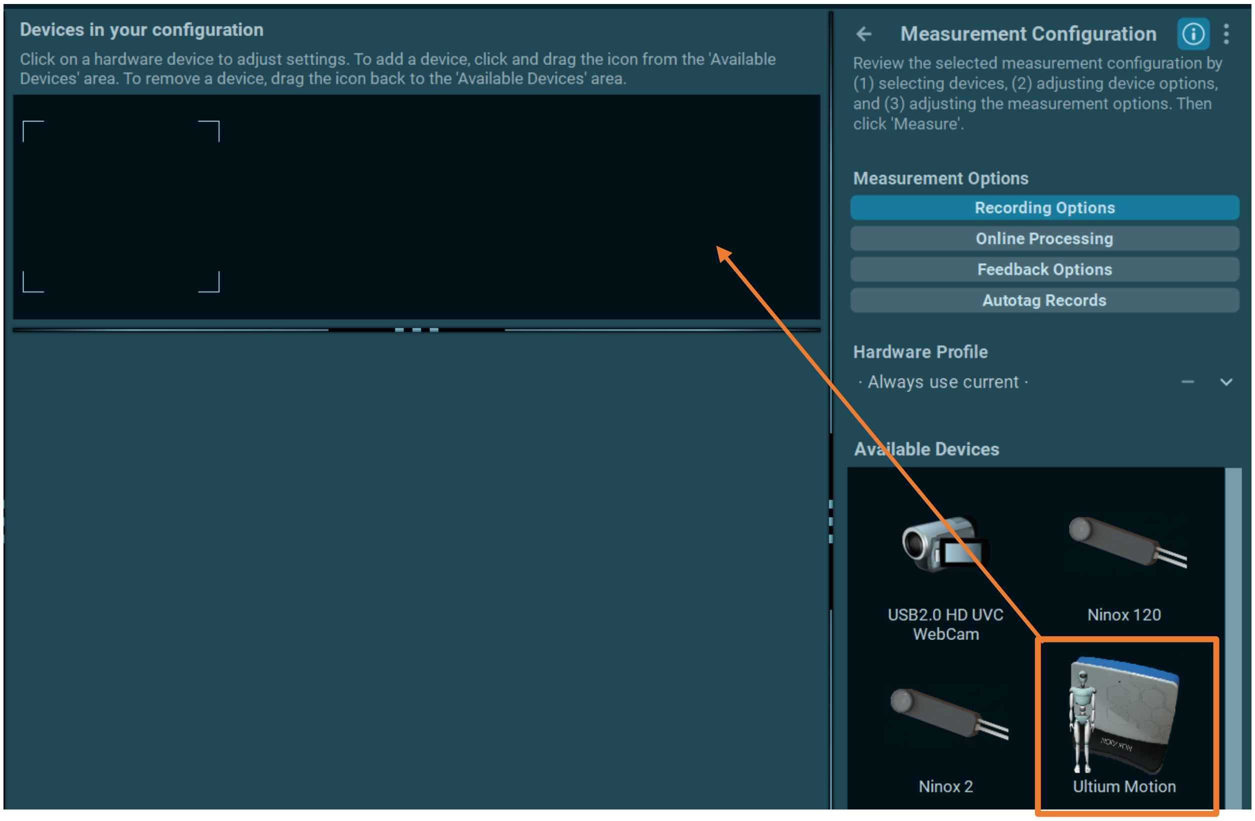
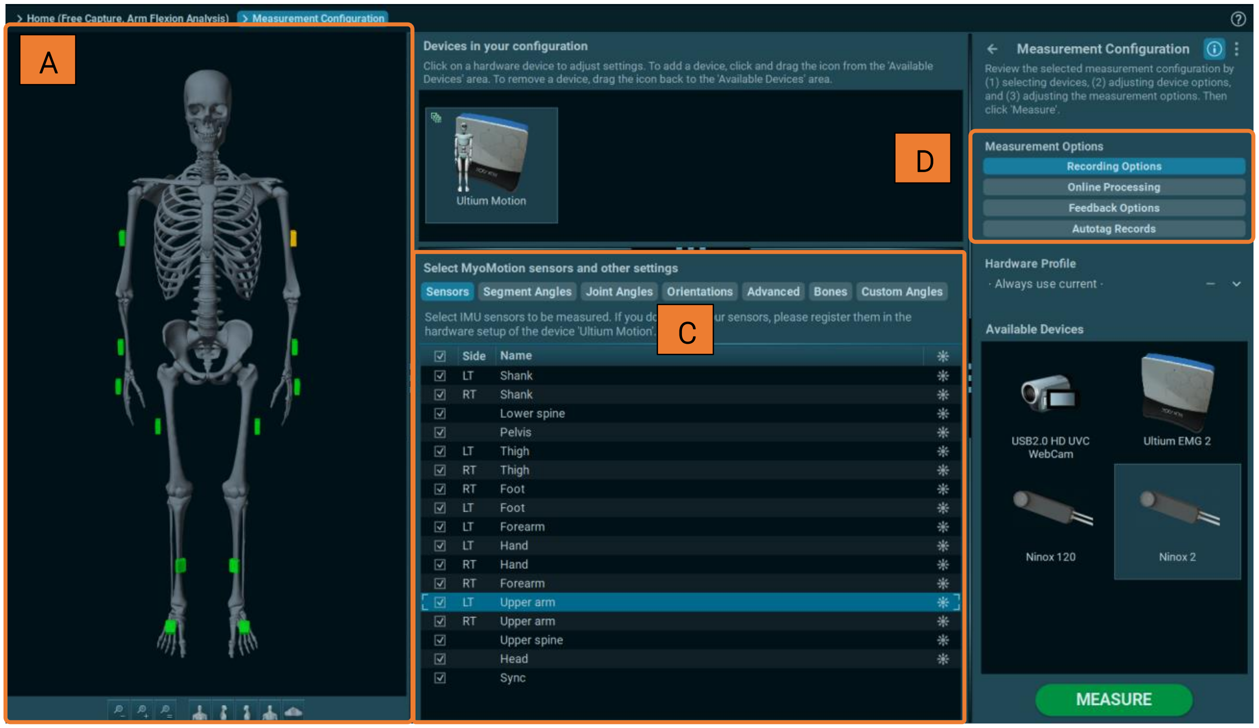
Motion Capture Systems

All measurement/recording devices that were inserted in the initial **Hardware Setup** dialog menu are listed on the right-hand side of the screen. Click on the **myoMOTION** icon and drag it to **Devices in your configuration**.



When done, the **myoMOTION** sensor selection screen will appear:



A) Sensor Location & 3D Zooming and Rotation Tools

The **IMU** map is a 3D free rotational map. Click and drag with the mouse to rotate the skeleton model and select a sensor location marked on the skeleton.

Click on the desired bone segment sites. Use the sensor location icons and click on all sensors that should be activated for the recording. For example, clicking on the pelvic sensor on the model will turn it **green**, and the check mark will be set on the right side of the **Sensors** tab list. Selecting any sensor from the list, enabled or not, will turn the sensor **yellow** to show where it is located on the skeleton map.

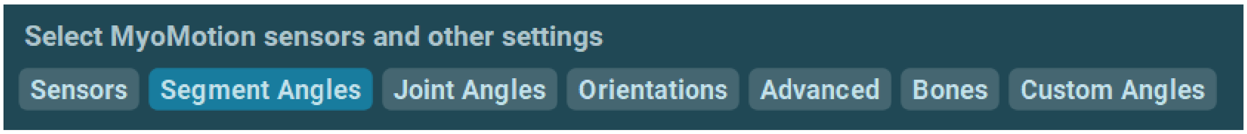
Note: Only the sensors assigned in the hardware device setup menu (see hardware manual) appear on the map. In the above example, all sensors from a 16-sensor full body system were assigned and are selectable on the map.

These tools can zoom, center, change view and freely rotate the skeleton. Alternatively, mouse buttons below can be used:

Left Mouse	Rotation
Middle Mouse	Move
Scroll Wheel	Zoom in and out

B.) Select myoMOTION recordings and Other Settings

This subsection is organized into 6 different lists.



Sensors Tab

Each available sensor assigned in Hardware setup will automatically be inserted to the list of selectable sensors. Sensors can be selected via skeleton map (see A) or directly here in this list. Once a sensor is selected, anatomical angles and other parameters are selectable in the other tabs.

If Object sensors were defined in the Hardware Setup menu of the Home Screen menu, they will be listed here. Object sensors are sensors outside of the biomechanical standard model consisting of 16 bones as seen in the 3D sensor selection map. Object sensors can measure the rotation angles of objects, tools, or sports equipment in relation to a Base sensor of the regular avatar.

To visually position object sensors in relation to the skeletal avatar, Placement and Distance can be defined.

Segment Angles Tab

The Segment Angles Tab contains a list of orientation-related data from enabled sensors. This information is the calculated orientation of the segment(thigh/hip). The key difference between Segment angles and Joint angles is that joint angles are calculated from the relative orientation between two segments(such as the thigh and hip in this case), while segment angles are the orientation calculated with respect to the environment, otherwise referred to as the progression frame. The Advanced tab contains options for modifying how the progression direction is calculated.

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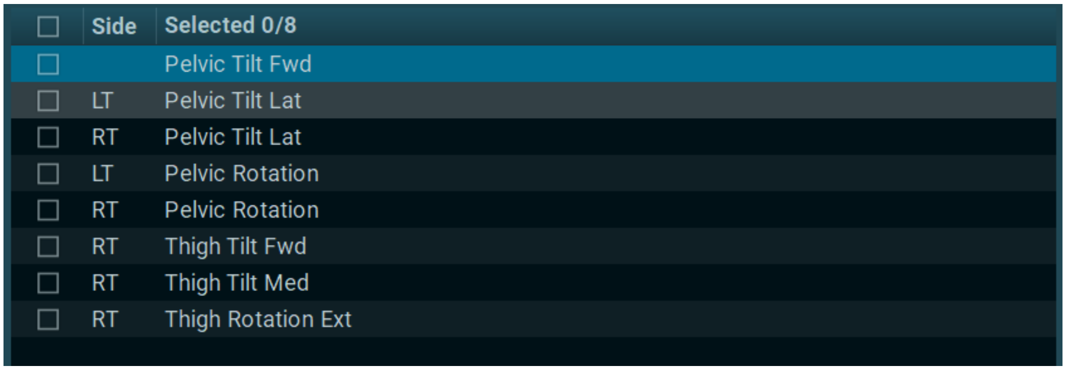
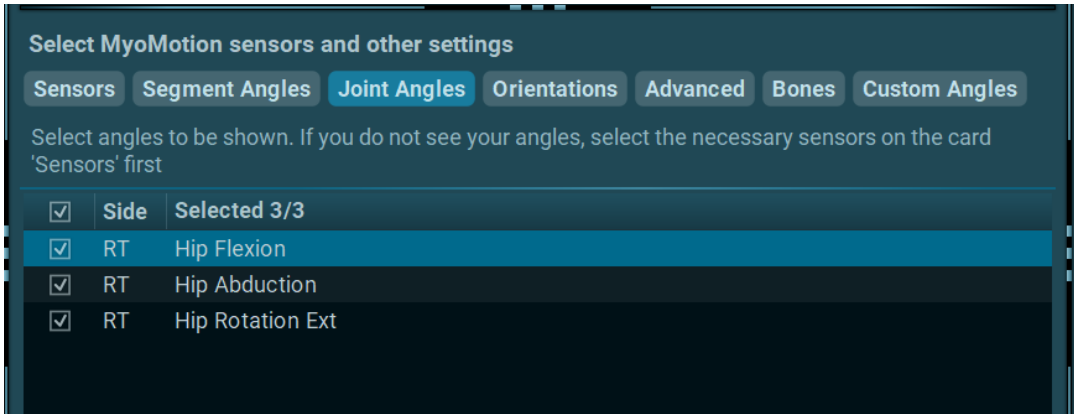


By default, segment angles are not enabled when sensors are selected. In general, joint angles are more commonly used for analyzing human biomechanics, but segment angles are available for more specific use cases.

Joint Angles Tab

As soon as two sensors surrounding a joint are selected, they automatically compile up to three joint angles.



In the example above, the Pelvic and Right Thigh sensor were selected. This resulted in right side Hip flexion, abduction, and rotation.

The **polarity** (positive or negative data in the angle curves) of the movement is dependent on the plane of motion. Each joint angle motion consists of a positive and negative component. The motion direction listed in each channel name is equal to the positive part of the curve. For example, “Cervical Flexion” means the flexion motion is positive, and extension motion is negative.

For body and trunk angles, the underlying Cardan rotation sequence follows the recommendations given by the International Society of Biomechanics (ISB). For example, the

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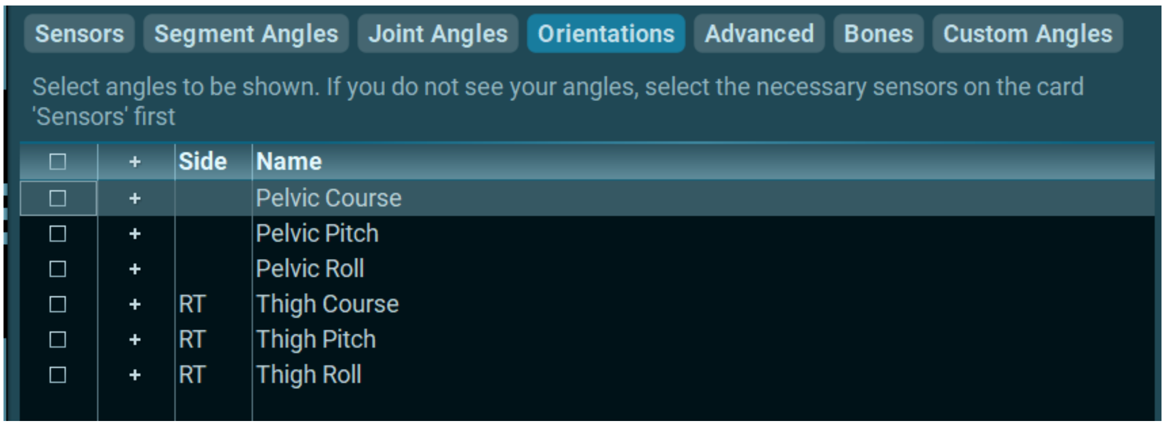
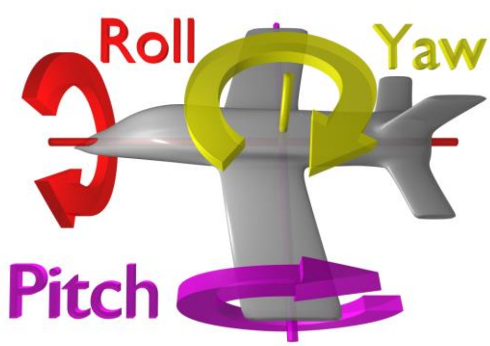
rotation matrix of the knee joint is flexion, abduction, and rotation. For the shoulder joint, the package uses planar angle calculations, since ISB recommends adjusting rotation matrices to the primary motion plane of a given activity. For more information around the biomechanical model and applied algorithms please refer to the inertial sensor hardware manuals available [on the Noraxon Website.]

Orientations Tab

If only one sensor is selected, there is no anatomical angle listed. Single sensor data can only be shown as acceleration and/or orientation angles **Course**, **Pitch**, **Roll**, as shown in the example below:



These orientation or navigation angles describe the *angular* displacement of the sensor in space. This can be visualized via an airplane flight maneuver:



Note: The course angle is known as Yaw or Heading.

By default, orientation angle recording is turned OFF and can be activated for each dimension via the check box. **Invert** changes the orientation polarity or sign of the recorded signal. By

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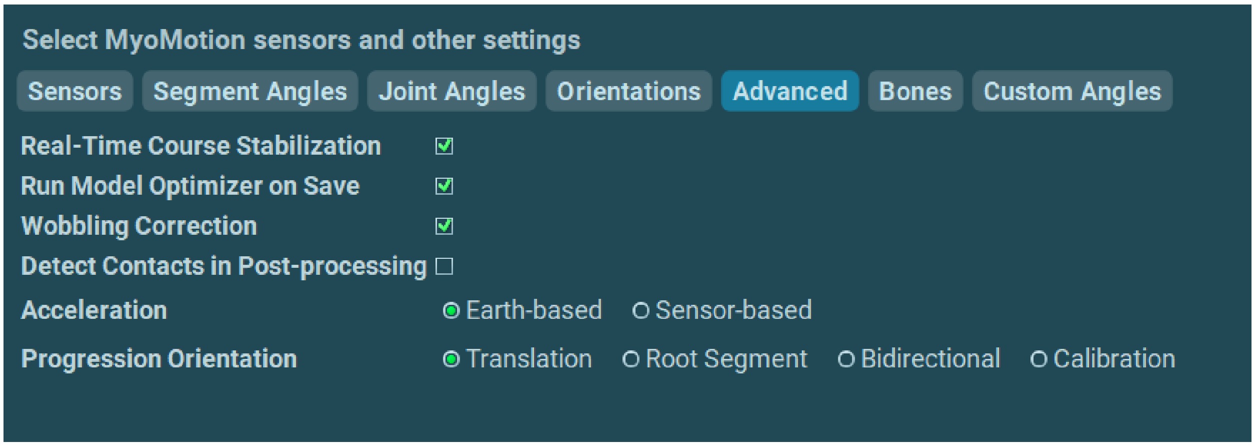
double clicking on the orientation names, the default names can be overwritten with user defined names. For example, the name “Pelvic course” can be changed to “Course.”

Advanced Tab

The **Advanced** Tab contains additional options for configuring the myoMOTION model:



Real-Time Course Stabilization	Enables real-time model drift correction while data is being collected – only affects real-time visualization of model and data. Enabled by default.
Run Model Optimizer on Save	The model optimizer automatically compensates for sensor artefacts upon save. Enabled by default.
Wobbling Correction	Reduces high-frequency motion which is likely not representative of body movements, e.g. vibrations of the sensor relative to the body. Enabled by default.
Detect Contacts in Post-processing	Enable automatic detection of foot-ground contacts.
Acceleration	**Earth-based** – Acceleration calculated in directions relative to the earth(X – East, Y – local magnetic North, Z – centre of the earth) **Sensor-based** – Acceleration calculated in directions relative to the IMU housing(marked on IMU).

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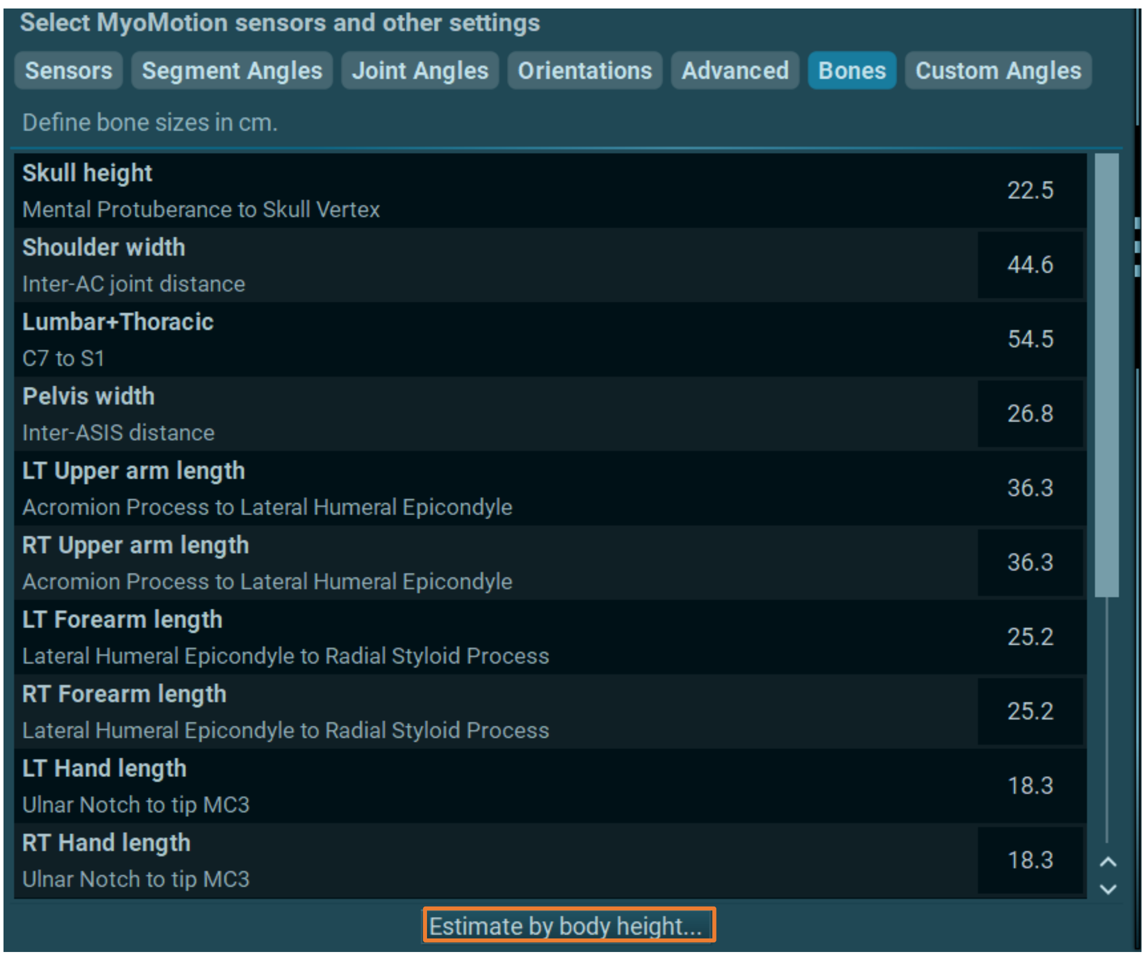
Progression Orientation	Determines how the progression direction is calculated from the subject's movement: **Translation** – Based on movement over the ground as the subject translates. **Root Segment** – Based on smoothed facing direction of pelvis, upper thoracic sensor if pelvis unavailable. **Bidirectional** – Based off of calibration direction, but polarity flips 180deg depending on direction(such as walking up and down a walkway) **Calibration** – Based off of direction during reference pose.

Bones Tab

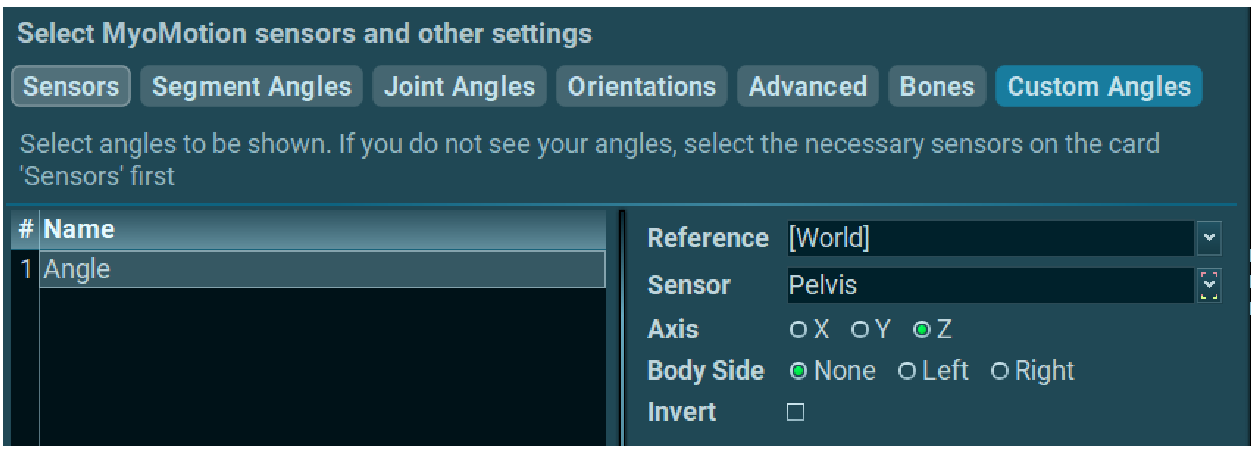
To allow for more realistic skeletal avatar animation, it is necessary to define the body height of the subject. Based on the height, the individual bone lengths are calculated. To enter body height, click on **Estimate by body height**:



Otherwise, individual bone data can directly be entered in the column Size (cm). Note: Bone lengths do not affect the angular calculations. The bone data is only used for more realistic avatar animation.

Custom Angles Tab

A special version of orientation angle is available in this tab section. Each available rotation axis (X, Y, Z) can be specifically assigned as a first rotation axis. This allows singularity points (which produce data clipping) to be avoided and for sensor data to freely rotate. The angles can be expressed in reference to the world or any available sensor in the given measurement setup. To create a custom angle, click the **New** button at the bottom of the local menu:



Axis defines the main axis and only this axis will be shown as an angular trace. The other axis can be shown by creating a new custom angle and selecting the same sensor again. Invert will change the polarity of the angle curve. An unlimited number of custom angles can be detected.

C.) Measurement Options

Some additional options to manage, control and operate a measurement are available here. For detailed explanations on each aspect of this section, refer to Measurement Options

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Pressure Distribution Systems

All measurement/recording devices that were inserted in the initial Hardware Setup dialog menu are listed on the right-hand side of the screen.

There are three types of pressure devices compatible with myoPRESSURE:

1) Small floor balance or foot roll-over plates (< 1 m size)
2) Floor Pressure gait plates (1.5 to 2m size) or pressure plate instrumented treadmills
3) Pressure insole system (Ultium or Medilogic)

Click on the myoPRESSURE device and drag it to Devices in your configuration.

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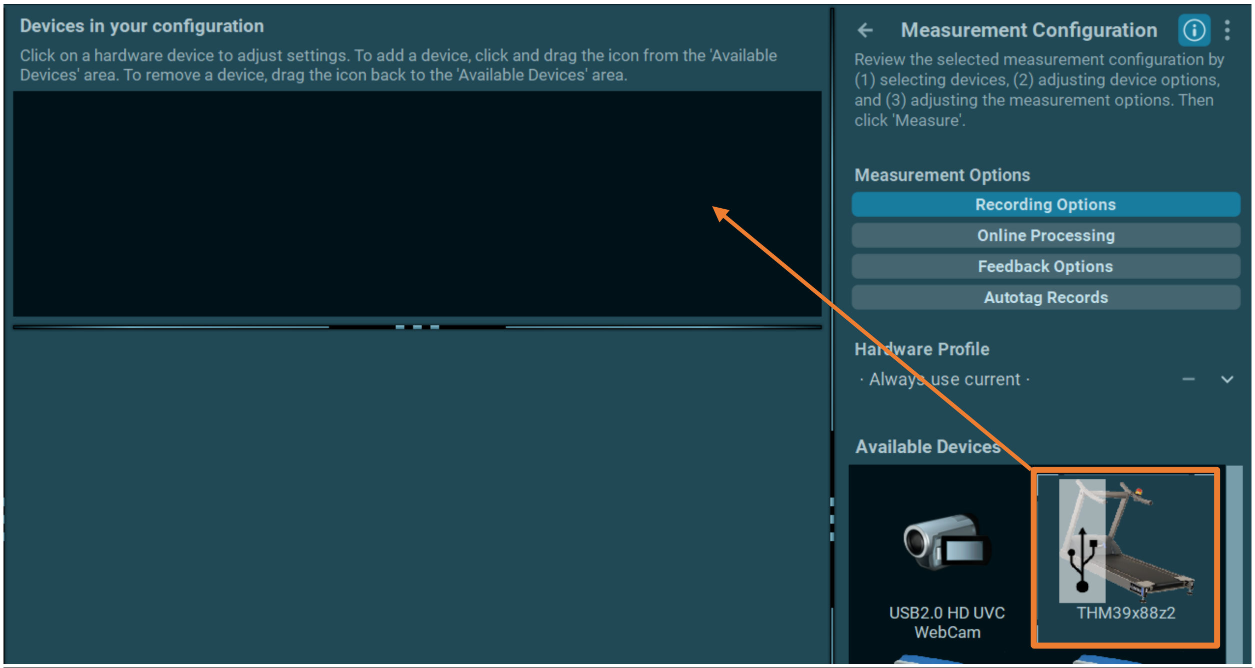
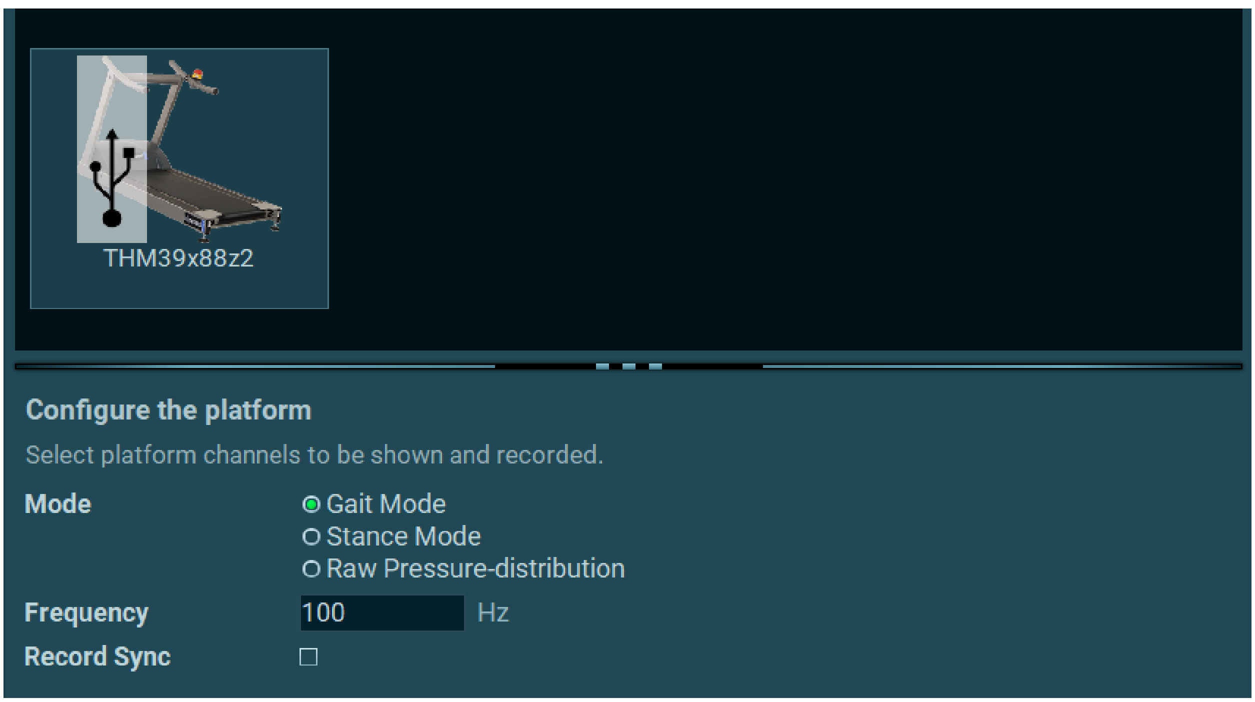


When done, the myoPRESSURE sensor configuration screen will appear:



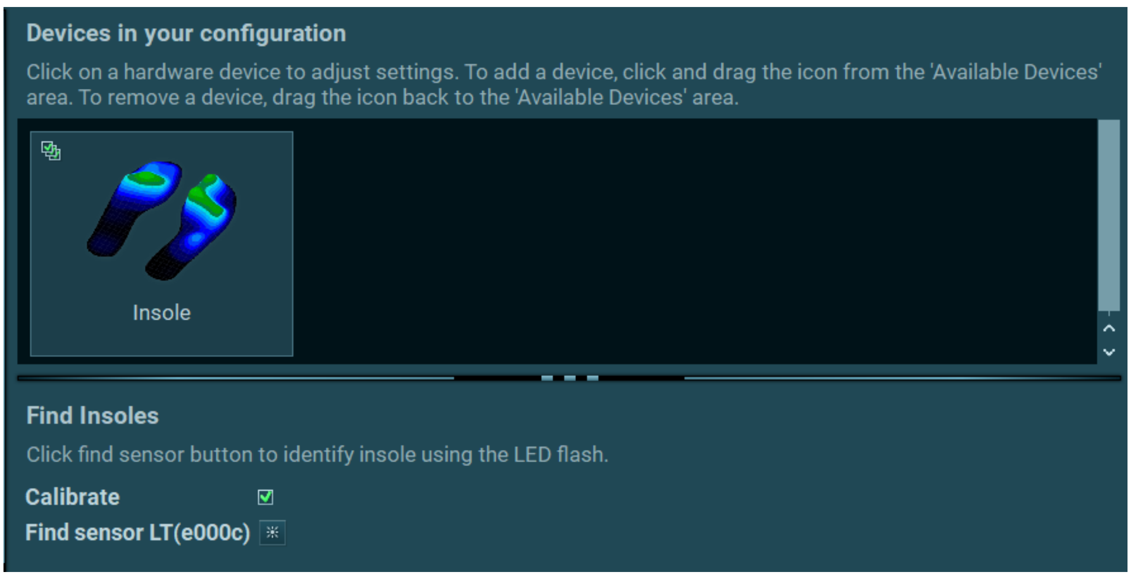
There are several different device modes to choose from when adjusting the settings for the pressure plate or treadmill. The available options for the selected device may vary depending on the type of device:

Gait mode	for bilateral gait and running analysis on instrumented treadmills
Stance mode	for bilateral stance analysis on instrumented treadmills and floor plates
Rollover mode	for bilateral rollover analysis on instrumented floor plates
Raw Pressure	for access to the raw pressure distribution for the device
Frequency	changes the poll rate of the device (Hz)
Record Sync	enables viewing the sync signal in a channel

Small balance or roll over plates do not have any special measurement setup settings and are operational as soon as they are dragged in as devices.

Setting up insole pressure system

After dragging the insole system from the right toolbar to the section Devices in your configuration, the following option to **Calibrate** the insole system is offered:



If Calibrate is checked, the insoles can be calibrated to the subject's body weight. The **Measurement** menu will prompt the subject to place all their body weight on their left foot and then their right foot. This process removes any in-shoe pressure of the unloaded foot and calibrates the existing pressure of the loaded foot to body weight – see the device-specific manual for more information.

If Calibrate is not checked, this calibration step is skipped, and the insole system can directly be used without any further preparation.

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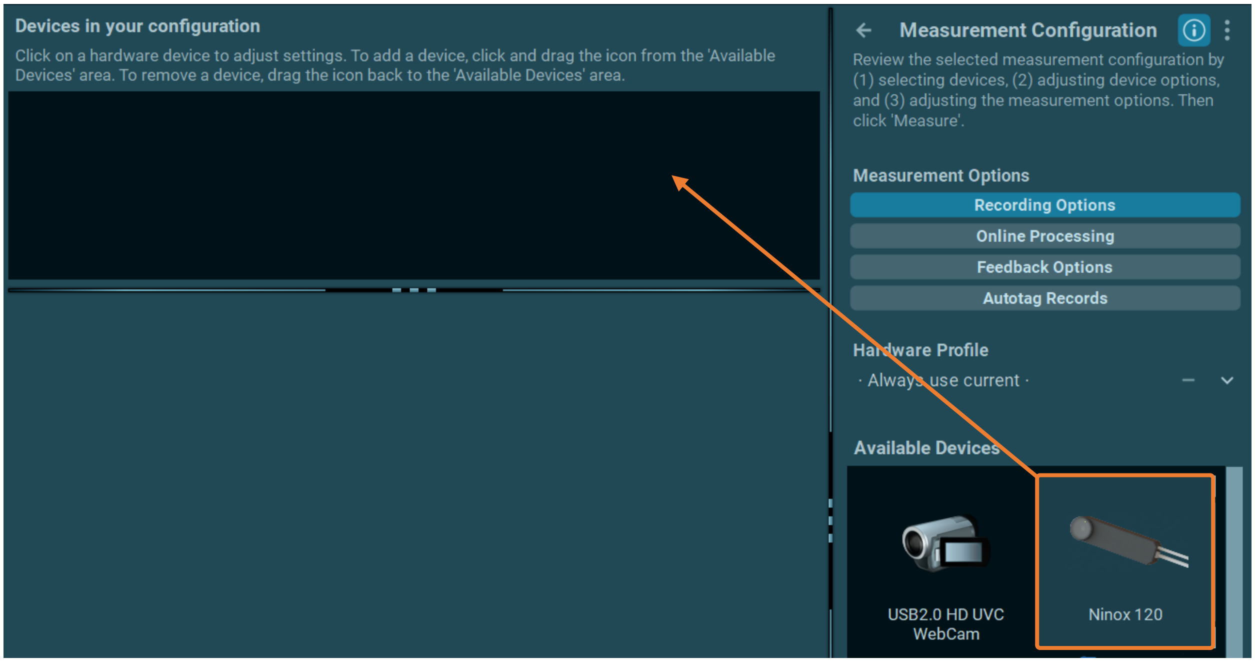
Ninox Camera Systems

All measurement/recording devices that were inserted in the initial **Hardware Setup** dialog menu are listed on the right-hand side of the screen. Click on the **myoVIDEO** device and drag it to **Devices in your configuration**.



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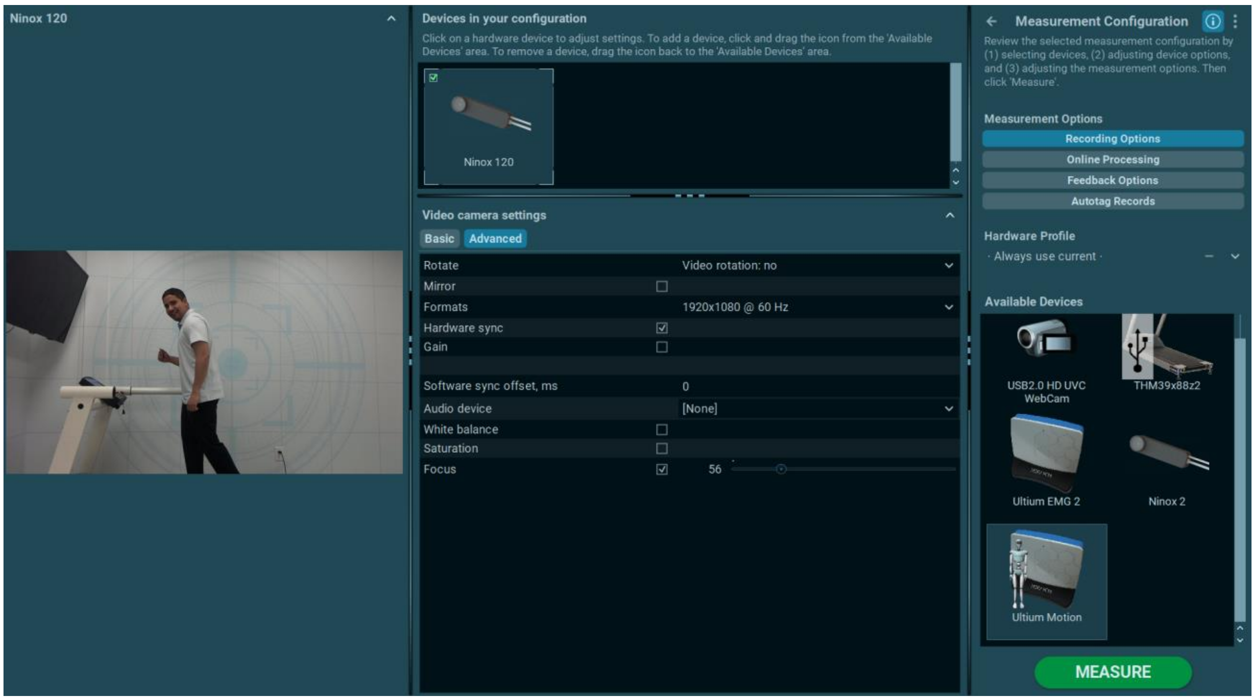
All measurement/recording devices that were inserted in the initial **Hardware Setup** dialog menu are listed on the right-hand side of the screen. Click on the **myoVIDEO** device icon and drag it to the section labelled **Devices in your configuration**. The **myoVIDEO** settings screen will appear:

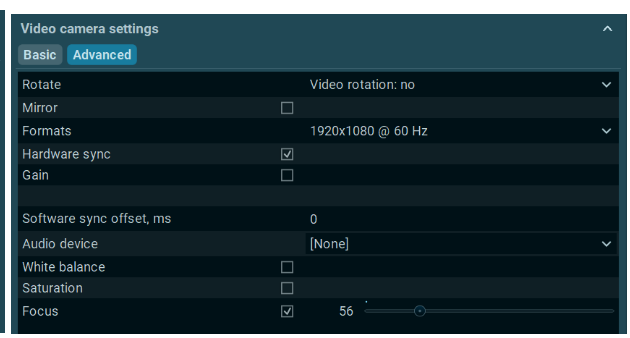


The screen is separated into two sections: the left side's camera preview picture and the two-step menu setup scheme on the right side.

Video Camera Settings

These settings are defined by the hardware setup menu located on the **Home Tab**. The camera settings can be adjusted in both the **Basic** and **Advanced** settings if the recommended settings do not fit the environment or given task.

Shown below is an example of possible Basic and Advanced settings windows (Ninox 120):



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The camera settings window may contain any combination of the following settings to adjust, depending on the device itself.

Rotate	If the camera position is rotated from landscape to portrait orientation, the video can be rotated clockwise or counterclockwise to match.
Mirror	Flips the camera feed across the vertical axis.
Format	The video resolution and framerate can be modified here, motion tracking performs better with higher framerates.
Hardware Sync	Hardware sync should be turned on when using multiple cameras or multi-device setups. It requires the myoSYNC or Video Sync Light systems.
Gain	This setting increases or decreases the overall light sensitivity.
Software Sync offset	This offset correction switch may be needed if there is a video delay between cameras and associated devices like EMG. Typically, 312 to 156 ms for 1500 and 3000 Hz, respectively. Please consult a Noraxon support specialist to learn whether this is necessary for your device. All cameras manufactured by Noraxon (the Ninox line) will synchronize automatically with Hardware Synchronization.
Audio Device	For Audio recording, within the video recording tab, select a microphone, usually the camera built-in mic can be used.
White balance	Corrects the color cast. Objects that appear white in-person are rendered white in the video.
Exposure	The exposure (shutter time) needs to be set to lower values when the auto tracking of markers is used (see below).
Saturation	Adjusts the intensity of color in the video.
Focus	The default automatic focus can be set to manual values. Auto focus should be turned off once focus is established.

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Hardware Sync for Video

The myoVIDEO module supports Noraxon's myoSYNC to synchronize hardware components within multi-device setups. The video cameras are synchronized when a compatible trigger is received from a MyoSync of 3rd party system.

The software and device setup can now be used to take measurements. The device setup can be modified at any time by going back to the hardware setup menu.

Force Plate Systems

myoRESEARCH is designed to measure data from 3D and 1D force plates. Models from **Kistler**, **AMTI** and **Bertec** are supported. The plate can be connected **digitally** via USB cable to PC or **analog** via BNC cable to Noraxon's **Analog Integration System (AIS)** analog input board.

Digital integration has the benefit of automatically presetting and prefilling all scaling and configuration data. Analog integration requires manually entering sensors and calibration data in the measurement setup menu.

Digital device setup and workflow will be explained in this section, with a **Bertec** force plate as an example. More detailed information regarding specific device manufacturers is linked at the bottom.

Analog system integration is also discussed in the following section.

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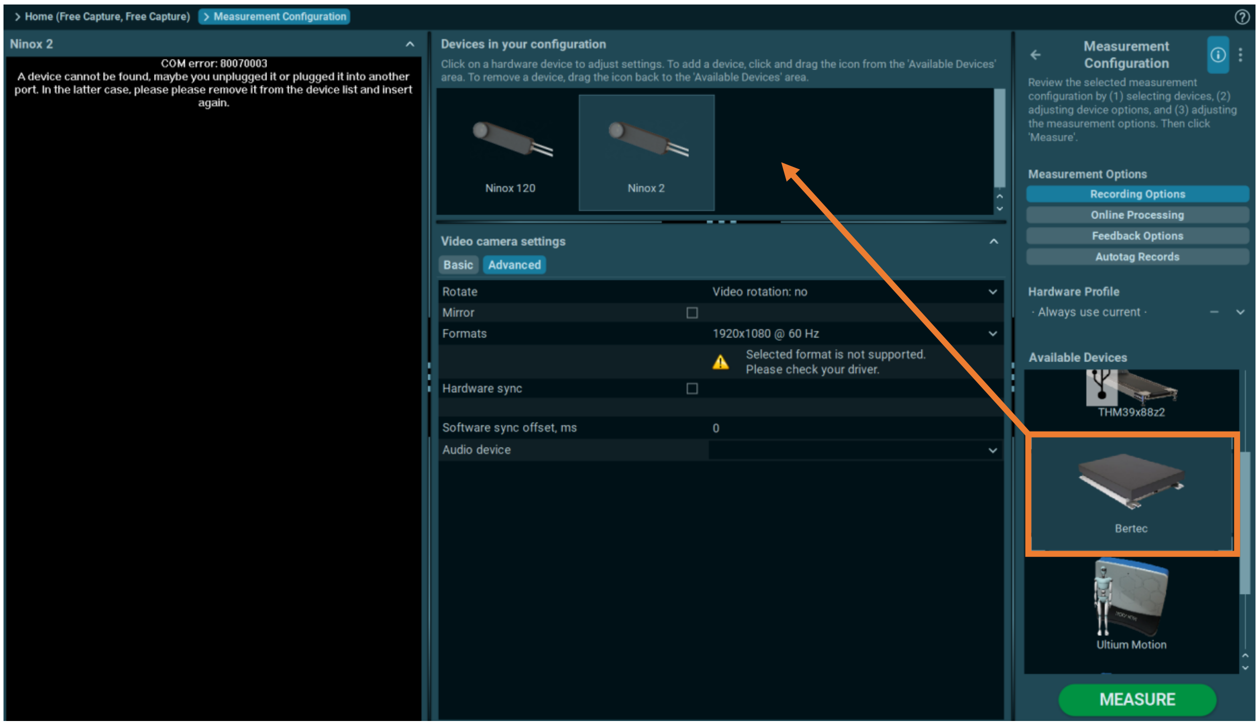
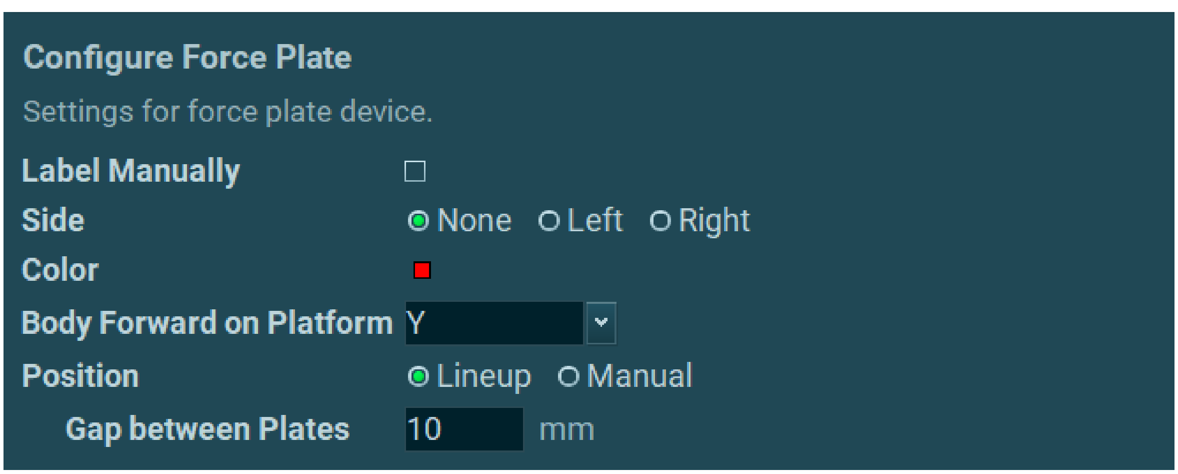
Create or Edit a Measurement Configuration (Digital integration)

Shown below is a demonstration of the Digital Integration of a **Bertec** plate. Drag and drop the plate icon into the section **Available Devices**:



Configuration settings are shown in **Configure Force Plate:**



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Label Manually	Allows the user to manually label the force plate with their own designation.
Side	Designate which side the force plate is located on in dual-plate setups.
Color	Modifies the color of any overlays tied to the force plate
Body Forward on Platform	Designate the subject's orientation relative to the force plate axes.
Position	Allows the user to determine the offset between two force plates – **Lineup** if the force plates are parallel to each other, **Manual** for two axis configuration.

Force plate setup is similar across the three different force plate brands – however, for more specifics, please view the relevant device/manufacturer-specific hardware manual.

Bertec Force Plate Setup Guide

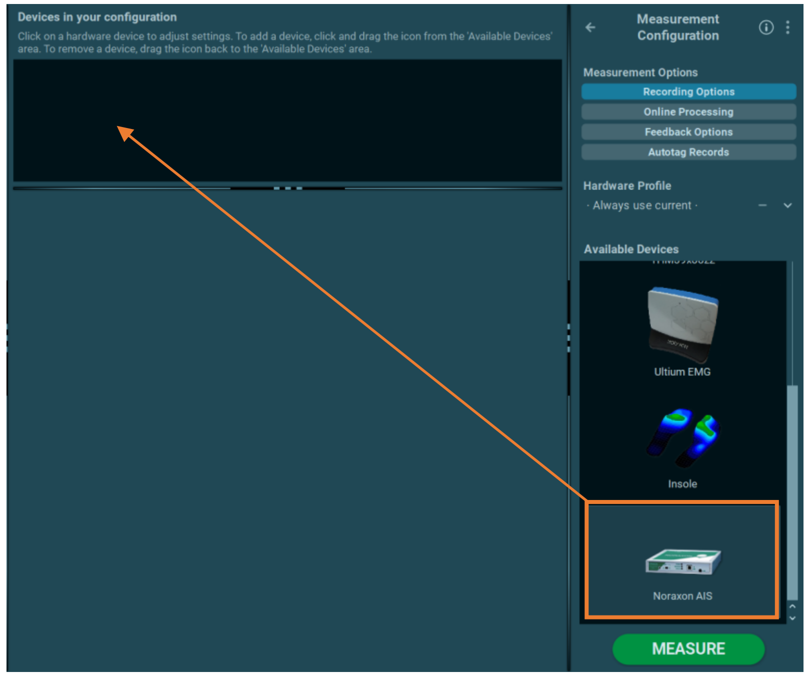
AMTI Force Plate Setup Guide

Kistler Force Plate Setup Guide

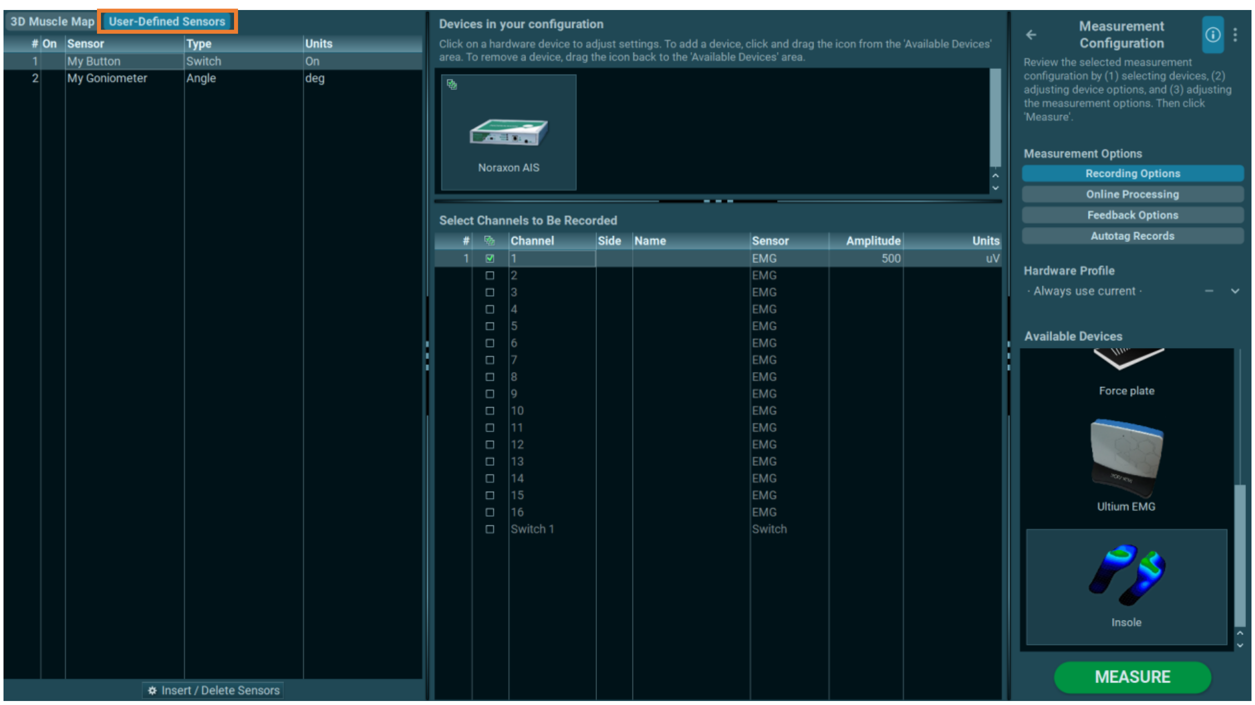
Analog Systems

myoRESEARCH can also be integrated with analog signals provided by compatible devices through the Noraxon Analog Input System(AIS). The most common application of analog systems is in the integration of analog force plate systems, as well as analog signals from devices such as isokinetic training devices. The following section demonstrates the **workflow** for adding an analog device for measurement in myoRESEARCH. For device setup specifics, consult your device-specific manual, as well as the Noraxon AIS manual for more information.

Enter any suitable file name and click **OK**. Select the **Noraxon AIS Board** device and drop it into the section **1) Devices in your configuration:**



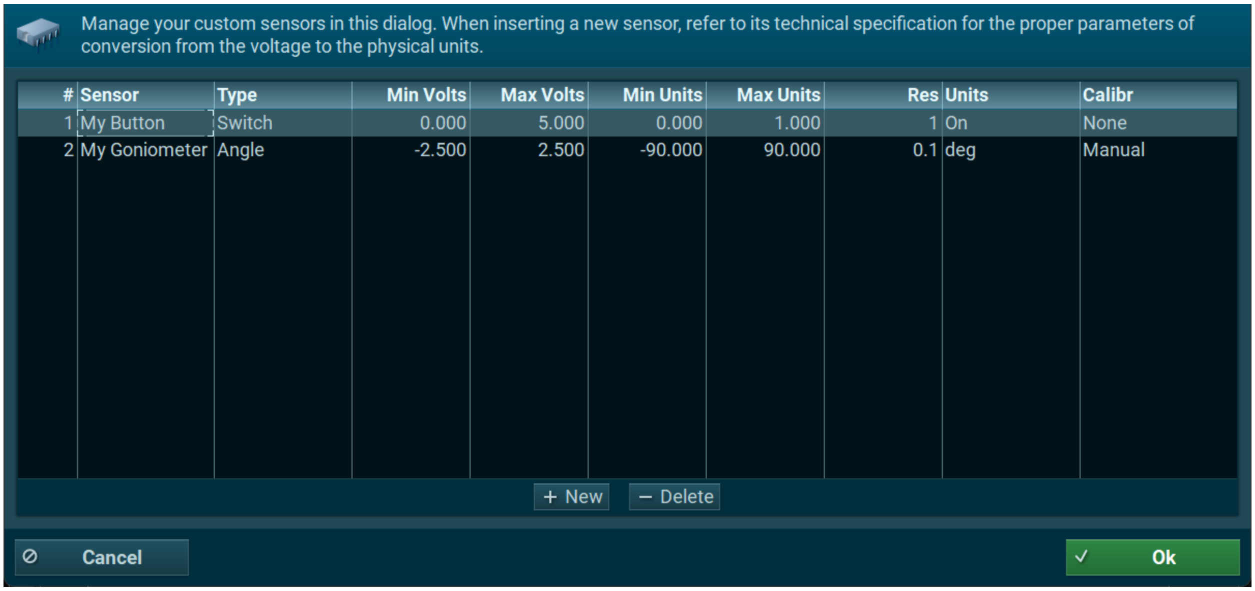
The following steps must be completed to set up the Noraxon AIS board:



1) Click on the device tab **User-Defined-Sensors.**

2) Click on **Insert/Delete Sensors.**

Each force channel (or sensor) requires a set of definitions and are operated on the next screen:



Click on **New** to add a new sensor and define:

Type	sensor type
Min/Max Volts	AD input range of sensor (up to +/- 10 Volts are supported)
Min/Max Value	calibration value corresponding to the AD range
Res	desired resolution of data in decimals
Units	the physical unit of the selected sensor type
Calibr.	defines the calibration mode to correct signal offsets. Automatic means that the start signal offsets are detected and automatically corrected within the measurement monitor. Manual means that the local calibration button must be pressed.

When done, all user defined sensors will be listed in the sensor selection list under **Section 3).**

3) Click on a sensor within the sensor selection list to assign it to a measurement channel under section **4) Select Channels to Be Recorded**:

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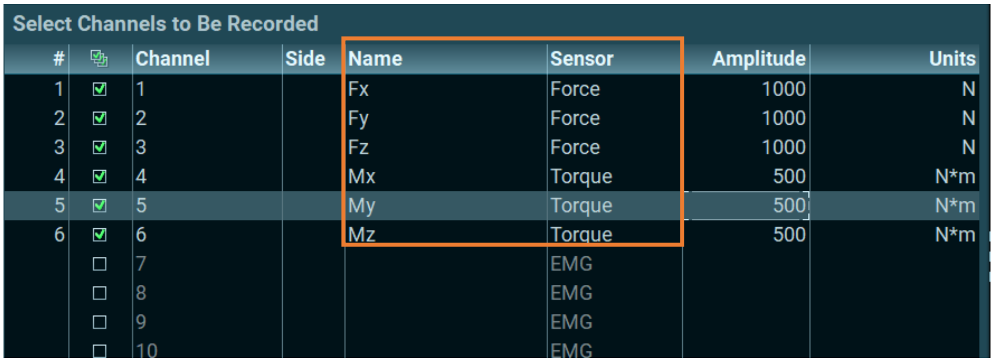
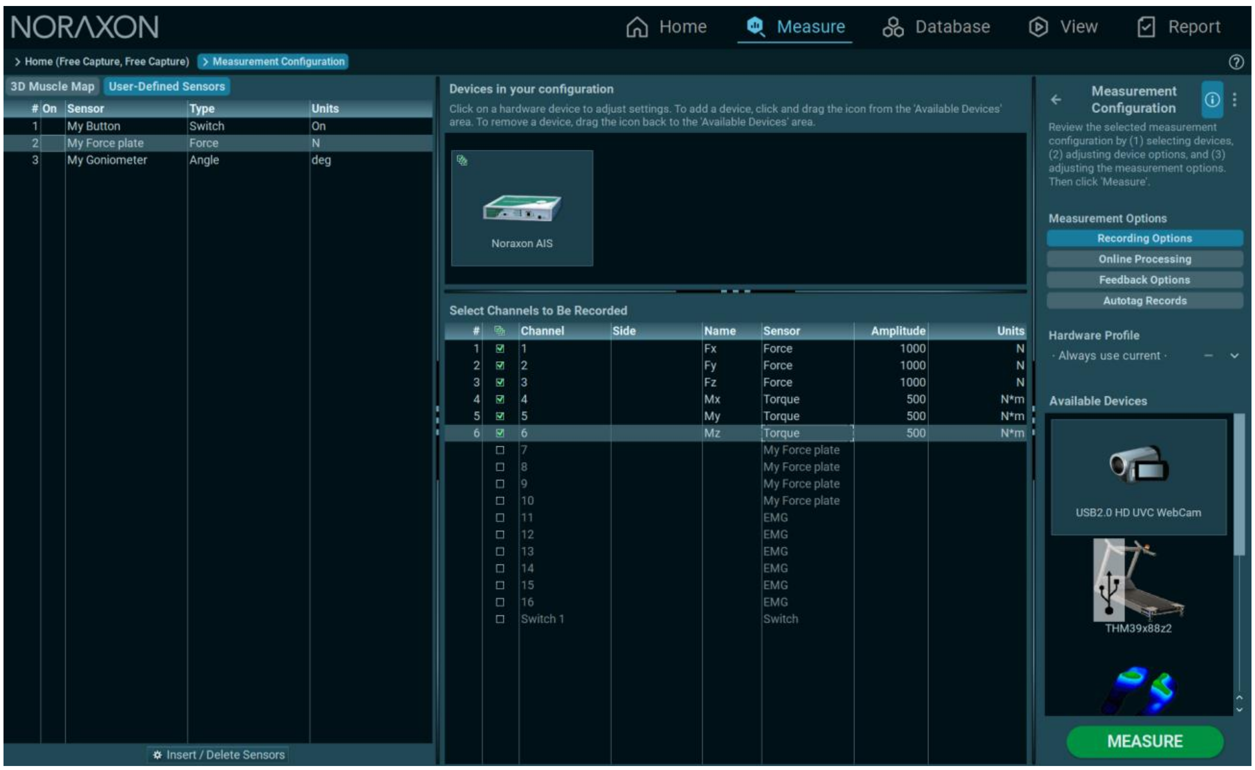
4) Each assigned sensor can be selected for a given measurement setup. Check or uncheck all sensors that should be recorded in this list. Enter the names Fx, Fy, Fz, Mx, My, Mz (green box) for the corresponding selected channels. *This is important for the force vector overlay function.*


Select Channels to Be Recorded

#	☑	Channel	Side	Name	Sensor	Amplitude	Units
1	☑	1		Fx	Force	1000	N
2	☑	2		Fy	Force	1000	N
3	☑	3		Fz	Force	1000	N
4	☑	4		Mx	Torque	500	N*m
5	☑	5		My	Torque	500	N*m
6	☑	6		Mz	Torque	500	N*m
	☐	7			EMG		
	☐	8			EMG		
	☐	9			EMG		
	☐	10			EMG		

The screen should now look like this for a setup with one force plate:



The setup is ready for measurement. Press Start Measure to start a measurement.

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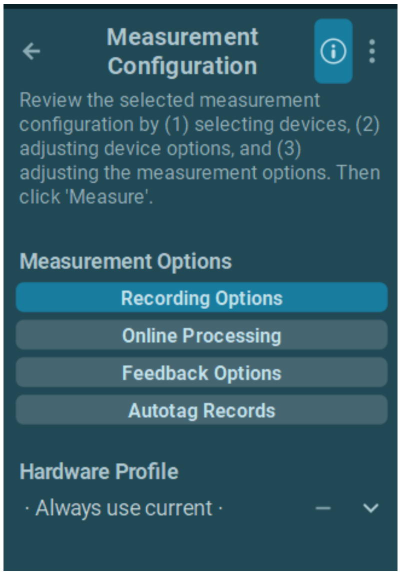
Measurement Options

Measurement Options are four categories of options that modify how data is collected, processed, and displayed to the user.

On the right side of the toolbar, the Measurement Options includes Recording Options, Feedback Options, and Online Processing.



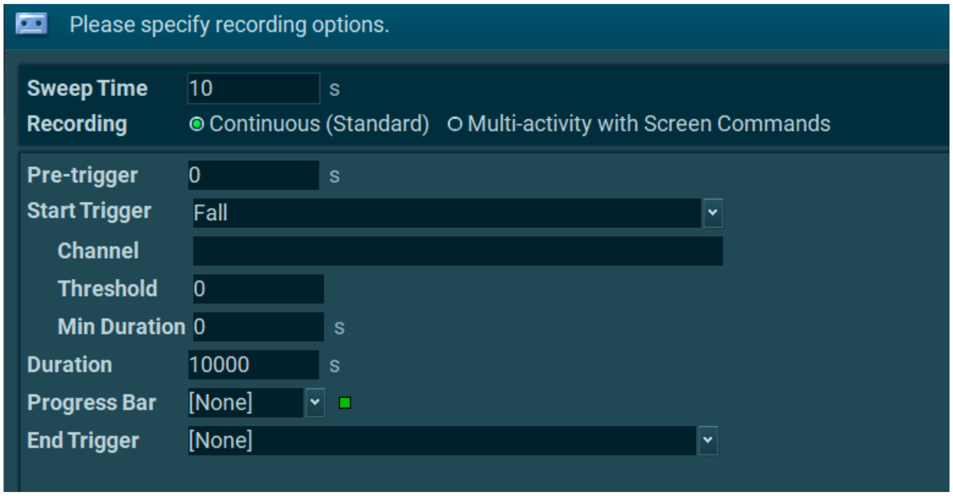
Recording Options

After selecting this option, two different recording modes can be selected:

- Continuous (Standard)
- Multi-activity with screen commands

Standard Recording Option

This sub menu manages the Trigger start and Pre-trigger settings. It cannot be combined with multi-activity recording.



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Sweep Time	defines the predefined time window shown on the measurement screen.
Pre-trigger	specifies the recorded portion before the trigger is initiated.
Start Trigger	defines the TTL type: Fall or Rise.
Channel	determines the trigger channel.
Threshold	defines the amplitude value that must be exceeded to initiate the trigger.
Min Duration	defines a minimum duration above or below the threshold to cause a trigger.
Duration	is a predefined time for the overall recording and works without a triggered recording start.
Progress Bar	activates a progress bar just below the signals screens to indicate the overall remaining time of recording.
End Trigger	defines the TTL type: Fall or Rise.

The recording can be set to have a starting and ending trigger, or it can record for a specified duration of time. The triggers are set according to the rise or fall of the signal on the specified channel.

Once the channel is selected, choose a threshold and a minimum duration that the channel must stay above to activate the trigger.

Multi-activity with Screen Commands

This menu creates and defines screen commands, assisting recordings containing a sequence of test activities. Voice commands can also be integrated. This function is meant to be used for standardized test sequences that have predefined activities, durations, and repetitions.

A more detailed explanation in video format is available on the Noraxon Website.

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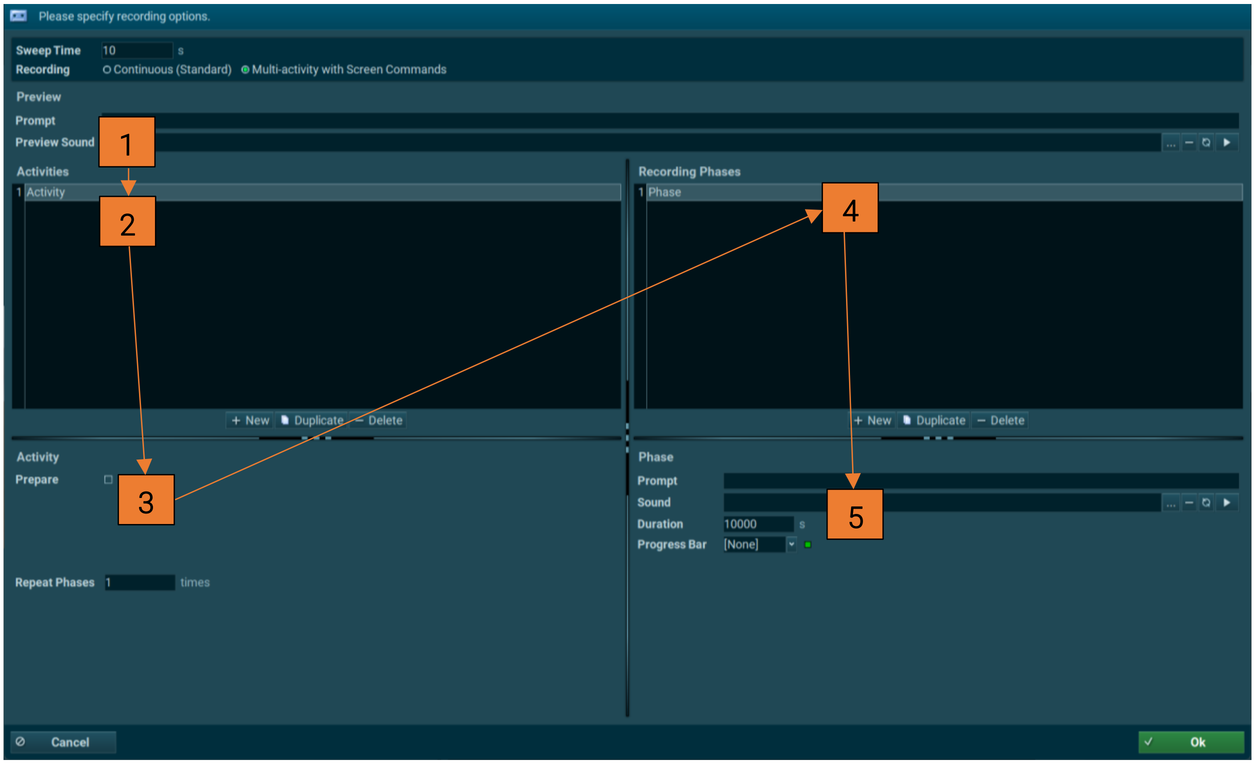

1) Preview/Recording
   A screen **Prompt** can be created and a **Preview Sound** can be selected for the initial "Preview" period prior to recording.

2) Activities
   A set of activities can be created here. Each activity consists of a sequence of motion phases such as extension and flexion. The controls **New**, **Duplicate,** and **Delete** edit the phase operations. Existing activities can be renamed by *double clicking* on their name.

3) Activity
   The **Prepare** option defines the unrecorded phase between activities. It is meant to be used as a preparational period to rearrange the subject's position, give task instructions, and create the other actions to continue. When checked, entries will appear to define a Screen **Prompt**, a **Sound** command, the **Duration** of the prepare period and selecting a **Progress bar**.

   Note: If the duration of the preparation period cannot be predefined, select a longer duration such as 30 seconds. The preparation period can always be interrupted if needed and the next activity can be started manually when ready.

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The entry **Repeat Phases** allows the repetition of the whole activity or just the motion phases a selected number of times.

4) Recording Phases

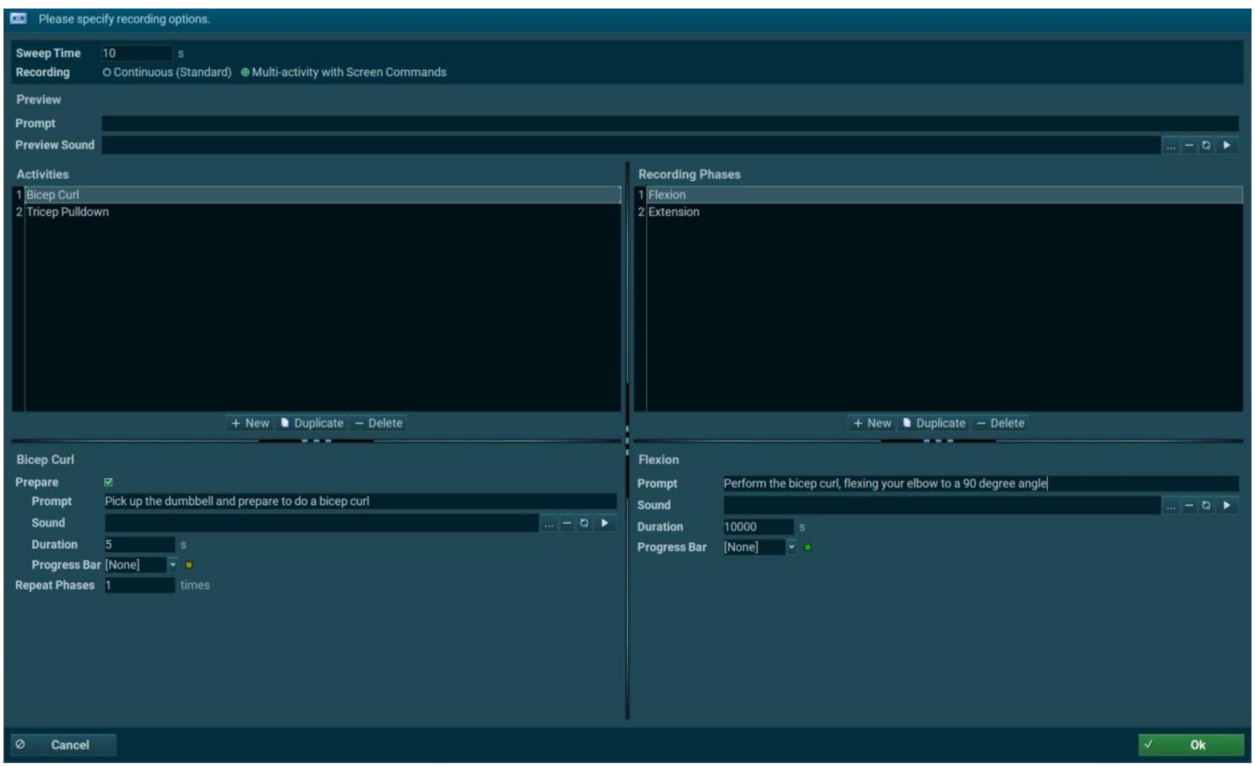
This section creates a set of motion phases for a given activity. The controls **New**, **Duplicate** and **Delete** are there to edit these phase operations. Existing phases can be renamed by *double clicking* on their name.

5) Phase

This section defines:

Prompt	screen command for the given phase.
Sound	sound for the given phase.
Duration	duration of the given phase.
Progress Bar	progress bar indicator and its direction.

Utilizing the previous settings, the following sample set-up can be created for a bicep curl and tricep pulldown measurement:



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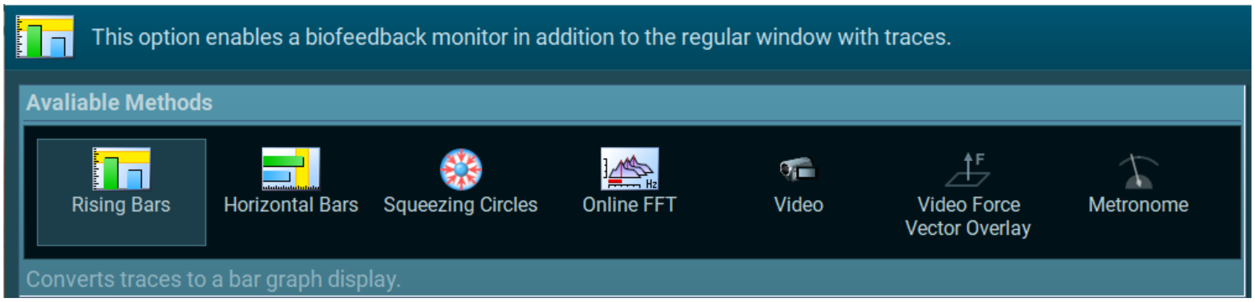
[1] Running Activity	name of the current running activity, as well as current/total repetitions
[2] Prompt	screen prompt of running phase (as defined in Prompt for phase).
[3] Current Phase	name of the current recording phase
[4] Progress bar	forward, as defined for this running phase.

Feedback Options

Feedback options convert signals into an intuitive visual format for users and subjects to observe throughout a measurement.

[See the Noraxon Website for a video on how to set up Biofeedback Training.]

There are currently seven different types of feedback available:



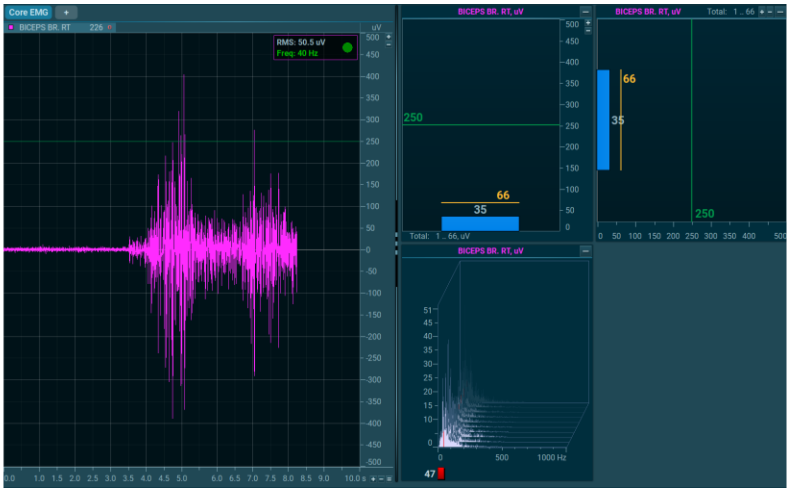
Note: Force on video is only available if myoFORCE Module is installed in myoRESEARCH.

The signal from any sensor can be used for biofeedback via rising/falling bar graphs, squeezing circles (designed for pelvic floor training), real time FFT analysis through cascading power spectrums, video playback, and a force plate vector overlay. A metronome can also be set up for appropriate applications.

The idea of biofeedback is to present the signal amplitudes in easy-to-read displays and threshold ranges that provide a target value to reach by the subject, such as an EMG activation value or joint angle. Two types of biofeedback categories include: Rising Bars, which can be used for general purposes, and Squeezing Circles, which is a specific signal presentation for pelvic floor muscle training.

The Online FFT is a special form of biofeedback because it cannot be directly controlled by voluntary efforts. It displays changes in the EMG frequency during contraction.

This figure shows all three signal related biofeedback modes.



To select a feedback mode, click on it and either click Insert, or double-click on the method icon, to load it to the list of selected operations:

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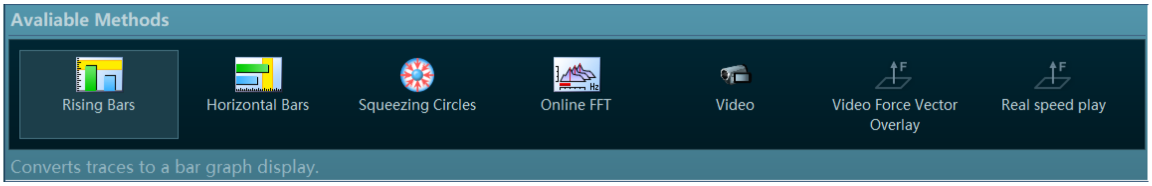
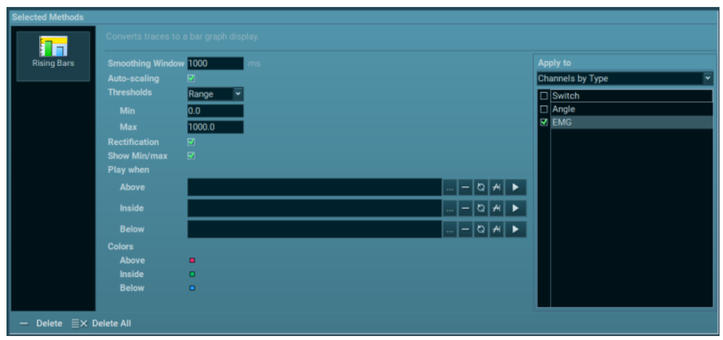


Note: Feedback methods can be loaded/inserted multiple times if needed for different channels.

Once a feedback option is selected, the settings for that method will be displayed. Certain options can be altered such as smoothing, thresholds, audio cues, and which channels to apply the feedback signal(s) to.

Rising Bars

The most popular biofeedback display, for any type of signal, is the Rising Bar Graph: a bar graph rises in proportion to the amplitude increase of the selected signals. Rising Bars can be used for general purposes, such as "Up" training for EMG innervation.



Smoothing Window

Typically, EMG signals appear highly variable, and it may be necessary to "smooth" them in both speed and amplitude. The Smoothing window defines a time range to apply a moving average to, which slows down the EMG spikes and makes it easier for the subjects to control the activation within a certain range.

Note: this setting is strictly a display feature for the bar graph and does not affect the recording and signal processing of the data.

Auto scaling

If active, the optimal amplitude scaling for the bar graph is calculated by the software.

Thresholds

Both range threshold and single line/value threshold are supported. Adding threshold range to the display defines minimum and maximum threshold range value. The threshold range is displayed as a yellow background area and typically serves as a target area for the EMG activation or amplitude level for any other selected signal type.

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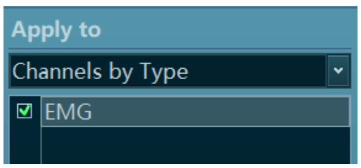
The **Play When** feature allows a sound to play when the signal is above, within, or below the threshold range or value identified in the **Thresholds** section. There are a number of pre-loaded sounds in the Noraxon directory, but any external file in the *.wav format can be played by uploading the file to the “sounds/object” folder, located in the myoRESEARCH package root level.

Rectification

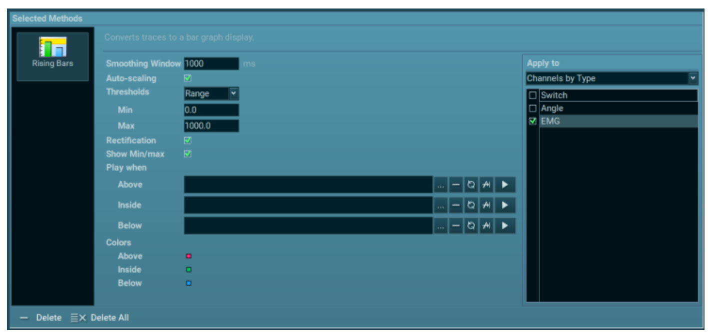
If checked, bipolar signals and their negative regions will be rectified automatically. The bars will only rise from zero to positive values. Uncheck this control to have biofeedback bars move to negative and positive ranges.



The **Apply To** control enables the biofeedback display for a given channel type or a physical channel number. To access specific channels, use the small down arrow on the right side to change from **Channels by Type** to **Channels by Numbers**.

Horizontal bars

Horizontal Bars can be used for applications such as left or right rotation angles of myoMOTION angles. For example, the signals obtained from myoMOTION trunk lateral flexion to the left and the right side may be more appropriately seen with a horizontal orientation of bars.



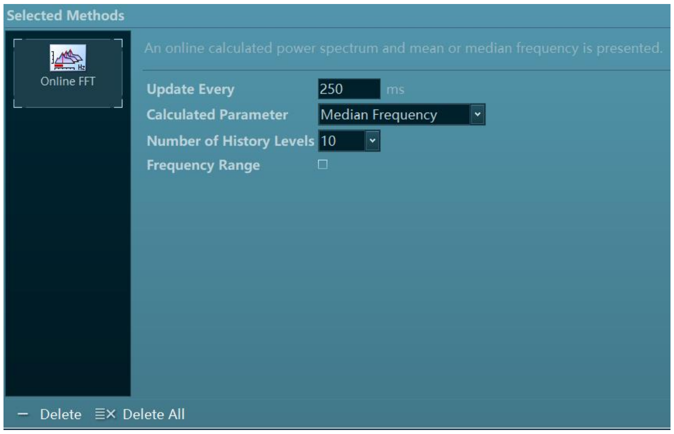
The setup menu for horizontal bars is the same as for Rising bars.

Squeezing Circles

This feedback option has the same functionality as the **Rising Bars** described above; instead, the amplitude change increases or decreases the diameter of the circle. This feedback style is designed for pelvic floor training (incontinence training) and should reflect the contraction of vaginal or anal muscle rings.

Real Time FFT

A real time frequency analysis is performed while monitoring/recording the data. Depending on the FFT window length, there will be a slight time delay for the calculation.



Window length

This is determined by the number of data points used for the FFT. The software can support anywhere from 62 to 4096 data points. The calculation time will increase with a larger number of data points.

Calculated parameter

The mean and the median frequency can both be used.

Number of 50%-overlapped windows

The amount of window overlapping can be specified here. This switch has a smoothing effect in the FFT display.

Number of history levels

This number defines how many power spectrums are shown in the cascading window.

The Apply To control applies the real-time FFT to selected channel types or channel numbers.

Video Feedback

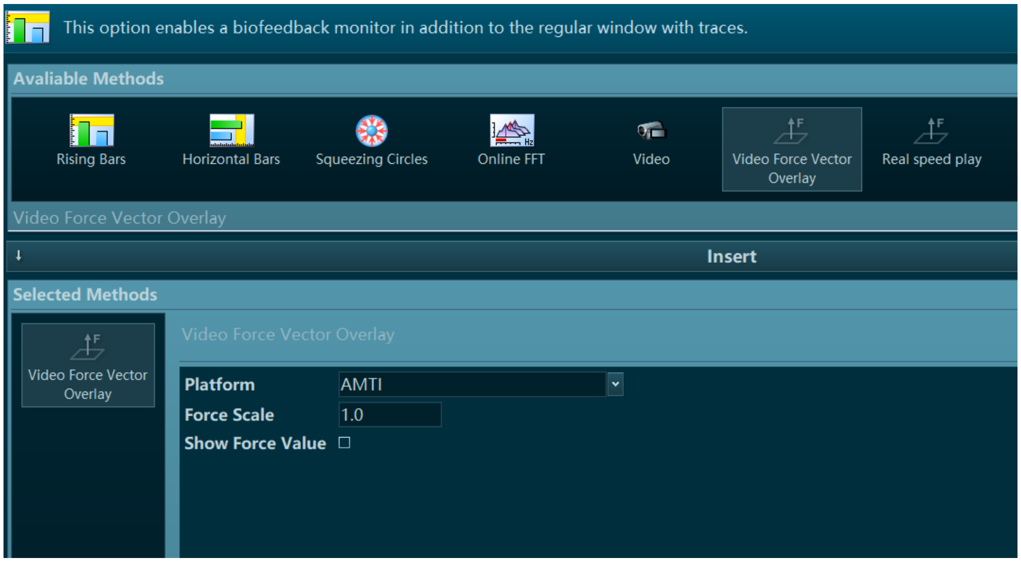
With this option, a video clip can be selected and played whenever the threshold criteria is fulfilled. If the signal leaves the threshold range the video was assigned to, then it stops playing and will continue as soon as the signal reaches threshold range again.

Force on Video Overlay

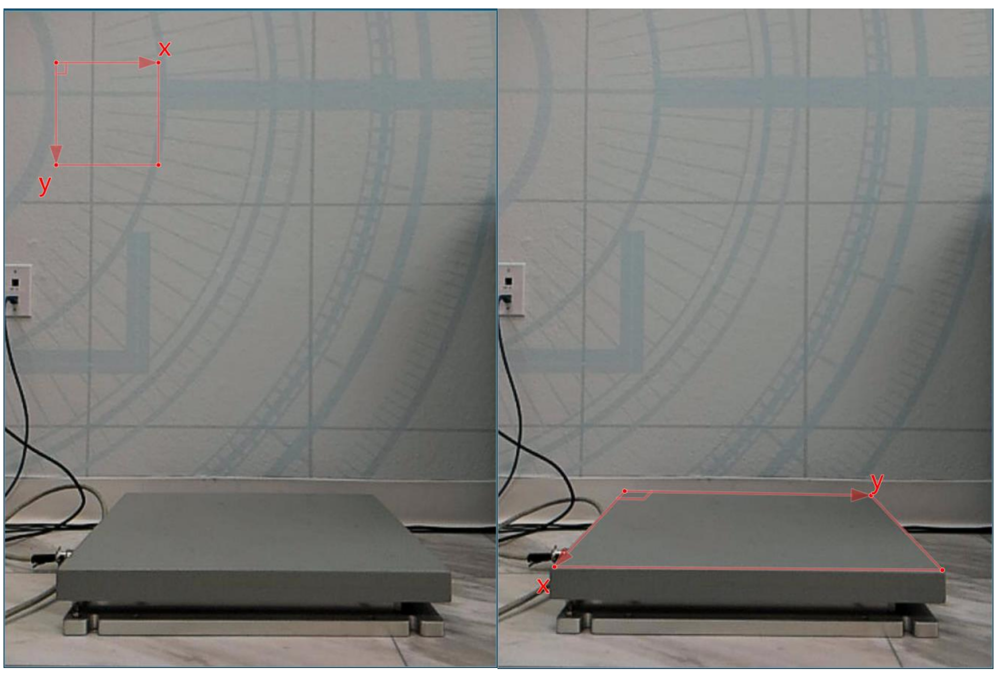
A real time force vector overlay (requires 3D force plates) can be displayed while recording data.

A video tutorial is available for this feature on the Noraxon Youtube Channel

Select this feedback option:



This feedback function also supports dual plates. Select the **Force on video** function a second time and assign it to the second plate. Click **OK** and go to the measurement monitor by clicking **Measure**. The video will show a calibration frame with an X and Y orientation. Then, using the left mouse button, drag the orientation frame to the origin, and then the corners of the force plate:



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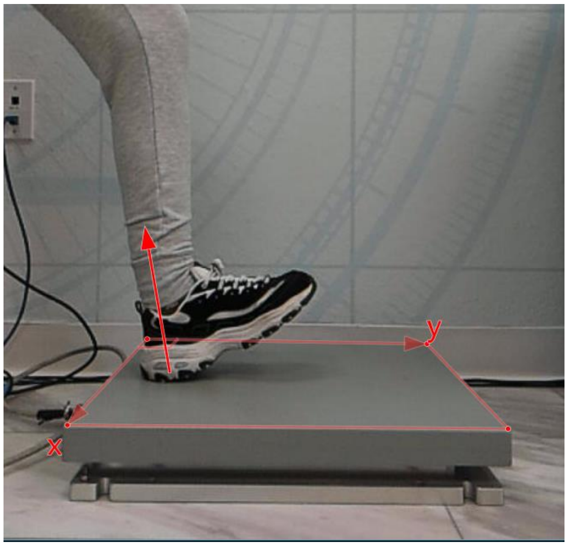
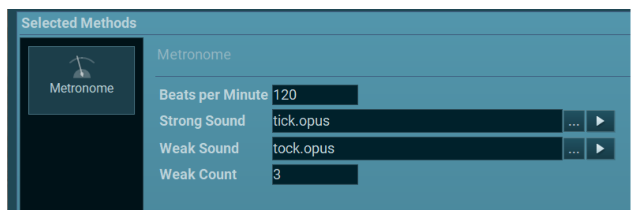
Note: The orientation of the X and Y dimensions for a given plate can be found in the manual of the force plate manufacturer. It may also be displayed on the force plate itself.



Test the accuracy of this force overlay by stepping on with a single leg, or using a wooden stick, and applying a vertical force:

Note: Force Vector Overlay setup is also explained in the appendix for analog integration of the different force plate models.

[*A video tutorial explaining how to set up the Force Vector Overlay is available on Noraxon's website*](#).



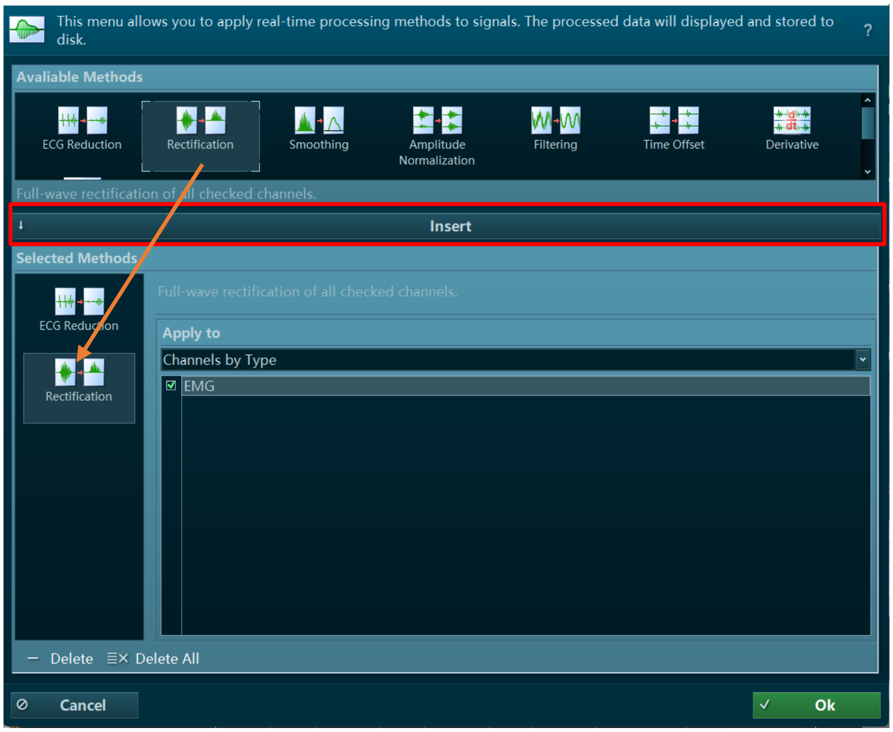
Metronome

With this option, a metronome can be set to play during the measurement. The bpm of the metronome may be adjusted for a specific speed, while the sounds played can be customized as well. The weak count adjusts how many beats will be played before the strong sound.

Online Processing

Online processing refers to real-time signal processing. When selected, the recorded signal channels will be displayed with the selected processing already applied. Real time processing also converts the biofeedback and circles explained in the Measurement Configuration chapter.

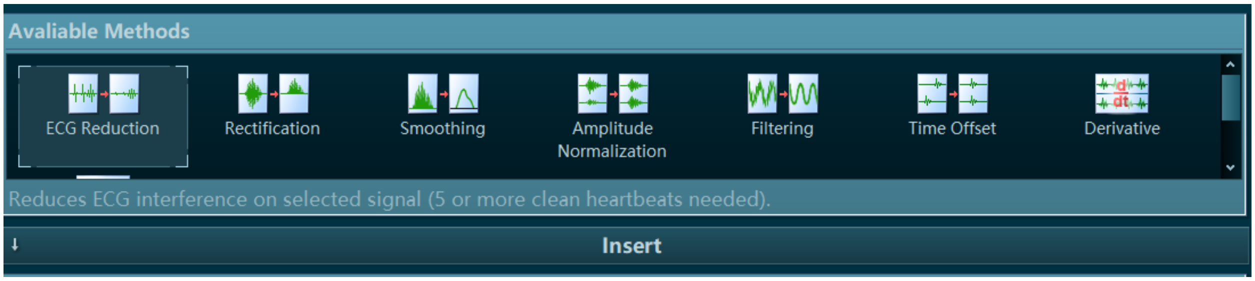
A video tutorial covering signal processing tools (in MR3) is available on Noraxon's website.



Available Operations

Eight Signal Processing methods, listed under **Available Operations**, can be loaded to a sequence of processing commands (**Selected Operations**). Any sequence or order of processing methods can be arranged, and any arrangement can be loaded multiple times.

Select a processing method and click **Insert,** or double-click on the method icon, to add it to the lower operation pipeline.



The **Apply To** control allows the processing operation to be applied to specific channels. Channels can be chosen according to channel type or specific channel numbers.

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ECG Reduction

myoRESEARCH is equipped with a unique algorithm to detect and selectively eliminate ECG artifact spikes from the EMG signal while maintaining the original EMG power. The algorithm responsible for this feature is a combination of pattern recognition and adaptive filtering.

By default, ECG reduction requires the user to input a time interval for the algorithm to look for clean heartbeats to establish a profile for the ECG waveform.

Select Interval By First Activity

Checking this box will instead use the first activity of a multi-activity recording as the interval to analyze for the ECG pattern.

Limit Duration To

This duration controls the time interval in the beginning of the recording in which the algorithm looks for the ECG pattern.

The quality of this "ECG cleaning" varies depending on the EMG recording quality and artifact conditions. The ECG removing function is useful in relaxation studies, especially when ECG spikes can greatly influence the outcome results.

Rectification

This popular processing method multiplies all amplitude values in a signal with +1, where negative values become positive. This operation achieves positive amplitude curves that can calculate parameters such as mean amplitude and area under the curve.

Smoothing

Typically, for amplitude-based calculations and analysis, the raw EMG is smoothed by digital filters, root mean square, or moving average algorithms. The effect is that non-reproducible EMG spikes are eliminated, and the mean trend of the EMG innervation is used. Smoothing also allows easier reading of the EMG patterns, which is useful for clinical tests and biofeedback-oriented treatments or training.

Algorithms

The following smoothing Algorithms are supported:

- RMS – Root Mean Square
- Mean – the moving average
- Mean absolute – the moving average with combined rectification

Window

This setting defines the window for each algorithm (in milliseconds) to calculate the smoothed value over. Select a window appropriate for the data being processed.

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Please refer to the ABC of EMG booklet for more information on how to use smoothing algorithms.

Amplitude Normalization

This feature allows the viewing of real-time MVC-normalized EMG amplitudes, which can be used for biofeedback purposes or for the study of the muscular demand of a selected activity.

A video tutorial for this feature is available on the Noraxon website.

Window

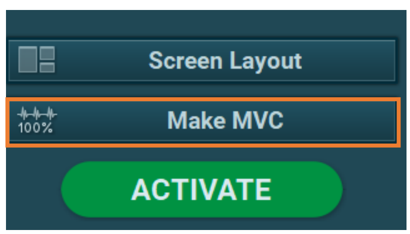
This entry defines the calculation window for the max value itself. Since it is not recommended to take single points as MVC values, the mean value of a user defined time window is presented here. A recommended value is around 500 ms for window width.

Amplitude

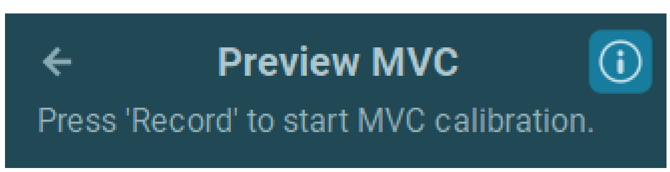
This entry defines the default Y-axis scaling for amplitude normalized data, which can be adjusted if needed.

Performing an MVC record

The normalization is based on the Maximum Voluntary Contraction (MVC) of a muscle, which must be performed prior to the test or training trials, typically as two static contractions for each muscle lasting 3-5 seconds. When activated, the online amplitude normalization routine adds a new Make MVC button to the right of the toolbar of the Measure menu.



When activated, the information panel of the operations menu will indicate that myoRESEARCH is prepared to perform an MVC calibration.:



Hit the Record button to start the MVC test recording. MVC recordings can be paused as many times as needed by pressing the Pause button located to the right of the toolbar on the

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measurement screen. In the record viewer, each pause will be indicated by a red line separating the total recording into sections.

When all MVC contractions are completed, press Stop to enter the Save Data dialog. Name the recording and if needed, create, or change the Subject in the second entry line:

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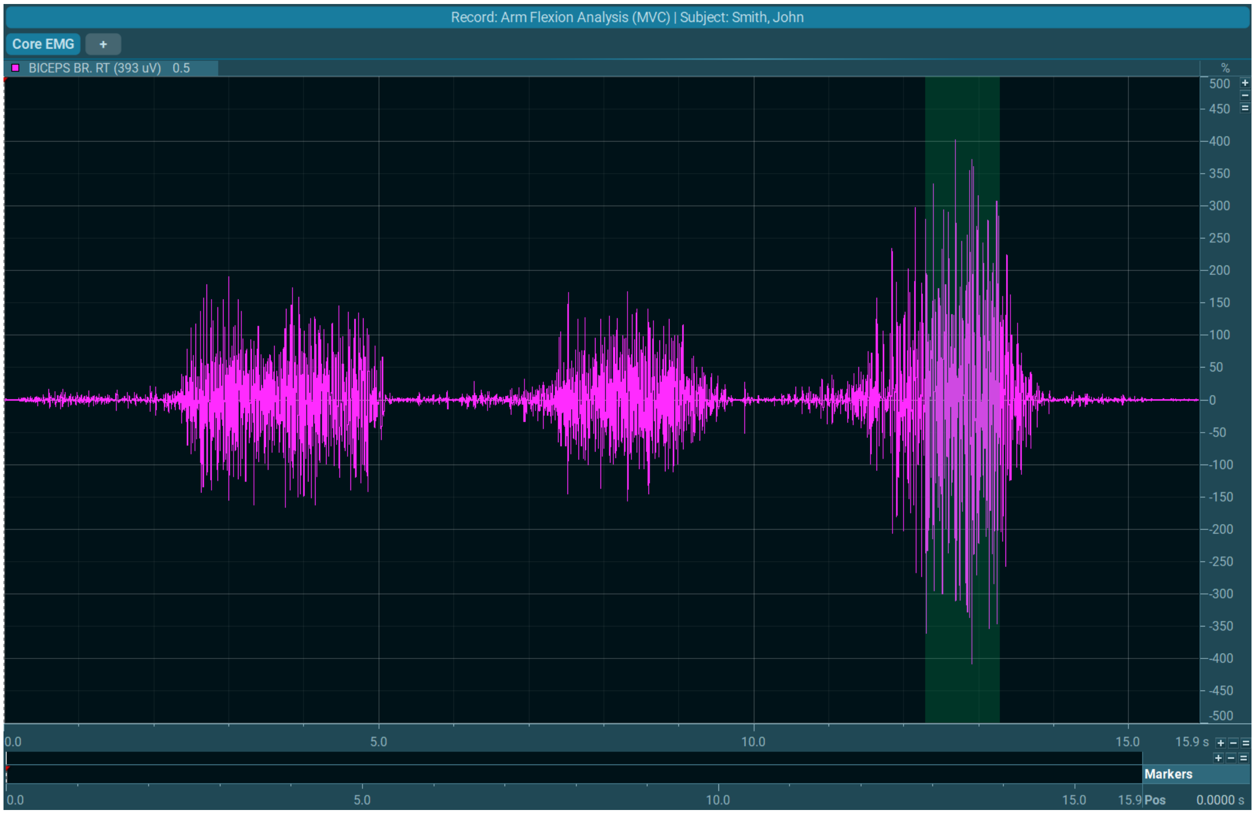
Click **Save & View** to continue to the record viewer.



The MVC algorithm automatically identifies the maximum EMG portion for each channel and indicates its position via a light green area. The numerical MVC value itself is shown in parenthesis in the upper left corner of each channel and to the right of the channel name(in this case, 393uV).

Using the MVC record data for online MVC normalization

After reviewing the MVC record in the **Viewer** menu, press **Measure** at the top navigation bar to return to the measurement screen. We can now see that the y-axis of the graph has been scaled to the MVC valued measured in the previous step, now presenting as a % of the previous maximum.

When finished with the recording, press **Stop** to enter the **Save data** dialog. After storing the data, review the recording in the **Viewer** menu and continue with the next recording by pressing **Measure** at the top navigation bar.

Filtering

A set of commonly used digital filters is available in this processing method:

<u>FIR – Finite Impulse Response Filter</u>

- **Window** – Number of points used to process the data. The quality improves with longer windows. Recommended default value is 79.

  Note: Longer windows improve the quality but increase the time for the real time processing.
- **Type** – The filter can act as a low, high, bandpass or rejector filter. Use the **Low/High** frequency controls to specify the filter range.
- **Windows** – sub-type selection to define the edge fading window.

<u>IIR – Infinite Impulse Response-Filter</u>

- **Bidirectional**: Use this mode to avoid phase lags created by the filter.
- **Type** – The filter can act as a low pass, high pass or rejector filter.
- **Frequency** – The edge Frequency can be entered in Hz.
- **Approximation** – Defines the subtype of the filter (not available in rejector filter).

Median – Filter

The median filter is an excellent spike cleaner for analog signals. It removes spikes from force/angle curves without affecting the original signal shape.

- Window – Defines the amount of data points used for the filter algorithm.

Time Offset

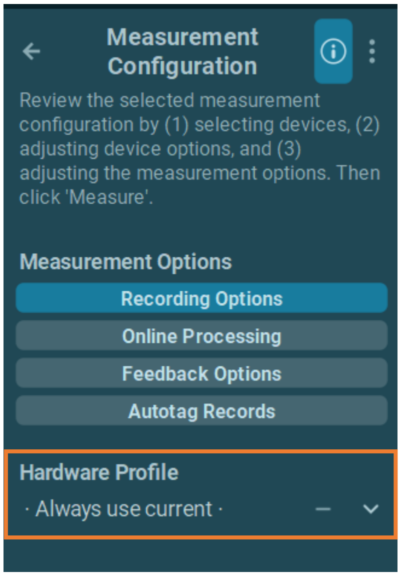
This processing time shifts incoming signals to correct the delay caused by telemetric transmission, pre-processing time at the device level, and similar delay sources. Enter the time delay correction factor in **Offset**. Positive values will correct delays.

Autotag Records

This option can be used to automatically tag records made with this particular measurement protocol. This can be useful to organize and sort records by a specific type of movement, or analysis (such as gait).

Hardware Profile

This menu section allows direct change to hardware profiles created in the Hardware Menu found on the Home Screen's right toolbar.



Always use current means that the latest hardware configuration in **hardware setup** will be used. Clicking on the **3-point** button allows direct access to the **Hardware Setup Menu,** which creates and modifies the hardware profiles. Create a new profile or select an already existing profile, then click **OK** to go back to the measurement setup menu.

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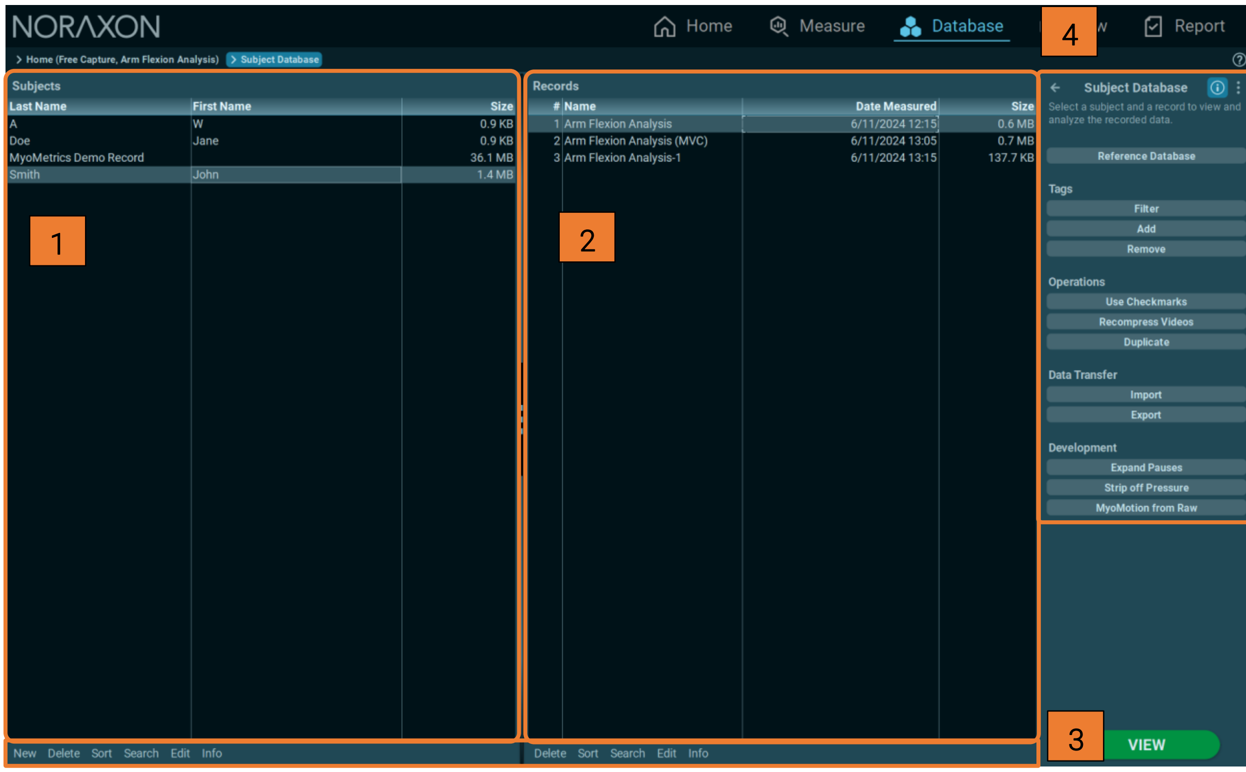
For more information on hardware profiles, refer to the [Home Screen section] or the [device-specific hardware manual.]

Database

Introduction

The **Database** tab manages all functions around data file management such as selection, editing, exporting, and importing. By default, the **Subject Database** records will be opened. A breakdown of the menu with descriptions of each section are explained below:



[1] Subjects	contains a list of all subjects in the local database
[2] Records	contains a list of all records under the selected subject
[3] Local List Functions	a toolbar of functions for each list
[4] Operations Menu	additional actions that assist with database management

List Sections

The screen is split into two sections: **Subjects** and **Records** stored under a given subject name. The width of a given list section can be modified by mouse dragging the sliders that separate lists.

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Local List Functions

At the very bottom of each list section is a toolbar of functions which may be performed on the subject or record:

New creates a new subject

Delete deletes the item and all the child folders under it.
Attention: Deleted subject data and records cannot be restored.

Sort sorts the database by first or last name (subjects), by name and measurement date (records), or by size (all) in either ascending or descending order.

Search searches for the project, subject or record of interest based on the input query. Helpful for large databases.

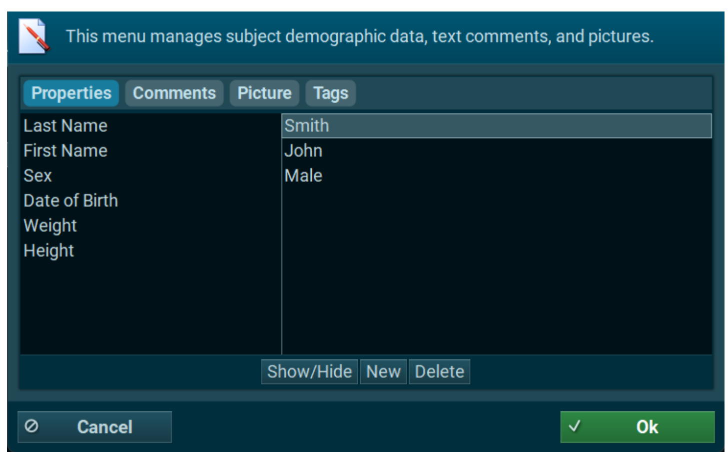
Edit changes the name of the item and adds additional information associated to the item like custom tags, comments, or pictures.

Info toggles the visibility of the item's properties window, with general file information, comments, associated pictures, tags, and additional helpful record information.

Subjects

The general information of **Subjects** is contained in a registration card format that manages details like name, sex, weight, height, and date of birth.

Double-click or click and press any key on an entry to edit the subject information. Click outside of the entry or press the Enter key to confirm or press the Esc key to cancel.



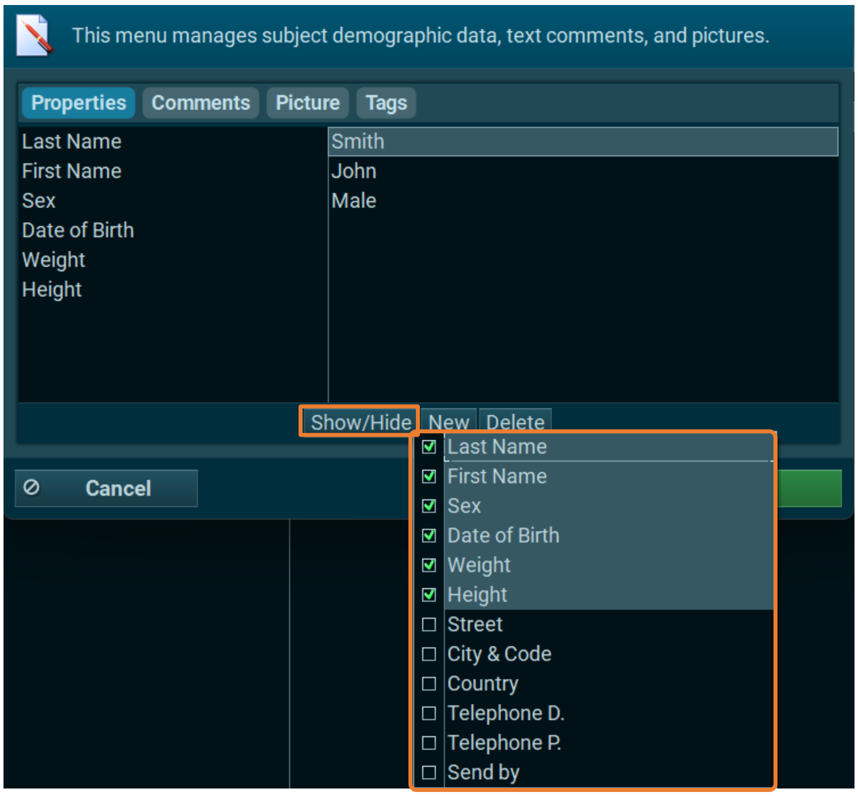
Properties basic identifying subject information like name, sex, birth date, weight, and height. More properties may be added to this list at the bottom of the window.

Comments — enter or edit patient comments. It is possible to use a text clip system or paste text from the clipboard.

Picture — enter a subject picture by pasting it from the clipboard or loading a picture file from the hard disk.

Tags — enter tags that allow filtering records via tag entries (see Filter by tag) function in the right toolbar.

When right clicking on the Basic or Subject Form tab, or clicking the Show/Hide button, this opens a context menu with to enable other types of subject data.



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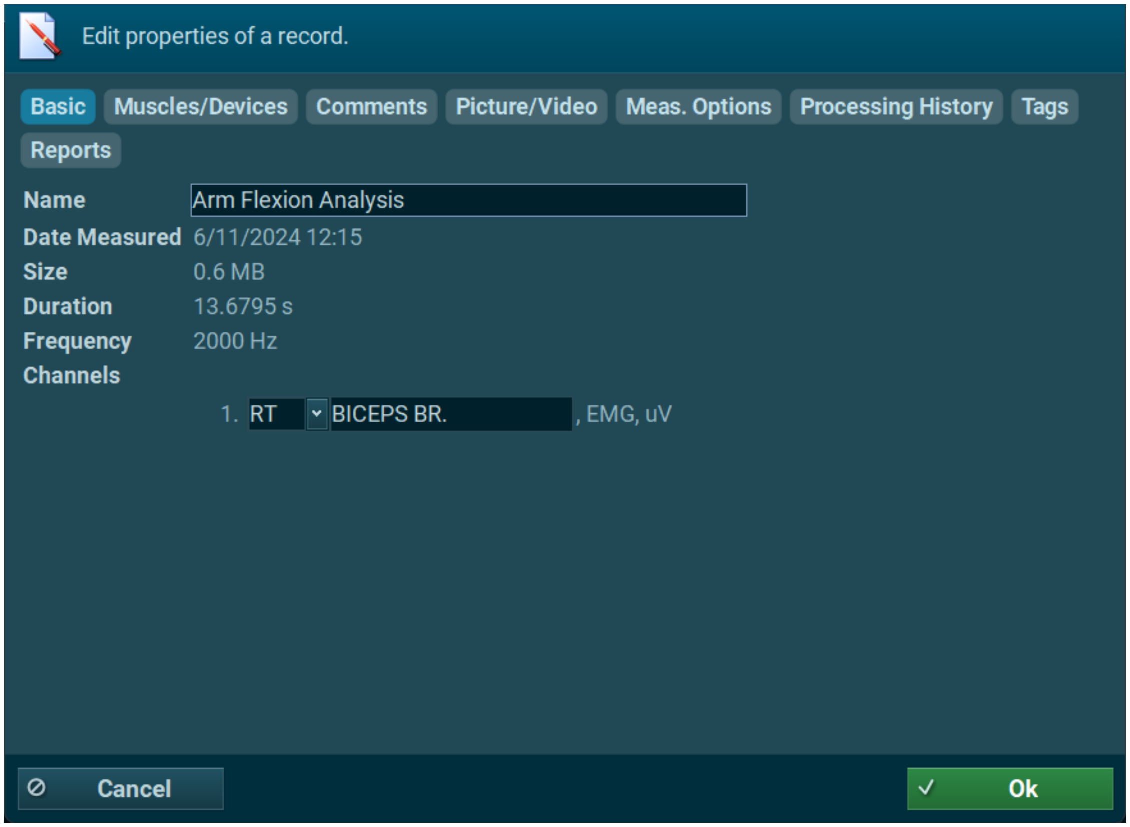
Records

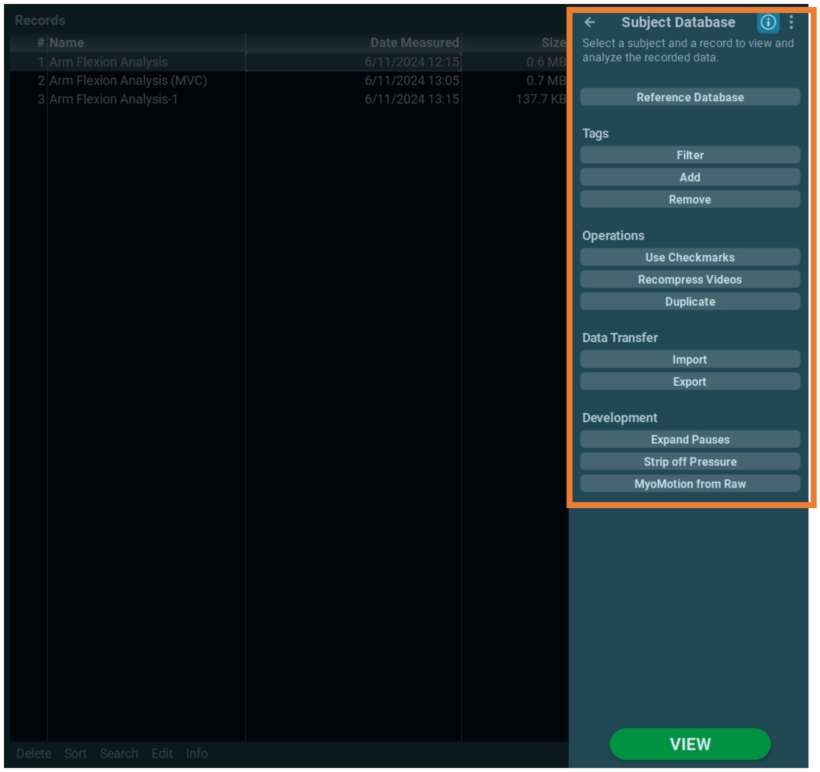
A similar set of list functions is available for the records list screen. The record's **Info** toggle button shows additional important property details. Double-clicking on any of the tabs in this section, or clicking **Edit** in the toolbar, opens a window to view all available information:



Basic	displays basic record properties like file name (editable), size, date, duration, and channels.
Muscles/Devices	shows a list of selected sensors (only applies to EMG sensors).
Comments	stores and displays record-specific comments.
Picture/Video	displays the first image or frame of a video. This tab is helpful to identify data associated with visual information.
Processing History	In case any signal processing was applied to the original record, all processing steps are documented here.
Tags	Each record can receive a tag that can filter records via tag entries. *(See the 'Filter by Tag' function in the right toolbar.)*

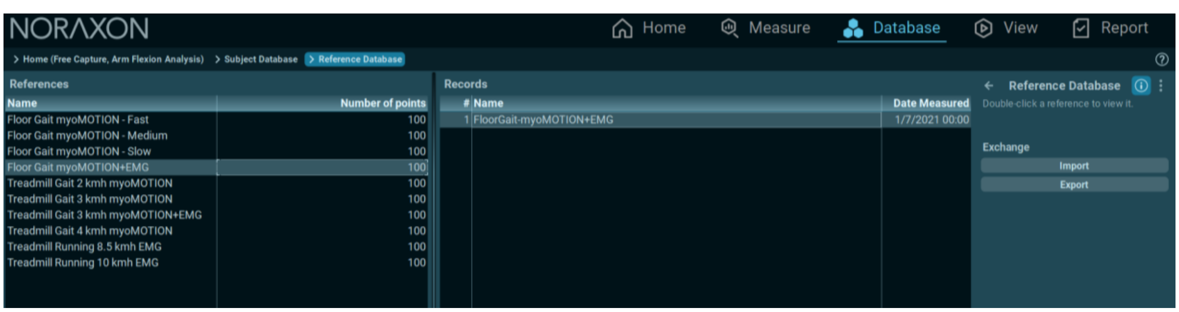
Database Operations Menu

The **Database Operations Menu** is located on the right side of the screen, listing relevant actions available in the **Database** section.



Reference Database

Another database section, called **Reference Database** (or the **Normative Database**), contains all reference data records that were compiled via the **Create or Update** Reference function within the **Report** menu. References are time-normalized averaged curves from several subjects and are typically based on repetitive motion patterns such as gait, running, and rowing, expressed in percent of (motion) cycle. The normalization vector length is expressed as the number of points. For more information regarding the Reference database, see the **Report** menu help documentation.



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All database functions described previously are also available for reference records. On the left in the **References** section, the reference name is shown (as entered in the report/reference section "Add to report"). On the right in the **Records** section, the records used to create the respective reference are listed. Individual files can be selected, viewed, and removed. Reference files can also be viewed or analyzed, such as through use of the Average Activation report template.

myoRESEARCH is shipped with a set of predefined references meant to be used for educational or training purposes. The contents of each listed reference record are described below. The amount, gender, and age of the healthy subjects differ between norm sets.

Note: The reference sample files are not meant to be used for any medical diagnosis or similar purposes. It only serves as an educational example on how to use or operate reference data.

Record	Contents
**Floor Gait MyoMOTION + EMG*	Natural floor gait and self-selected speed overground Contains MyoMOTION angles and EMG activation from several lower body muscles.
**Treadmill Gait 3.0 KmH MyoMOTION + EMG*	Treadmill gait at 3 km/h speed, contains MyoMOTION angles and EMG activation from several lower body muscles.
Floor Gait MyoMOTION – Fast	Overground walking with instruction: "walk fast". Averaged group velocity = 5.5 km/h.
Floor Gait MyoMOTION – Medium	Overground walking with instruction: "walk at normal speed". Averaged group velocity = 4.5 km/h.
Floor Gait MyoMOTION – slow	Overground walking with instruction: "walk at normal speed". Averaged group velocity = 3.5 km/h.
Treadmill Gait 2 KmH MyoMOTION	Treadmill walking at a preselected speed of 2 km/h.
Treadmill Gait 3 KmH MyoMOTION	Treadmill walking at a preselected speed of 3 km/h.
Treadmill Gait4 KmH MyoMOTION	Treadmill walking at a preselected speed of 4 km/h.
Treadmill running 8,5 KmH EMG	Treadmill running a preselected speed of 8,5 km/h.
Treadmill Running 10 KmH EMG	Treadmill running a preselected speed of 10 km/h.

Attention: These norms can only be used if the name of a given measurement channel matches with the one selected for the reference curve. This is the case if the predefined muscle naming system of the measurement setup menu (3D muscle map) is used.

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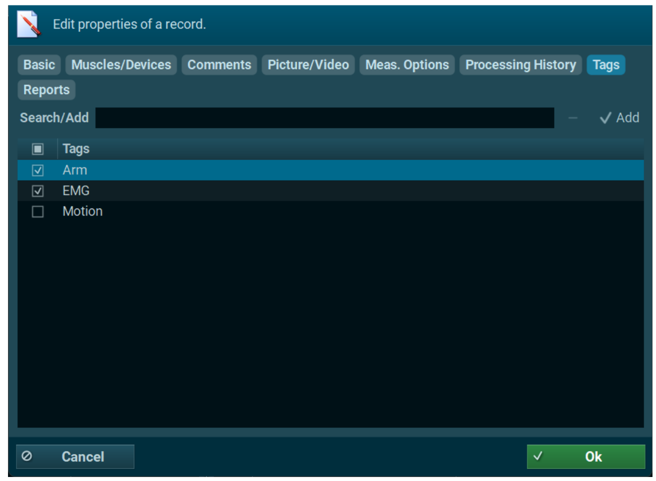
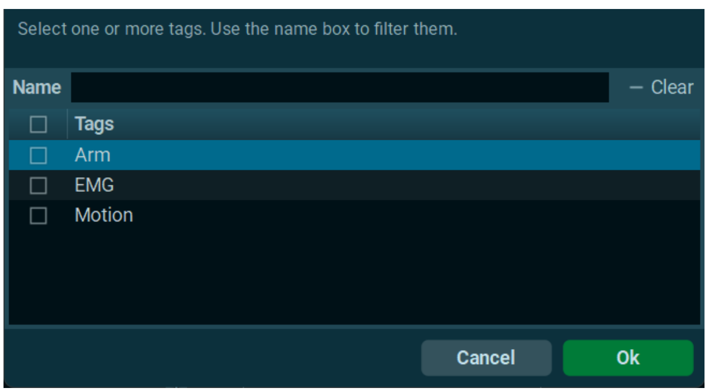
Tags

Tags can be defined for projects, subjects and records. The tagging system is helpful to sort/select records by certain criteria and export them in one run.

Tags can be created and added in the edit mode of subjects and records. To start, press the local **Edit** button or double click on the **Info** screen when opened. The **Tag** tab can be selected to search, create, or place a tag in the registry. The example to the right shows the Tag submenu on record level.



Filter by Tag



Tags can be filtered using the **Filter** button in under the **operations menu** on the right. This menu allows the selection of **tags** – when returning to the database, only records with the selected **tags** will be shown.

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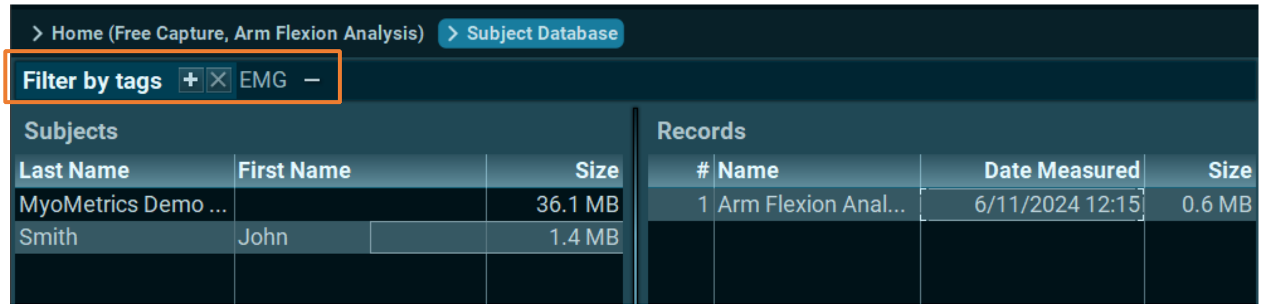
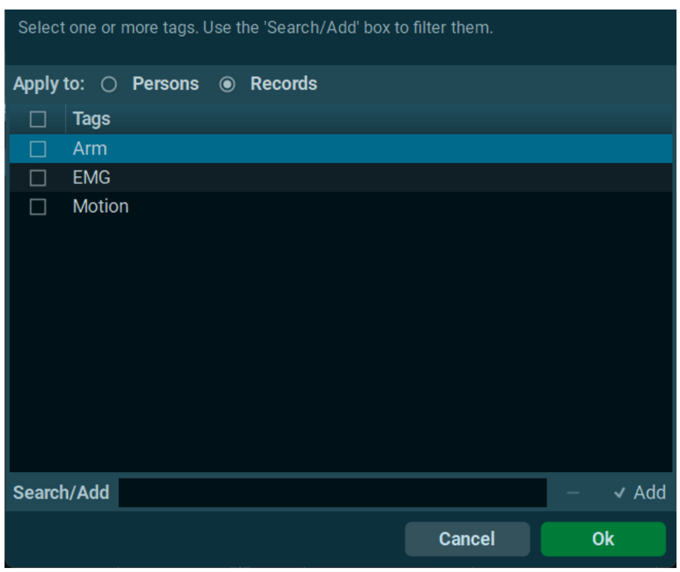
Now single or multiple tags can be selected. When clicking back to the myoRESEARCH main database screen, the tag filtered list of records will be shown. On the top left corner above the Projects list, there will be a **Filter by tags** line, listing the tags being used:



The **plus (+)** button allows the addition of tags to the filtered search, the **minus (-)** button will delete the tag preceding the sign, and the **X** button will delete all selected tags and show the unfiltered database.

Add Tag

Tags can be added to selected records from this window – select whether you are modifying subjects or records, and then select which tags that need to be added to the data.



Remove Tag

This window allows you to remove tags from records.

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Operations

Recompress Videos

Recorded videos can be compressed in real time while saving a record, or later in the **Database** menu. The mode is defined in the Software Settings at Home Screen Menu.

Recompress can compress uncompressed records at any time after recording. Compression may take a while depending on file size. It is strongly recommended to compress videos, as the file size may be reduced by up to one tenth of the uncompressed size!

Select records as needed. If all records are checked, myoRESEARCH will automatically find the uncompressed ones and compress them.

Duplicate Record

Since the function creates an identical copy of a record, this is helpful if certain processing or editing techniques need to be developed or compared.

Import Records

myoRESEARCH records can be imported from external locations, such as a previous version on the PC, backup data from external hard drives, or network folders. Additional import options allow for importing external videos, CD3 files and other data.

To import an external myoRESEARCH record, select **Import**, then **Import from External location** and set a path to the external directory such as a net drive or external hard disk directory. MR4 data is in a sub directory called **Noraxon MR Data**. This directory is automatically created when exporting existing records to external locations.

The import screen has 2 sections: **Subjects** and **Records.** All data for a subject can be imported by clicking the checkmark to the left of the subject. Alternatively, the **Records** tab displays records from the selected subject, so that only specific records can be imported.

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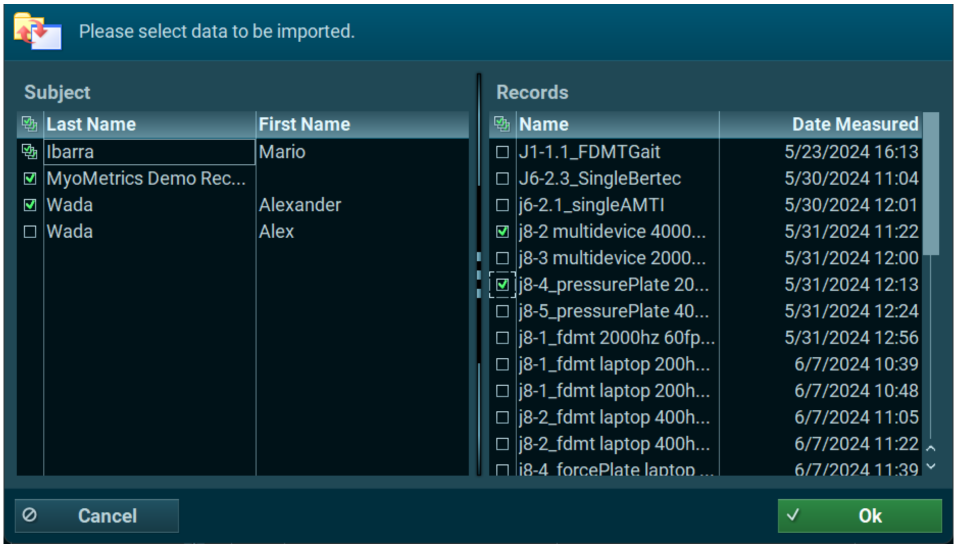

Other import options include:

Import video file	Externally recorded video files in avi format can be imported.
Import MRXP data	Import of records from the former myoRESEARCH XP version.
Import record from text file	Imports Ascii or .txt formatted records. *Attention: Only myoRESEARCH ascii data or similar arrangements are supported (see export file to ascii).*
Import of C3D file	C3D format used by most Motion capture companies is supported and allows importing of kinematic, force, and EMG data.
Import record from Noraxon Datalogger (*.tm2)	Data logged files from the Telemyo G2 and Telemyo DTS belt receiver can be imported here.
Import Zebris16 Data	Imports pressure data recorded with old Zebris 16Bit systems (FDM, FDMT).
Import MyoPRESSURE data	Data recorded with the MyoPRESSURE Software from Zebris can be imported here.
Import record from MyoMOTION Datalogger (*.mml)	Imports data recorded with MyoMOTION inertial sensor data logger.

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Import record from Ultium Datalogger (*.nudl)	Imports data recorded with Ultium inertial sensor data logger.
Offline data recovery	Import both myoMOTION and Ultium EMG onboard data which was saved in the sensor onboard chip memory due to high Wi-Fi traffic or large transmission distance (lossless function of sensors). For more details about this process, reference the user manuals for Ultium Motion or Ultium EMG.

Export Records

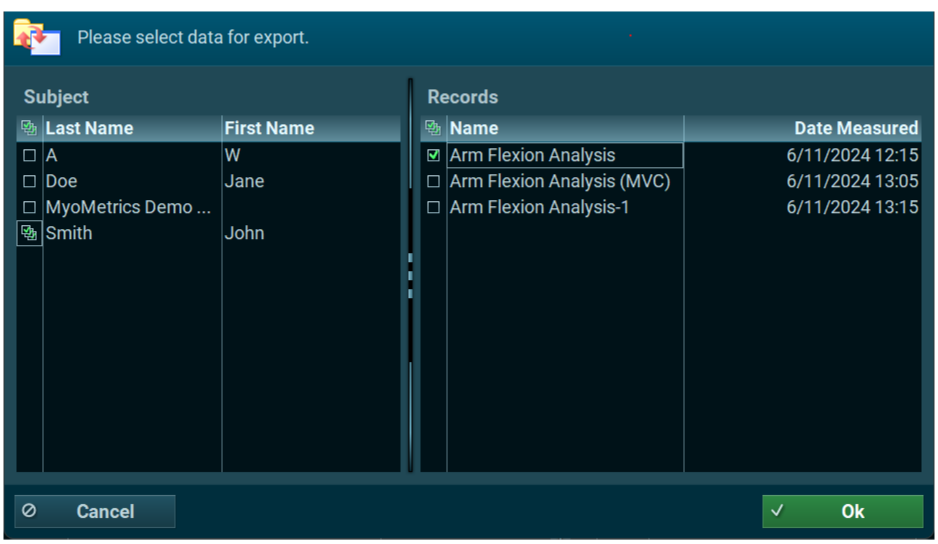
myoRESEARCH supports several different methods of record exporting.

Export Data in Native myoRESEARCH Format

myoRESEARCH records can be exported in their native format to an **External location** on a PC or to another Subject within the running version. The different options are described in more detail below:

Export to an external location	exports the chosen subject from myoRESEARCH to a location outside of myoRESEARCH
Export records to another subject	copy a record from one subject directory to another inside the myoRESEARCH version.

A dialogue menu will open, from which the specific subjects and records to be exported can be chosen. Check the desired subjects and records shown in the list.



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Export Data in Other Standard Format

Other supported standard export formats are listed below:

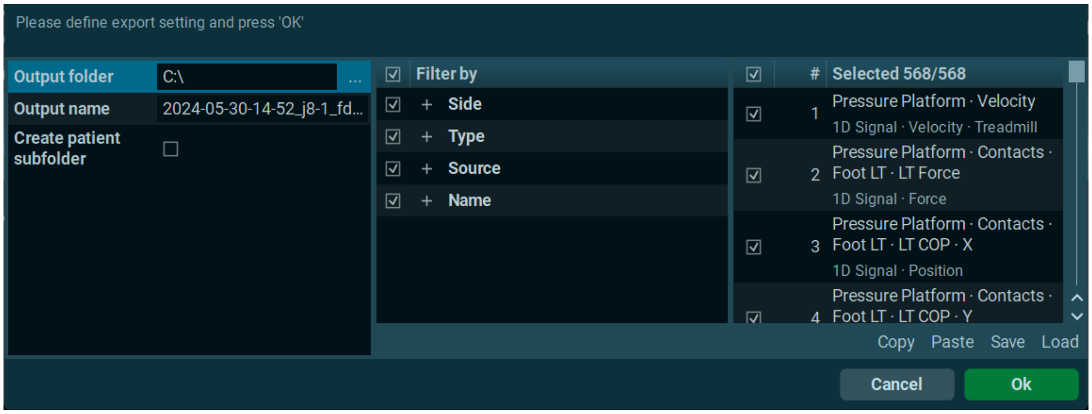
Export record to text file	Numerical Data (signals) can be exported to an ascii compatible format
Export to CSV file	Numerical data can be exported to Excel compatible CSV format
Export record to Excel (.slk) file	Numerical Data (signals) can be exported to excel compatible SLK format
Export record to MATLAB file	Numerical Data (signals) can be exported to MATLAB format
Export pressure data to XML	Export of pressure data to XML (access to all single cell data of each step)
Export MyoMOTION data to BVH	Export of MyoMOTION data (full body required) to Biovision (BVH)
Export Medilogic data to CSV	Export of Medilogic footprint data to CSV
Export Report Statistics to CSV	Export data from report ran on a record to CSV. Multiple records can be selected with checkmarks to batch export reports. A video tutorial on this process is available on Noraxon's website.

Export of data to CSV and C3D with extended export and filter options

The last three export methods support extended export and filter functions for signals with 1D (EMG, foot switches), 2D (Goniometer, inclinometer), or 3D components (inertial sensors, 3D accelerations, trajectories, quaternions). Each option is described below:

Export data to separate CSV files	each signal type will receive its own file and will export it at its native sampling frequency
Export data to single CSV files	all signal types will export one file – signals are up sampled to the highest sampling rate available
Export record to C3D	all signals will be exported into one C3D compatible file

The filter menu is separated into two major sections:



The left side section manages general export settings. Use the given entry line to define the Output Folder, Output Name, and whether to create a Patient Subfolder.

The right side displays extended filter functions and are segmented into several categories:

Side	selects the body side (RT/LT) or no body side for a given signal
Type	selects based on the dimensional nature of the signal (1D, 2D, 3D)
Source	select based on the origin/device that generated a signal
Name	select using the given channel name of the signal

Note: The order of filters can be changed by clicking on the small right side arrow buttons that show up when moving the mouse over the filter title line.

The export function supports batch mode. All selected records will automatically be exported with the selected mode and signal filter/selection settings.

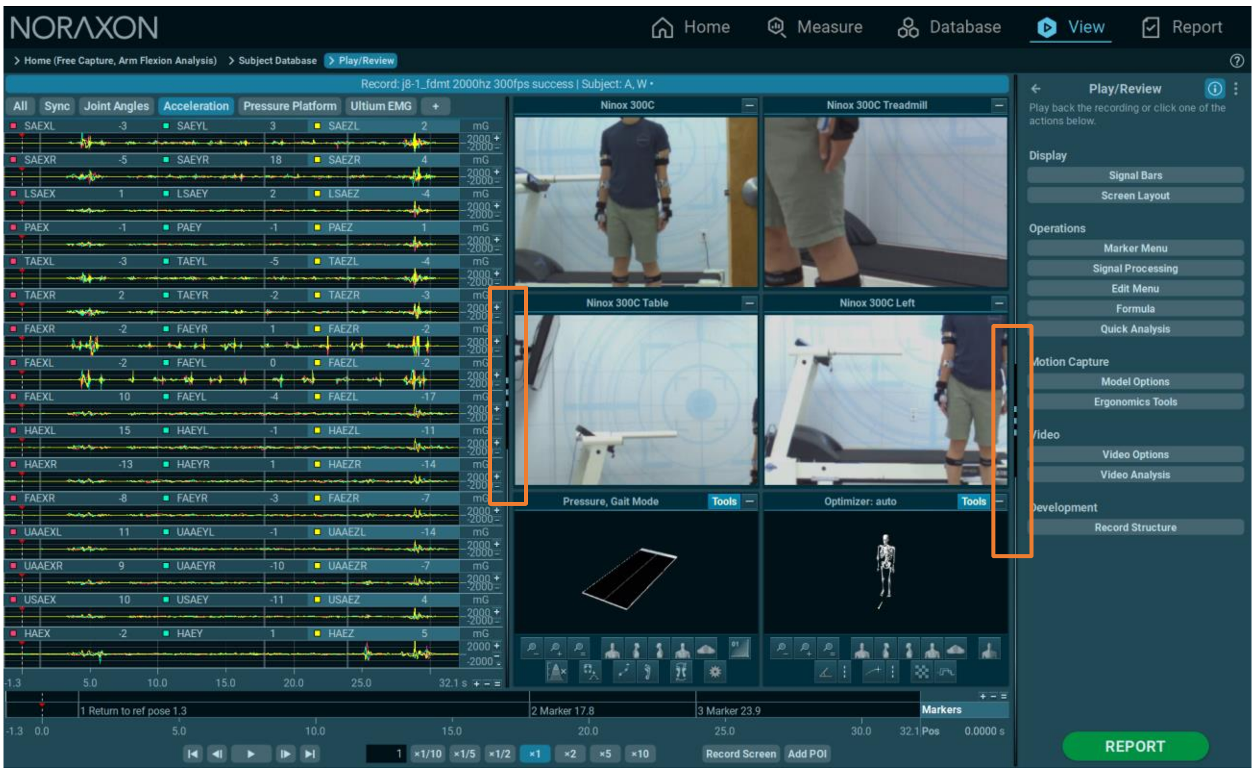
Attention: Do NOT use Windows Explorer to move or send myoRESEARCH data files directly from their data directory or hard disk. The important index files and data may become unreadable. To copy data, always use the ***Export to external location*** *function of the database, located in the right toolbar.*

View

Introduction

Each successfully stored recording can be loaded to the record **Viewer**. Data will be presented in the same channel layout scheme that was used for recording. The main goal of the **Viewer** menu is to inspect the quality and details of a record, apply signal processing, place markers (for analysis period definition), and prepare the record for further analysis in the **Report** menu.

Depending on the multi-device setup used in measurement, the record viewer will show appropriate curves and elements. An important display tool of the **Viewer** menu is the splitter lines that separate the screen sections within the viewer screen:



To resize the given screen element, the sidelines can be dragged by clicking and dragging the left mouse button.

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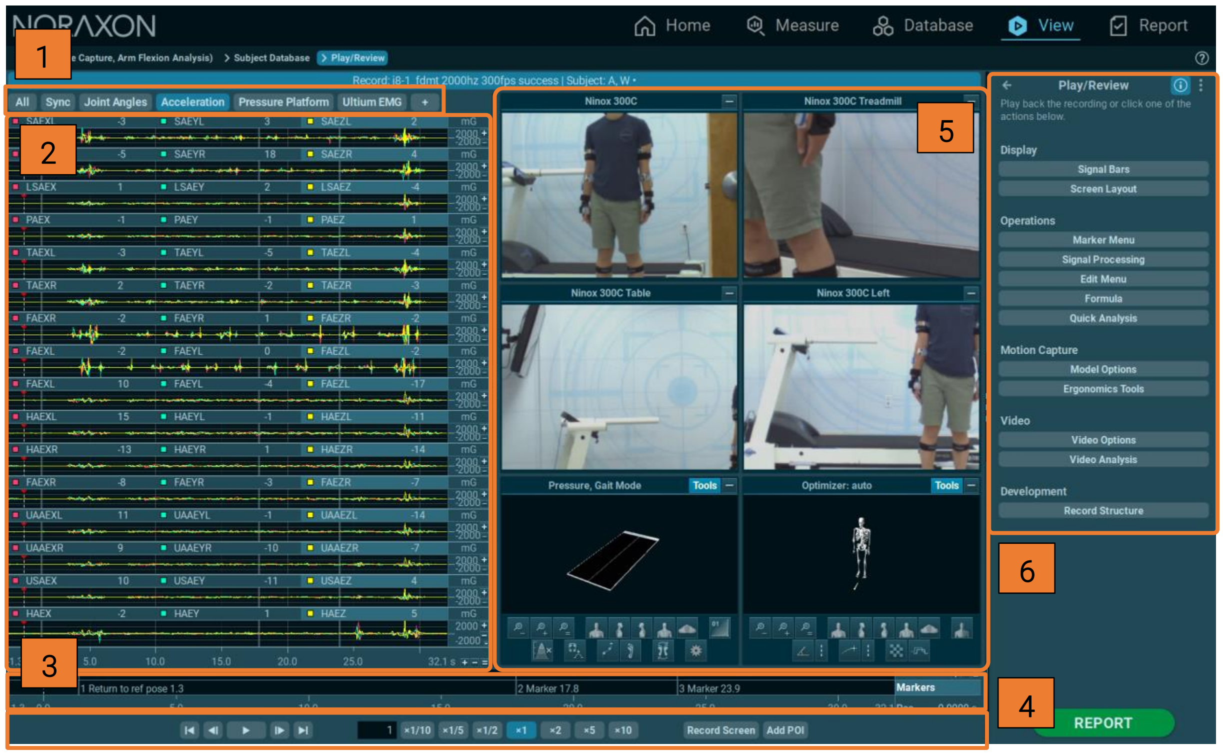
The following menu elements are available with the complete set of Noraxon modules. Some features are only accessible depending on which myoRESEARCH modules were installed to the software.



Record Viewer Layout:

- [1] Register Tab system to separate/sort devices
- [2] Channel and View Functions
- [3] Timeline
- [4] Video control/replay and POI Analysis
- [5] Video and animation windows
- [6] Toolbar menus

Note: *In the example above, a multidevice setup with Ultium Motion (myoMOTION), Ultium EMG (myoMUSCLE), FDM-T Pressure Platform (myoPRESSURE), and Ninox camera (myoVIDEO) is being used. Different categories of devices in a recording will populate the* ***Operations Menu*** *with different options. For example, myoVIDEO based video recording adds two more submenus in this case:* ***Video analysis*** *and Video* ***Options****.*

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Device Register Tab

By default, the measurement signals of all connected devices are grouped by the given device. Click on the desired Device tab to see the measurement signals. Drag to the channel name or tab name. Signals/tabs can be moved into other tabs or new tabs.

The **Viewer** menu presents all recorded data exactly in the same screen layout arrangements as they were measured. Because of the multi-device nature of myoRESEARCH, numerous signals, animation video and biofeedback screens can occur which require sophisticated screen element management. In the case of multi-device setups, each device will be shown in its own tab section.

The following channel display functions, **zoom** and **overlay functions** are available:

- Local X and Y axis +/-, = buttons allow for zooming operations of amplitude and time.
- Double click on the **channel name** to maximize a channel to a pop-up window.
- CTRL Double click to add a second or third (….) channel to the pop-up window.
- Pop-Up windows can be moved to a second monitor.
- Dragging the channel name to another signal creates an overlay of both signals.
- Any number of signals can be dragged to a new tab.
- Any given screen layout arrangement can be stored as a configuration available in the right toolbar.

For more detailed descriptions and examples of the Zoom and Overlay functions, please see the help documentation for the **Measurement** menu.

Mouse-based Channel View Functions

Note: More details about convenient mouse and key features in the View section are shown in the MR4 Hot Keys & Shortcuts article on Noraxon's website.

Left Mouse Button

Double clicking on the signal screen places a marker on the timeline.

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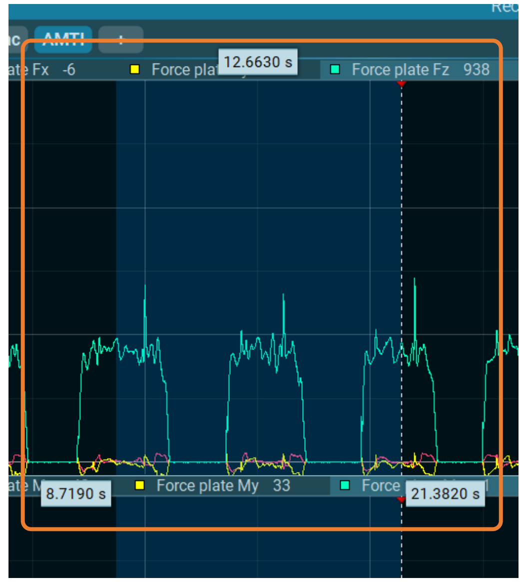
Clicking and holding the left mouse button, or holding shift while left clicking, marks an area to zoom in to or for period definition when generating a report.

While dragging an interval with the left mouse button, three small fly-out windows will appear, indicating the time point of the beginning and end (lower values), and duration (upper value) of the interval:



Middle Mouse Wheel and Button

If the mouse wheel is scrolled up, the signal is zoomed out. If the wheel is scrolled down, the signal is zoomed in.

Pressing the middle mouse wheel button allows the mouse to pan along a zoomed-in signal recording when the mouse is moved to the left or right.

Pressing the mouse wheel button and Shift key together allows the mouse to pan along the amplitude/zero-line position when the mouse is moved up or down.

Scrolling the mouse wheel while pressing CTRL button zooms in or out on the amplitude of the channel below the mouse arrow.

Right Mouse Button

Clicking with the right mouse button opens context sensitive commands that are available. This feature is currently in development. Depending on the position of the cursor, several direct commands are supported.

Mouse over marker	add to Report. add Comment. remove marker.
Mouse over signal screen	create Report marker.
Mouse over marked area	add/Remove selected markers from/to report. remove markers.

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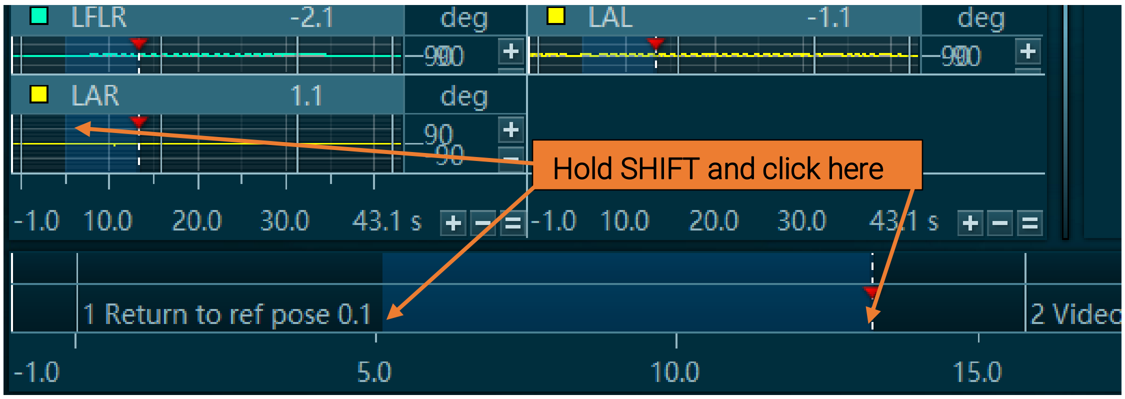
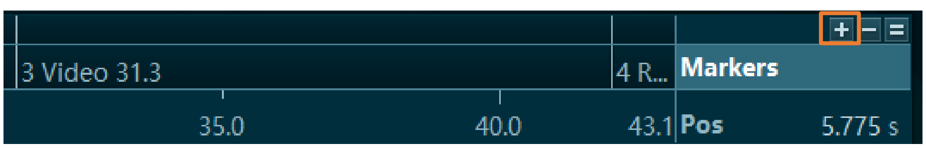
Timeline

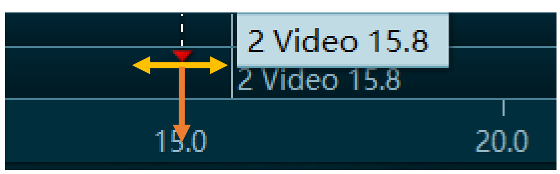
The timeline indicates the cursor and marker time positions, marker labels, and the location of the analysis periods for more information, see the Report chapter for details. The timeline can be used to mark areas within the record and zoom up on them with the plus (+) button on right side of the timeline:



The marked area is **highlighted** both on the timeline and signal screen. The plus/minus and equal sign screen icons can be used to **zoom in and out** of the period selection and to show the full recording:



Double clicking or pressing **enter** places a marker to the record. Its position is shown both on the signal screen and on the timeline. By left clicking, the marker can either be moved to the left or right side (shown in **yellow**), or dragged down to the next line to delete it (shown in **orange**):



The timeline can also be rescaled to a highlighted section of the recording. To do so, mark an interval on the timeline or signal screen, then zoom in by clicking the plus button on the right side.

The page up/down buttons are used to scroll through the measurement.

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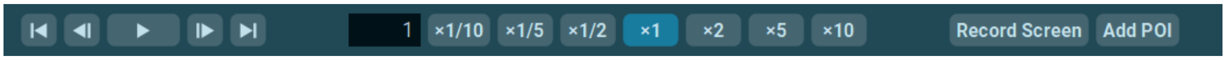
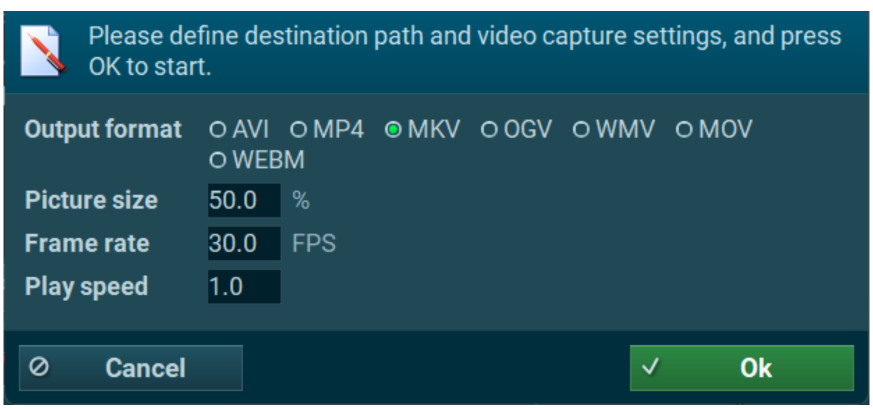
Video Control/Replay and POI Analysis

Videos and all available animations, such as the myoMOTION avatar or pressure Isobars of myoPRESSURE, can be replayed at any time by using the video controls:



Play	plays all visible videos on the screen at the same time
Numerical Entry field	allows the entry of any numerical value for play speed
$^1/_{10}$x, $^1/_5$x $^1/_2$x, 1x, 2x, 5x, 10x	allows the entry of any numerical value for play speed between – 10x to + 10x speed
Record Screen	myoRESEARCH's video/screen recorder function allows the recording of any video, screen animation, or time synchronized measurement signals, to an .avi, .mp4, .mkv, .ogv, .wmv, .mov, or .webm file. A small menu dialog defines the file destination directory, video size, frame rate, and playback.

The screen recorder function **Record Screen** requires setting a path through **Browse**. It defines the **Picture size** which is preselected to 100%, **Frame rate** which is recommended to be 30 Hz, and **Play speed**.



Attention: do NOT operate any other action while the screen recorder is running.

Add POI

Clicking this button on a selected marker will mark it as a **point of interest (POI)**. Any POI created by this function receives an **orange** Report marker label in the marker/timeline shown below.

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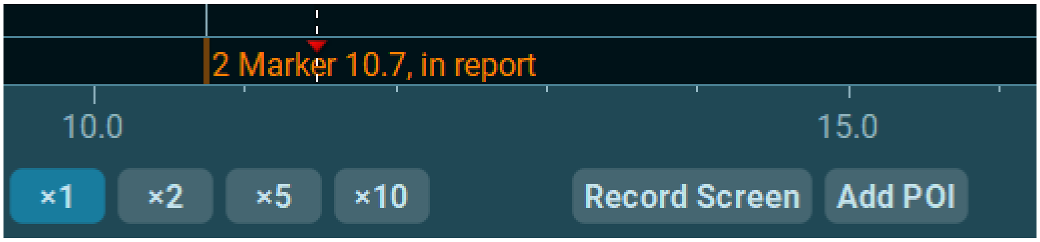
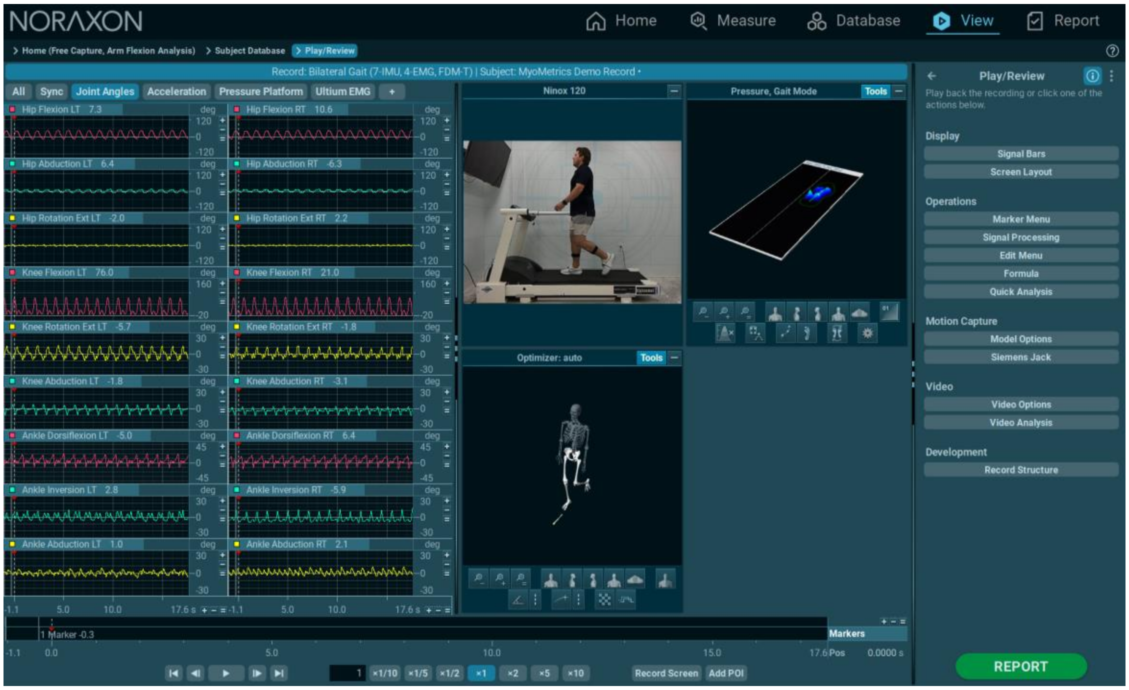


The timeline shows a seconds-based time scale and a marker label line. myoRESEARCH operates an intricate system of markers related to biomechanical event triggering, manual marker labeling, or POI based "Add to report" labeling. Any created marker line is shown here and can be used to identify events, motion phases, interesting analysis sections, and many other features.

This function can be useful to identify key points in a recording to be included in a report. In reports, video from POIs will be displayed in the POI section.

Video and Animation Window Section

In the case of multi-device setups including myoMOTION and myoPRESSURE, data is visualized with 3D animation graphs:



Several view and perspective buttons located on the bottom side are available and already described in detail on the measure menu. Each of these animation windows has local tools which are described below.

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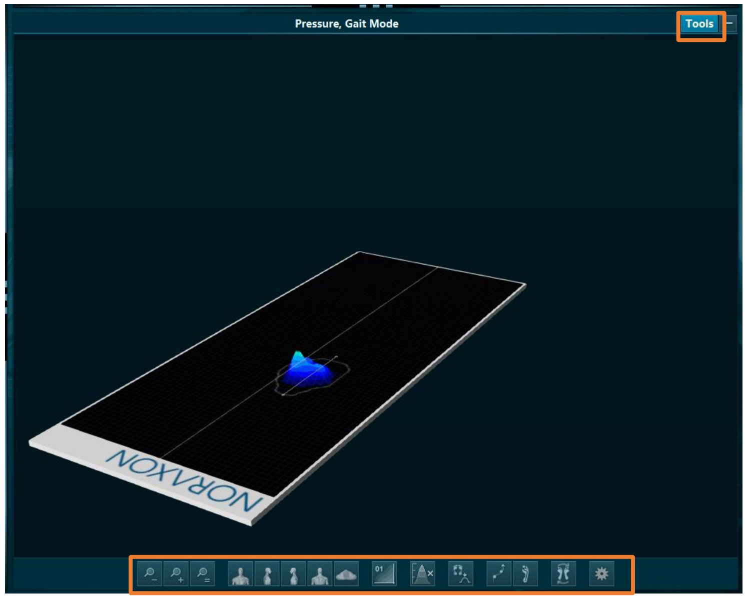
Pressure Animation Window - Local Toolbar

This visual animation of the isobars view can be changed using the mouse buttons. The animation can be rotated by clicking and holding the left or right mouse button. The mouse wheel zooms in and out of the 3D visualization. When pressing and holding the mouse wheel button, the animation can move in all directions within the video animation screen. Alternatively, animation tools at the bottom of the animation window can be used. To activate the animation tools, click on **Tools** in the upper right corner.



The following **Tool** buttons/functions will appear:

Arrow circle	rotates avatar with the left mouse button.
Arrow square	moves avatar up/down and left/right with the mouse wheel button.
Magnifying Glass +/-	zooms in or out with the mouse wheel.
Magnifying Glass =	resets the avatar when it is moved out of view
Human Profiles	left/right and back/front views.
01 Plot	the projected top view of pressure data, which can be zoomed in on to view the pressure data of individual cells.
Pyramid with X	color intensity slider. Move to the left to increase sensitivity.
Foot/Peak Symbol	shows or hides the max pressure of each cell in that given step.
Line with 2 dots	shows or hides the COP line from step to step.
Foot with COP line	shows or hides the gait line
Reverse foot symbol	changes all left contacts to right contacts and vice versa

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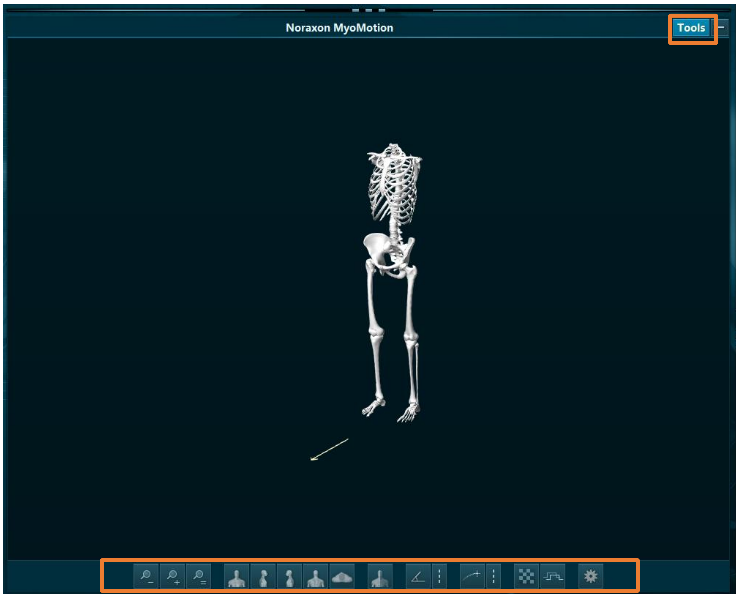
myoMOTION Skeletal Animation

The main function of the avatar animation is to visualize the angle measurement curves on the left side of the measurement screen. The avatar moves in relation to the measured curves. It may be helpful to adjust the view and perspective. If it is not visible on the lower section of the screen, click on **Tools** in the upper right corner to make the avatar view tools visible:

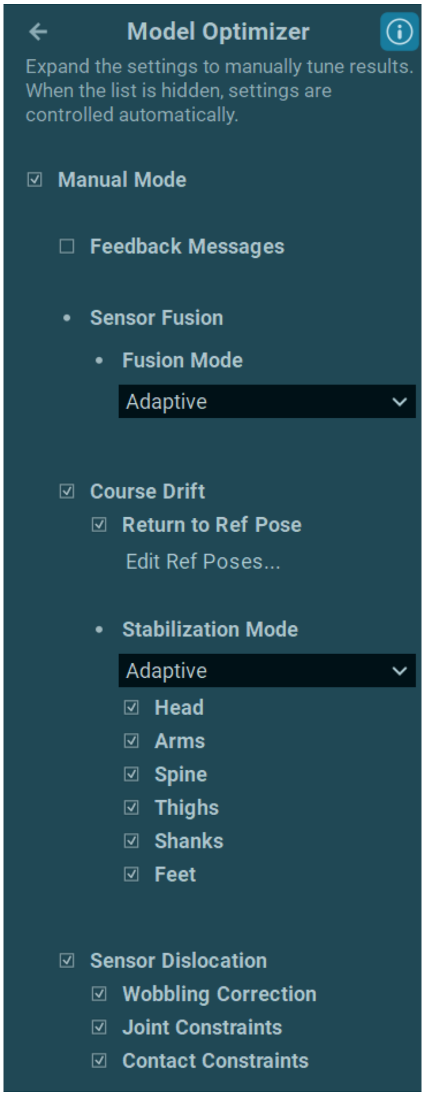


The following **Tool** buttons will appear:

Arrow circle	rotates the avatar using the left mouse button.
Arrow square	moves avatar up/down and left/right via the scroll wheel button.
Magnifying Glass (+/-)	zooms in or out using the mouse wheel.
Magnifying Glass (=)	resets the view when the avatar is moved out of view
Human Profile Symbols	orients to different views (left/right, back/front).
Half Human Profile Symbol	can choose between all, measured, or selected bones. Undetected segments are grayed-out and do not move.
Angle Symbol	toggles joint angle overlay
Trace Symbol	toggles trajectory visualization (COM, etc)
3 Dots Symbol	selection screen for bone or joint specific trajectories.
Chess Board Symbol	turns on the floor and translational animation of the avatar.
Square Wave Symbol	enables floor contact as signals in channels.

Model Optimizer

This submenu is designed to fit data in case of external magnetic distortion or muscle wobbling during impact conditions. It is recommended to operate the Model Optimizer in auto mode, which is automatically used. Consult with a Noraxon support specialist (support@noraxon.com) for further advice on if using the Model Optimizer in manual mode is applicable for your application.



Toggle Manual Mode to access model options:

Feedback Messages	Enables model optimizer feedback (e.g. “more reference poses recommended”)
Sensor Fusion Mode	Choose between different sensor fusion modes to fit data: Adaptive(recommended) – magnetometer data is only used per sensor in sensor fusion depending on usage of other drift correction options. Mags On – magnetometer data is enabled on all sensors Mags Off – magnetometer data is disabled across all sensors Onboard – uses sensor fusion output computed in realtime during data collection by sensors – limited by hardware
Stabilization Mode	Determines behavior of constraint usage for course correction Adaptive(recommended) – Implements constraints based off of posture and activity Standard – Implements constraints uniformly

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Return to Ref Pose	Use reference poses to correct drift

Course Stabilization

IMU measurements can be subject to external magnetic distortion which may influence or alter data in an uncontrolled way. To overcome this limitation, segment optimization modes are offered to mitigate this effect.

Head	velocity constraints relative to kinematic chain
Arms	velocity constraints relative to kinematic chain
Spine	postural constraints in lower spine segment relative to pelvis and upper spine
Thigh	postural constraints in thigh segments relative to the pelvis position
Shank	shank segments stabilized by knee joint
Feet	Drift of the foot in the horizontal plane relative to the shank is removed (soft hinge constraint at the ankle joint)

Sensor Dislocation Settings

IMU measurements can be affected by artifacts caused by sensor movement and imprecise placement of sensors on the subject's body. myoRESEARCH contains three automatically enabled settings which mitigate these effects artifacts related to soft tissue movement.

Wobbling Correction	removes high-frequency motion from sensor quaternions that are unlikely to represent body movements.
Joint Constraints	reduces soft-tissue artefacts in the upper arm and thigh segments using known constraints of the elbow joint and 1D knee joint.
Contact Constraints	mitigates thigh soft-tissue artefacts by constraining foot-floor interaction during squats and lunges

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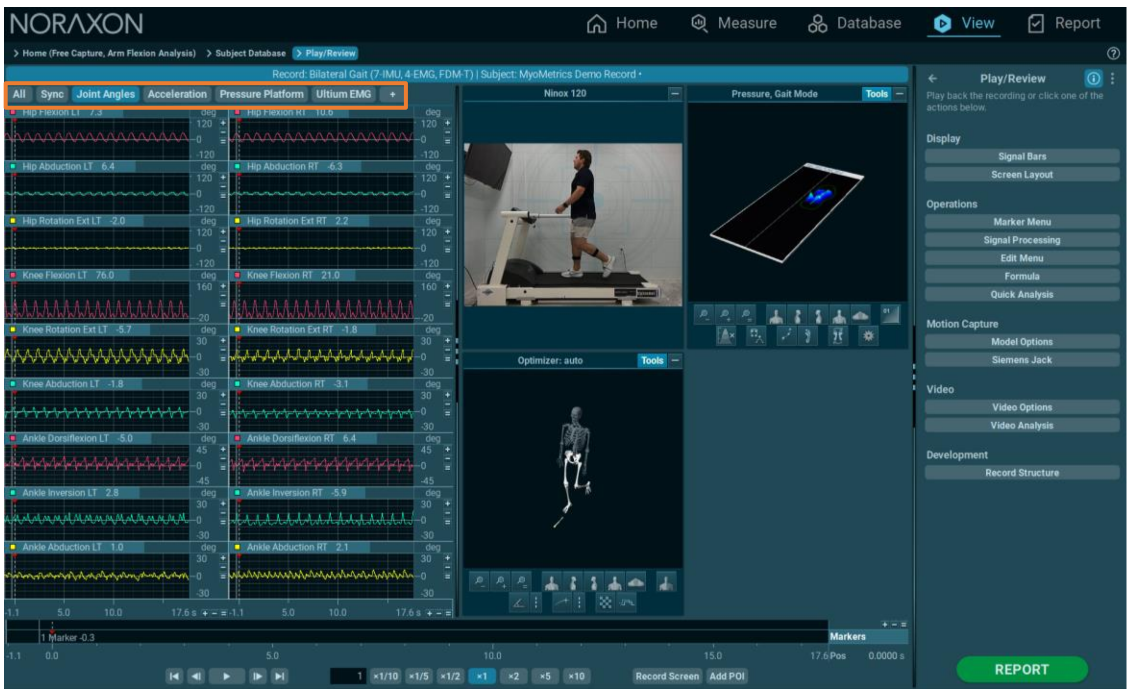
Screen Layouts in Record Viewer

For multi-device setups, it may be helpful to use the customizable screen layout system of the **Measure** menu, which is nearly identical to the **Record Viewer** layout system. The layout system is explained by using a multi-device setup including myoPRESSURE, myoMOTION, 3D analysis, and EMG.

Generally, each device will have its own specific tab:



As an example, myoMOTION can have up to 4 tabs:

- Anatomical angles
- Orientation angles
- Accelerations
- Contacts

Any signal from any tab section can be dragged into other existing device tabs or into the empty **"+ "Tab.** Any tab can be renamed by right-clicking on the tab name and clicking "Rename" on the fly-out popup. A new dialog box will open; then, enter any new/suitable name for continuous use.

This screen layout system allows the user to create as many signal selections or new user views as needed, and a similar system is available for the **Video/Animation** window. Double clicking on a given signal name moves this signal into a pop-up window:

The small red **X** button on the upper right corner of the window will close the pop-up window and move back the signal to its original tab location, and no data will be lost.

Holding the **CTRL key** and clicking on a second channel will allow the viewing of multiple channels in one window. Double-clicking the second title will pop out the channel in a separate window. To

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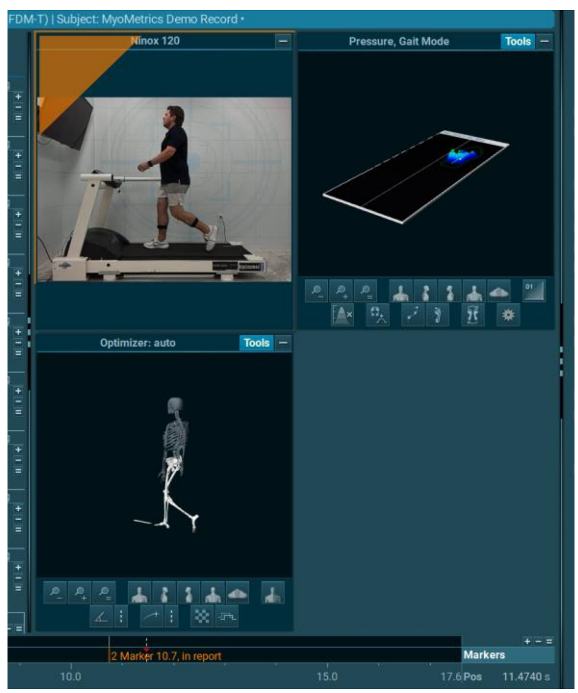
allow multiple channels to be viewed in the pop out window, click and drag the desired channel name into the pop out tab.

Clicking on the channel name and dragging it over to another channel will cause orange arrow buttons to appear on the side, which will prompt an overlay of the dragged channel with the existing one:



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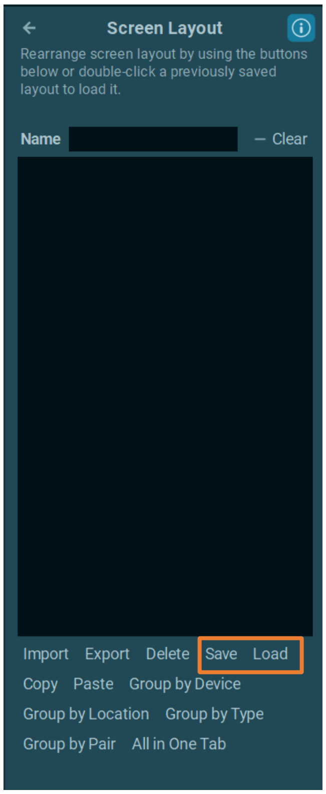
To insert a channel above the existing one, move the mouse up to the upper left corner of the channel window until a corner arrow appears. The same procedure applies to moving a channel below an existing channel by dragging the name to the lower right corner.



Whenever a certain screen/window layout arrangement is established, it can be saved in the **Screen Layouts** section on the right toolbar. Enter a suitable name for the channel layout configuration in the **Name** box and click on **Save**. It will then appear in the space below:

To load a screen, double click on a listed layout name, or click **Load**. The lower button bar can **Export/Import** or **Delete** a screen layout. The following options are also available in **Screen Layouts**:

All in one tab	all signals from one tab will be shown on one shared screen.
Group by device	separates devices and signal categories and displays them in different tabs. This is the default screen.
Group by pair	will overlay all left and right channels from a given device.
Group by location	will overlay all available signal dimensions of a given sensor.
Copy	will copy a layout to the clipboard.
Paste	will paste a layout.

*Note: The channel layout created in the **Measurement** menu will be the same in the **Viewer** menu. All layout and channel arrangement functions are still available in the **Viewer** menu.*

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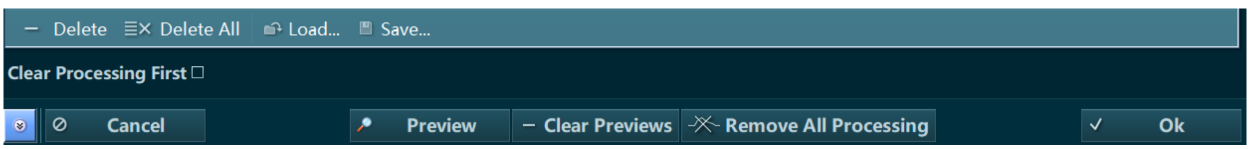
Signal Processing

Signal processing is a collection of commonly used processing modes for biomedical signals. Bipolar raw EMG recordings require post processing prior to analysis. The menu operates in 4 basic steps:

1) Select a processing method.
2) Click insert to add it to the processing pipeline.
3) Apply a channel filter to select specific channels for the selected processing.
4) Confirm selected processing by clicking **OK.**

Operation Tools

The basic operation tools are independent of the selected signal processing mode and are very similar to the tools found in the **Measurement** menu under **Online Processing**. The **Operation Buttons** found at the bottom of the window are described below:



Delete	removes a method from the list of Selected Operations.
Delete All	deletes all selected operations from the list.
Load	opens a window showing all available signal processing configurations.
Save	saves any sequence of Selected Operations to be stored and loaded for repeated use. To save a configuration, give it a name and optional icon, and click **OK**.
Clear processing first	removes all previously applied processing operations(applied during online processing) and reloads the original raw data.
Preview	minimizes the window to a toolbar and gives a preliminary view of how the signals will be processed with the currently selected operations.
Clear previews	returns the signals to their original state prior to the applied preview.
Remove All Processing	restores the raw data and removes ALL processing operations from the signals.

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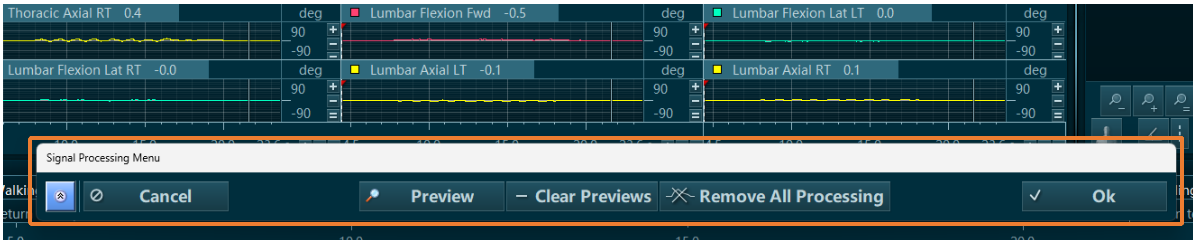
When the signal processing operations are previewed in the viewer, the signal processing menu is minimized to a toolbar line to give full view to all processed signals:



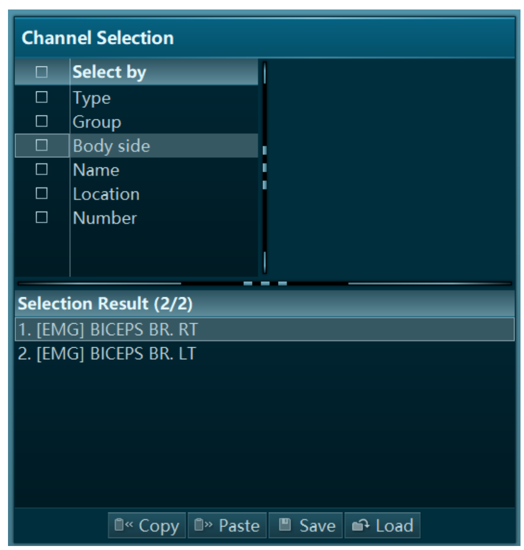
To maximize or minimize the signal processing menu, press the blue arrows button. Press **OK** if the preview of processing shows satisfying results.

Available Operations

Eight Signal Processing methods, listed under **Available Operations**, can be loaded to a sequence of processing commands (**Selected Operations**). Most of these operations are identical in function to the **Online Processing** methods found in the **Measurement** menu. Any sequence or order of processing methods can be arranged, and any arrangement can be loaded multiple times. Select a processing method and click **Insert,** or double-click on the method icon, to add it to the lower operation pipeline.

Filtering by Channel Selection

Each signal processing operation can be applied or restricted to certain channels, which can be selected in the **Channel Selection** menu. This feature is like the **Apply To** control in the **Online Processing** options in the **Measurement** menu.



If no box is checked in the **Select By** section, all channels are selected for processing. If the filter **Type** is selected, it offers a sub selection of all channel types in the record. If the subcategory **EMG** is selected, only the **EMG** type channels will be shown in the **Selection Result** list.

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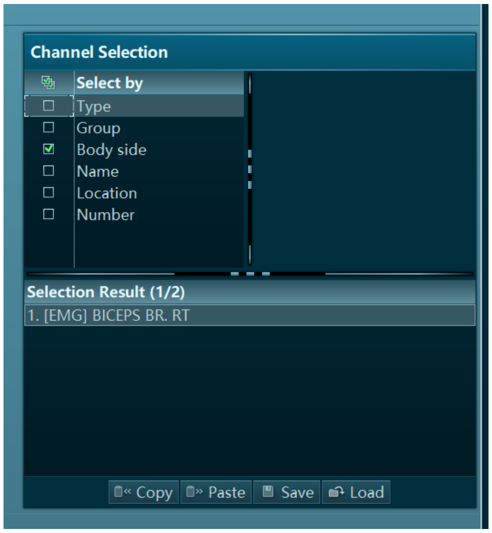

Filters work in cascading mode. For example, if the RT type channels are also selected under the **Body side** filter, only the EMG channels of right body side are selected.

*Note: the order of filters in the **Select Channels** section can be changed by mouse dragging a filter to a higher or lower position.*

ECG Reduction

myoRESEARCH is equipped with a unique algorithm to detect and selectively eliminate ECG artifact spikes from the EMG signal while maintaining the original EMG power. The algorithm responsible for this feature is a combination of pattern recognition and adaptive filtering.

By default, ECG reduction requires the user to input a time interval for the algorithm to look for clean heartbeats to establish a profile for the ECG waveform.

Select Interval By First Activity

Checking this box will instead use the first activity of a multi-activity recording as the interval to analyze for the ECG pattern.

Limit Duration To

This duration controls the time interval in the beginning of the recording in which the algorithm looks for the ECG pattern.

The quality of this "ECG cleaning" can vary depending on the EMG recording quality and artifact conditions. This ECG removing function can be helpful in relaxation studies, especially where ECG spikes can greatly influence the outcome results.

Rectification

This popular processing method multiplies all amplitude values in a signal with +1, where negative values become positive. This operation achieves positive amplitude curves that can calculate parameters such as mean amplitude and area under the curve.

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Smoothing

Typically, for amplitude-based calculations and analysis, the raw EMG is smoothed by digital filters, root mean square, or moving average algorithms. The effect is that non-reproducible EMG spikes are eliminated, and the mean trend of the EMG innervation is used. Smoothing also allows easier reading of the EMG patterns, which is useful for clinical tests and biofeedback-oriented treatments or training.

Algorithms

The following smoothing Algorithms are supported:

- RMS – Root Mean Square
- Mean – the moving average
- Mean absolute – the moving average with combined rectification

Window

This setting defines the window for each algorithm (in milliseconds) to calculate the smoothed value over. Select a window appropriate for the data being processed.

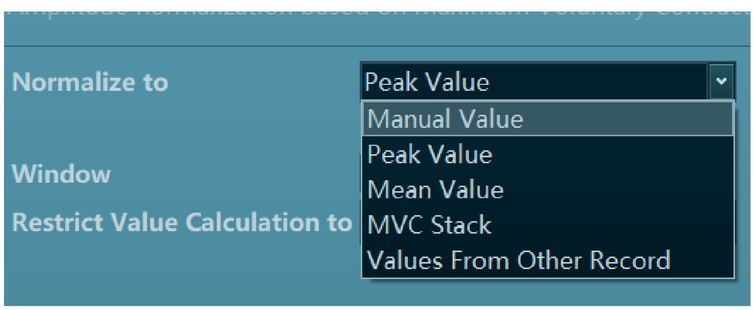
Refer to the ABC of EMG booklet for more information on how to use smoothing algorithms.

Amplitude Normalization

This feature allows amplitude normalization of microvolts based on EMG recordings. Amplitude normalization post-processing is like the online processing offered in the Measurement Configuration setup, with a few additional options. Usually, amplitude normalization is applied after rectification and smoothing of the data.

Normalize to

This entry allows for the selection of different normalization routines available in the pull-down list, as shown below:



- Manual Value - Here the normalization value can be entered manually. Manual values can be applied per channel– see the Filtering by Channel Selection section.
- Peak value - This is the main method of finding MVC (Maximum Voluntary Contraction) values within MVC records. An automatic algorithm separately calculates the MVC of each

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selected channel within the chosen interval. The highest mean window is in the channel and highlighted in green, and the user is notified that the MVC stack was updated. The **MVC stack** is an internal memory that archives the MVC values if it is filled by the next peak value normalization operation.

  - **Window**: This entry defines a window for the software to locate the highest mean MVC value in use. A smaller window typically means average MVC values will be higher than with larger windows.
  - **Restrict calculation to**: Click **Pick** to use the mouse to mark a signal portion in the record. The MVC calculation is only done in this marked area. This function may be helpful in measurement designs, where an MVC recording period is part of a given record. Click **Clear** to clear any previously selected areas.

- **Mean Value** – The amplitude mean value is calculated for each channel and taken as a reference value for the normalization. This type is often used in gait analysis where time-normalized EMG patterns are amplitude normalized to their own mean value.
- **MVC Stack** – This method loads the current MVC values from the stack and can be repeated as often as needed to normalized records and trials done after the MVC record of a subject. The MVC values remain in the stack, even if the myoRESEARCH is closed.
- **Values from other records** – This method accesses normalization values found or applied from other records for the same subject only. The normalization routine performed on this other record should be verified for its appropriateness for this function. See **Processing History** in the **Record** section in the **Database** help documentation for more information. Typically, this is the MVC trial of one subject, which is used to calculate/find the MVC values within the maximum contraction series. If a record that was not normalized by any method was chosen, this operation cannot be performed.

In summary, normalizing to **Peak value** creates an MVC value; normalizing to **MVC Stack** loads an MVC value to normalize other records from the same subject; and normalizing to **Values from other records** uses the MVC value from another record.

When amplitude-based normalization is loaded to the signal processing pipeline, the processing saves the MVC data to the MVC stack. These MVC values will be available for the trial records and the amplitude normalization mode normalized **"To MVC Stack".**

Restrict value calculation to

This operation applies amplitude normalization to a certain section of the record by highlighting the section using the mouse.

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Filtering

A set of commonly used digital filters is available in this processing method:

FIR - Finite Impulse Response filter

- Window: Number of points used to process the data. The quality improves with longer windows. Recommended default value is 79.
- Type: The filter can act as a low, high, band pass or rejecter filter. Use the Low/High frequency: controls to specify the filter range.
- Window: subtype selection and defines the window edge fading.

IIR - Infinite Impulse Response Filter

- Bidirectional: Use this mode to avoid phase lags created by the filter.
- Type: The filter can act as a low pass, high pass or rejecter filter.
- Frequency: The edge Frequency can be entered in Hz.
- Approximation: Defines the sub type of filter.

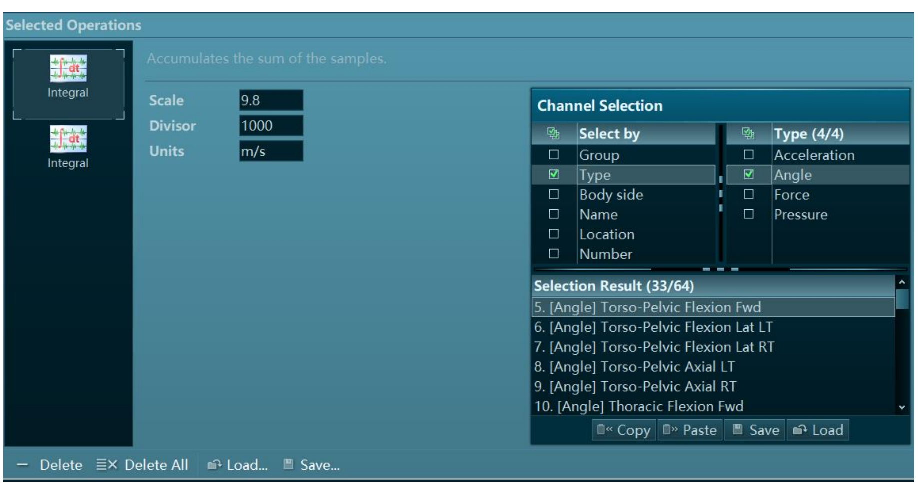
Median - Filter

The median filter is an excellent spike cleaner for analog signals. It removes spikes from force/angle curves without affecting the original signal shape.

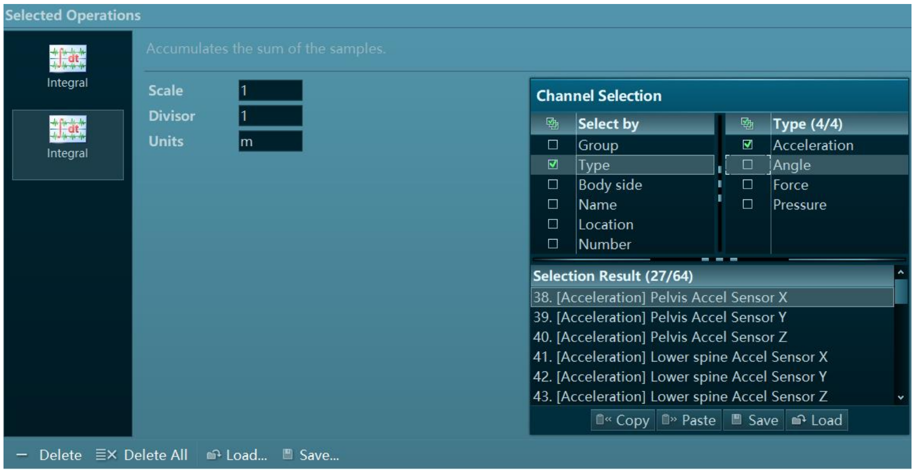
- Window: Defines the amount of data points used for the filter algorithm.

Integral

This operation calculates the integral of a given signal. For example, to calculate velocity out of a myoMOTION acceleration signal, the following settings can be entered. Source data is expressed in mG:



To operate a double integration from acceleration to distance, enter these settings in the second integral operator:



This is a preview screen of a double integrated myoMOTION acceleration signal:

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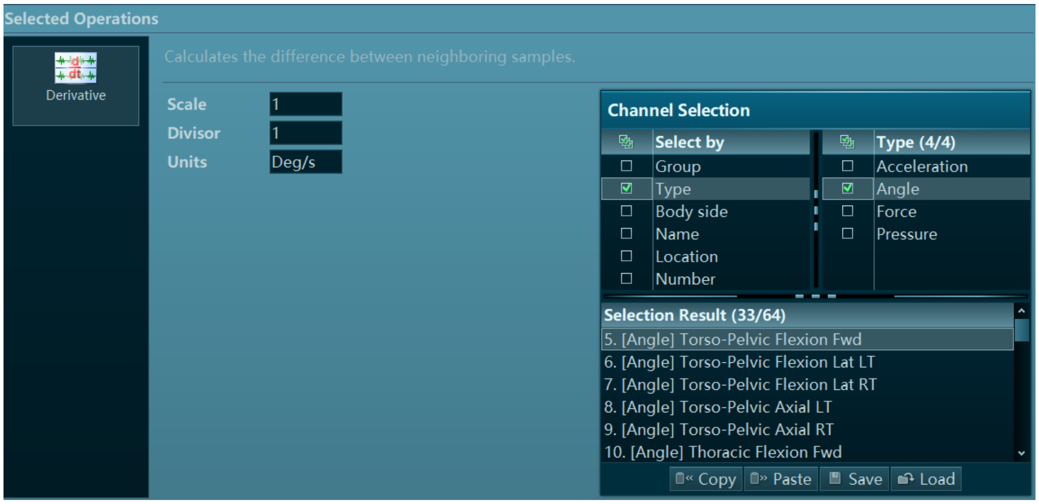
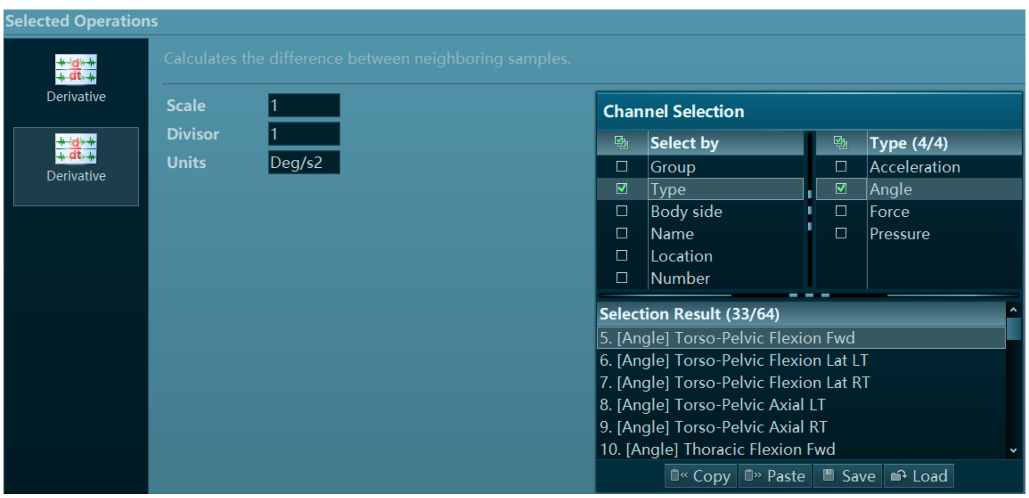
Derivative

Complementary to the **Integral** operation, the **Derivative** operation can be used to calculate the rate of change based off an angle signal to reach angular velocity:



The acceleration is derived if this is done a second time:



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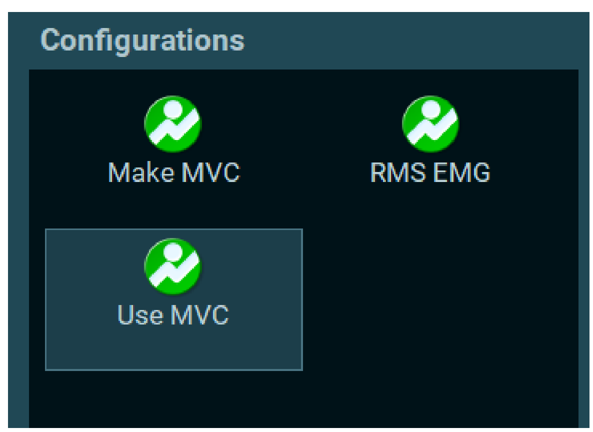
Here is an example of a double differentiated angle signal to angular acceleration:

Time Offset

This processing time shifts incoming signals to correct the delay caused by telemetric transmission, pre-processing time at the device level, and similar delay sources. Enter the time delay correction factor in **Offset**. Positive values will correct delays.

Configuration List

Any signal processing configuration saved under a user-defined name will be listed in this section in alphabetical order. Click on the desired configuration in the right toolbar to automatically apply it to the displayed signals. The signal processing menu will be minimized so the previewed processes can be reviewed:



If the preview shows the desired results, confirm the processing by clicking **OK**.

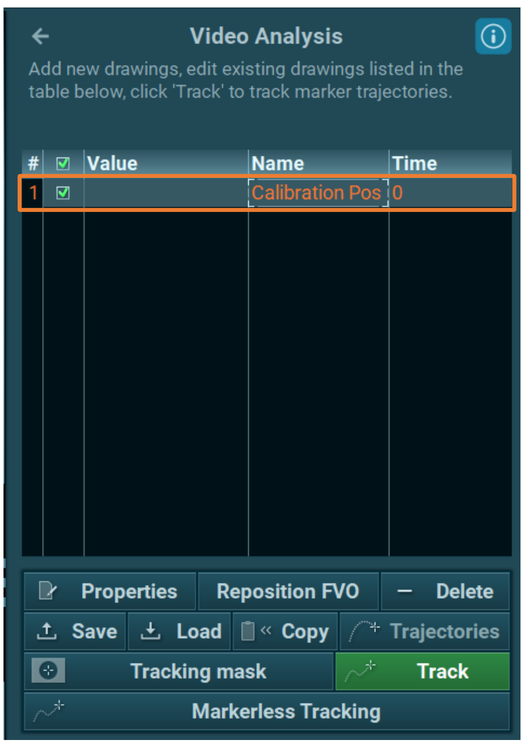
Three predefined configurations are installed by default:

Make MVC	finds and stores MVC values in MVC records.
RMS EMG	produces smoothed/rectified EMG curves based on RMS at 100 ms.
Use MVC	smoothes at RMS 100 ms and normalizes the records to the MVC values loaded to the normalization stack.

Video Analysis

The **Video Analysis** function contains the main video analysis tools for myoVIDEO and consists of a set of angle and distance drawing tools, a text label function, a distance measurement tool, crosshair function, and trajectory analysis based on auto-tracked angles or single markers.

All drawings except **Trajectories** can be used on any video picture position for an unlimited number of times. For example, there could be 4 sets of 3 marker angles for a total of 12 markers being tracked.



When a drawing is placed into the video after clicking **OK** on the setup window, it can be dragged and positioned to another place within the video. Any drawing will be automatically displayed in the drawing list, below the drawing tools, as shown in the image to the left.

The checkbox to the left of the drawing can be used to toggle its visibility within the record. Additional functions can be found in the toolbar below the drawing list:

Properties	reopens the setup screen to reconfigure the settings for a drawing. This action can also be performed by double-clicking on a drawing name.

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Reposition FVO	adjusts the force vector overlay markers. This function is only available with an active force plate channel in the recording.
Delete / Save / Load	deletes, saves, and loads a drawing. Hold **Ctrl** to select multiple at a time. ***Note**: The green checkmark toggles the drawing for viewing purposes only.*
Copy	copies of a drawing to a clipboard so that it can be reused in another frame of the video. This function also pastes all drawing data to other Windows applications like Excel.
Trajectories	display the traces of auto-tracked dots/angles in the video.
Tracking mask	excludes areas of the videos from tracking. This function is associated with reflective marker tracking and will be discussed later in this section.
Markerless Tracking	automatically tracks angles included in the integrated Media Pipe tracking engine. No reflective markers are required for this process. Reference Markerless Tracking Tips to ensure your measurement and camera setup is optimized for markerless tracking.
Track	automatically tracks dots or angles that are defined just above reflective markers using an auto-tracking algorithm. It can be used to compile marker-based angle curves or trajectory traces. ***Attention**: Auto tracking requires certain camera settings and tracking markers, which is explained in more detail in the chapter Auto tracking.*

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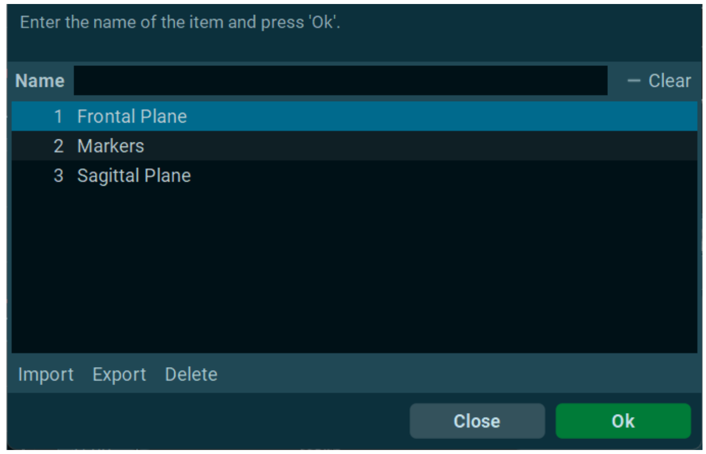
Templates

If a given set of angle drawings or distance measures applied to reflective markers is needed for other recordings or subjects, they can be stored as a drawing configuration and loaded to a new video record. Select the drawings to save and click **Save** under the drawing list to store the set:

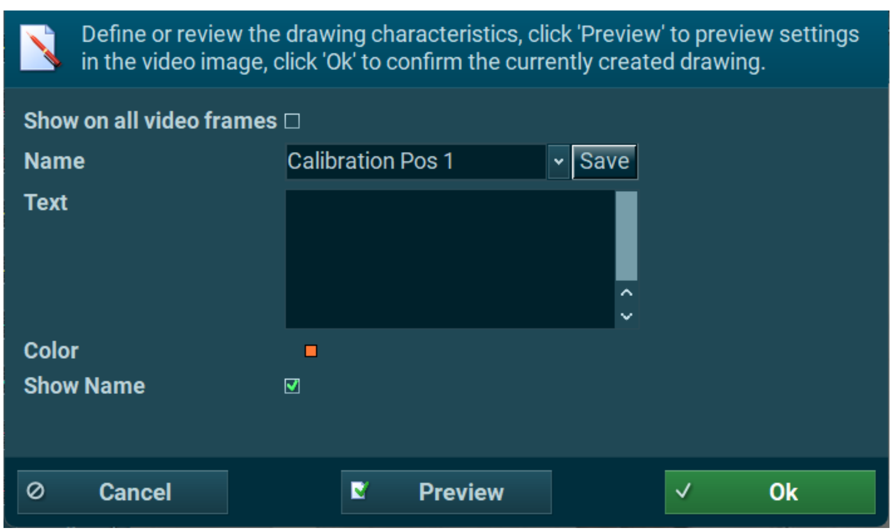


Enter a name and click **OK** to save the configuration and return to the **Video Analysis** menu. The templates can also be deleted, exported, and imported to other myoRESEARCH installations. To load a template, click on **Load** to apply it to the selected video.

Note: This function requires the same marker set, camera settings and positioning, and good video quality without reflections to work optimally.

Text

This tool places a text string on the video picture. When opened, click on the desired point where the text should be shown:



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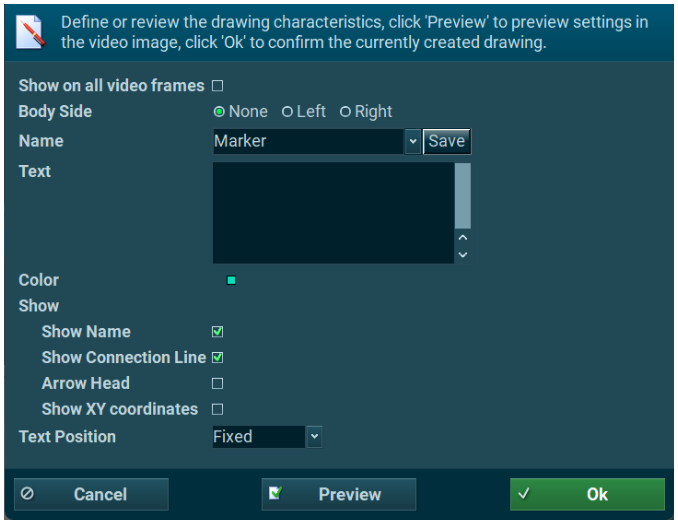
If **Show on all video frames** is checked, the text string will be visible over all individual video frames for the entire record. The **Name** entry serves as the label for the optional text box entry below. The **Color** entry defines the text color and **Show label** will toggle the text label.

Title text can be saved by pressing **Save,** or a preexisting one can be selected from the drop-down list. Clicking the **Preview** button at the bottom of the window previews the text on the video. The text font size can be changed in the **Video Options** menu.

Marker

This tool can be used in two ways.

Its main purpose is to mark or label a point of interest in the video. If this point is placed on a reflective marker, it can be auto tracked.
To set a marker, click two points in the video picture. The first will be the arrow end or dot. The second one will be the position of the label/text:



*Note: To better view a marker or point, use the mouse wheel to zoom into the video picture. Alternatively, holding down the **Ctrl** key while hovering the mouse pointer over the video will temporarily zoom in on the mouse pointer.*

Each entry in the marker setup window is explained below:

Show on all frames	makes the marker visible on all frames of the video.
Body side	can be defined based on which side of the body the marker is attached to.
Name	defines a new or preexisting label name for the marker. Previously saved names can be accessed through the drop-down menu.
Text	stores additional, optional text for the marker.
Color	defines the color of the marker.

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Show Name	turns on/off the label text.
Connection line	turns on/off the line between tip and end of the marker's arrow.
Arrowhead	places an arrowhead to the first dot.
XY coordinates	creates XY coordinates. If auto tracked, the continuous data is available in the export file but not the record **Viewer**.
Text Position	changes the position of the text in relation to the point

All markers, like text labels, are placed into and managed in the drawing list within the **Video Analysis** toolbar.

Distance

This drawing tool measures the distance between two points in the video. This calculation is based on a reference line that determines a known distance within the same video such as a 1-meter scale. The reference line will automatically appear near the corner of the video frame and can be moved around by clicking and dragging the line or its endpoints to match the known distance. The properties of the reference line can be adjusted in the **Video Options** menu.

After selecting this tool, click two points in the video to determine its distance. To better view a marker or point, use the mouse wheel to zoom into the video picture. Alternatively, holding down the **Ctrl** key while hovering the mouse pointer over the video will temporarily zoom in on the mouse pointer.

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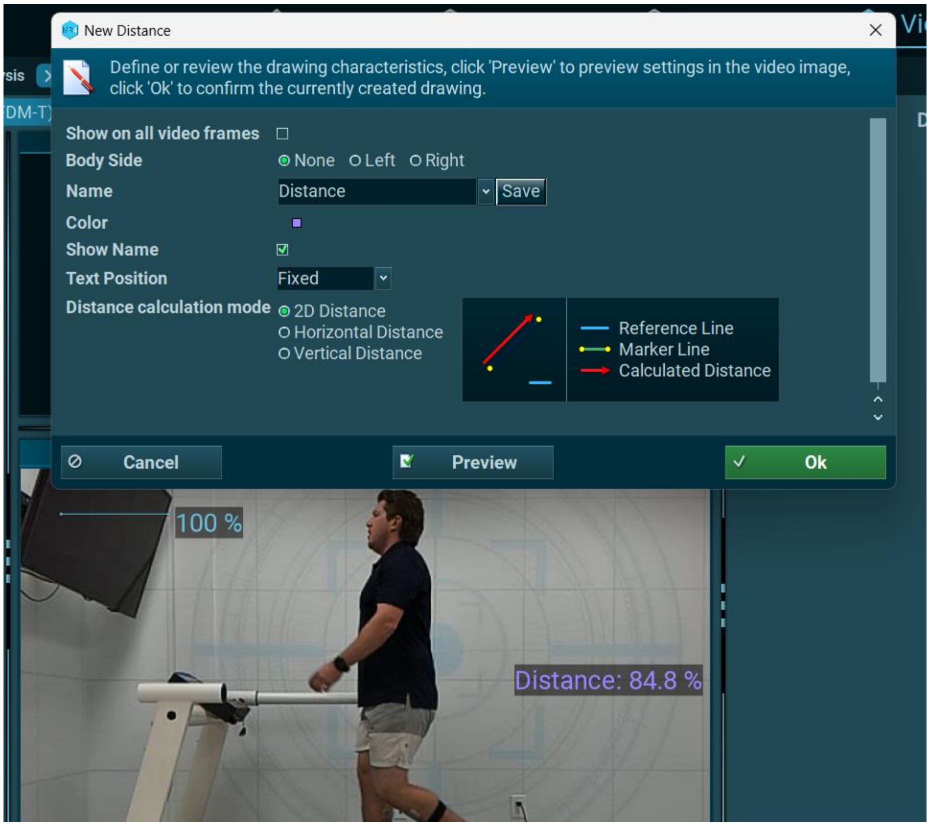

When done, the Distance property settings screen automatically pops up and determines body side, name, color, polarity, and distance calculation mode. The latter offers 3 modes, each of which is explained in greater detail in the following sections.

2D Distance

This mode measures the distance independent of the orientation to the reference line.

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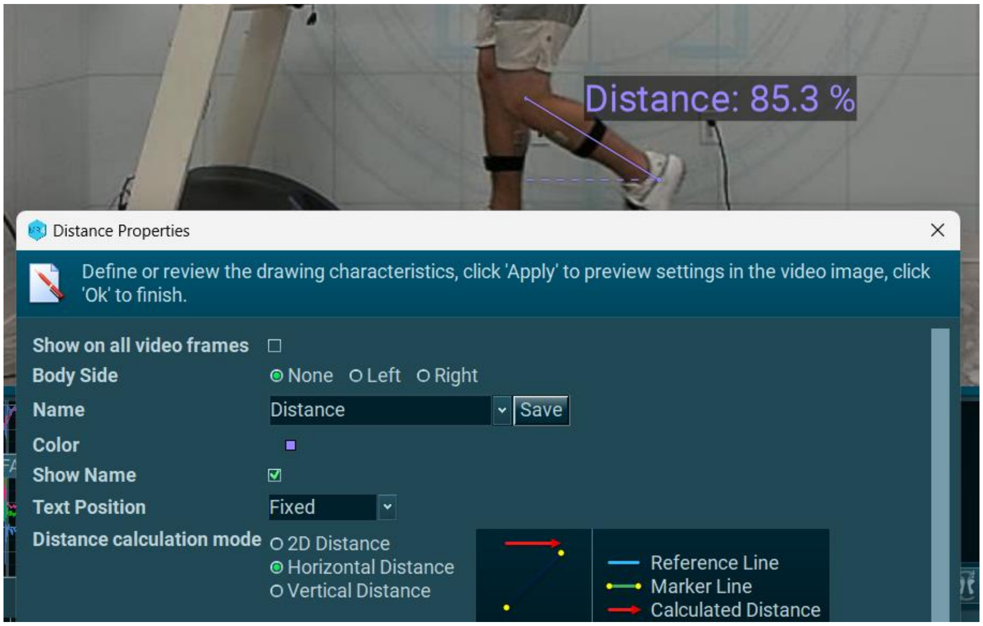
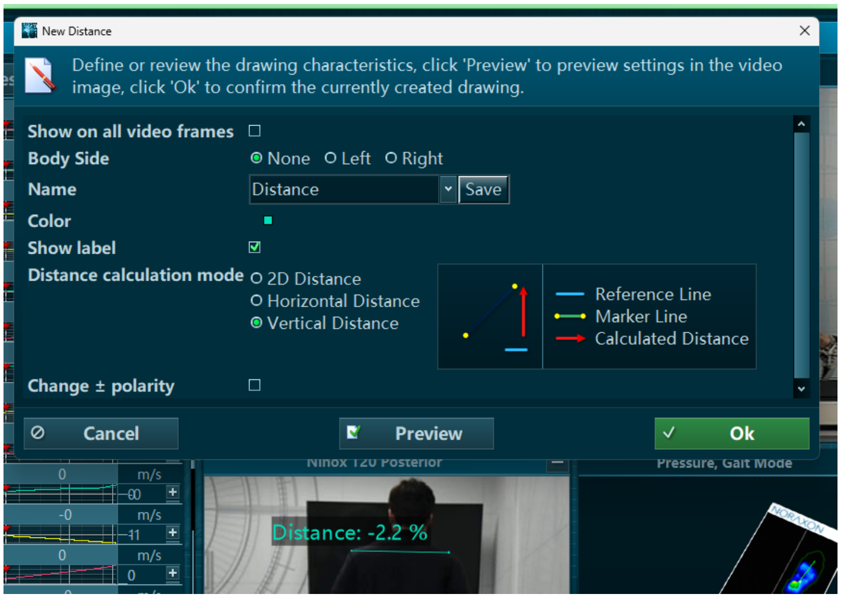
Horizontal Distance

This mode measures the horizontal distance between the 2 markers. The horizontal line is defined by the position of the reference line. In the following example, the horizontal distance between the knee and ankle marker is calculated and displayed as a horizontal dashed line:



Vertical Distance

This mode measures the vertical distance between the 2 markers. The vertical line is defined as perpendicular to the position of the reference line. This tool can, for example, be used to measure the shoulder height difference between the left and right side of the body. The vertical difference is calculated and displayed as a vertical dashed line:



All distance drawing settings can be stored under a configuration by clicking **Save** to the right of the **Name** entry. If placed on reflective markers, the distance tool can auto-track dynamic changes, as explained in a later section of this chapter.

The final entry in the **Distance** setup window is **Change ± polarity**. When checked, this function simply inverts the polarity of the calculated value, which may be necessary depending on the configuration of the video or markers.

2-Marker Angle

This tool is designed to measure the tilt against the vertical or horizontal orientation line of a video frame. To match the true vertical or horizontal orientation, the camera needs to be positioned so that its vertical and horizontal frame matches the true world orientation.

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To define a 2-marker tilt line, click two points in the video. This can be selected by zooming in using the scroll wheel or holding **Ctrl** to temporarily zoom into the mouse pointer.

The drawing option menu will open, offering several adjustment possibilities. Each entry is described in greater detail below:

Show on all video frames makes the marker visible on all video frames of the video.

Body Side defines the polarity of an angle based on which side of the body is selected. For example, a knee flexion motion changes polarity if this motion is seen from the other side.

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Name	defines a new or preexisting label name for the angle. Previously saved angles can be accessed through the drop-down menu. The myoVIDEO module comes with a set of commonly used angle names and their correct setup.
Color	changes the color of the angle and text.
Save	Any label name and all settings done for this given angle can be stored for permanent use by pressing this Save button.
Change ± Polarity	inverts the signal polarity when toggled. Depending on the body side, a limb movement can create a negative or a positive polarity curve and may need to be switched.
Advanced	opens a more comprehensive setup system for anatomical angle definition based on the neutral – zero method. This is typically used in the medical and healthcare field.

If the 2 dots marked with this tool are on reflective markers, they can be auto tracked using an algorithm by clicking on the green **Track**. If successfully operated, it will create a new angle channel to the right of the video displaying the calculated dynamic angle curves. This operation will be discussed in further detail in a later section of this chapter.

Advanced Settings - Anatomical Zero

These settings define the anatomical zero reference angle, which can be selected in increments of 90° or manually entered by selecting **Other**. Generally, anatomical angles are defined to be zero when a human body is standing upright in what is known as the anatomical position.

Mechanically, most angles naturally show an angular offset in this position, excluding the shoulder joint. To correct for these offsets, use the **Advanced** settings offset correction factors listed under **Anatomical Zero**.

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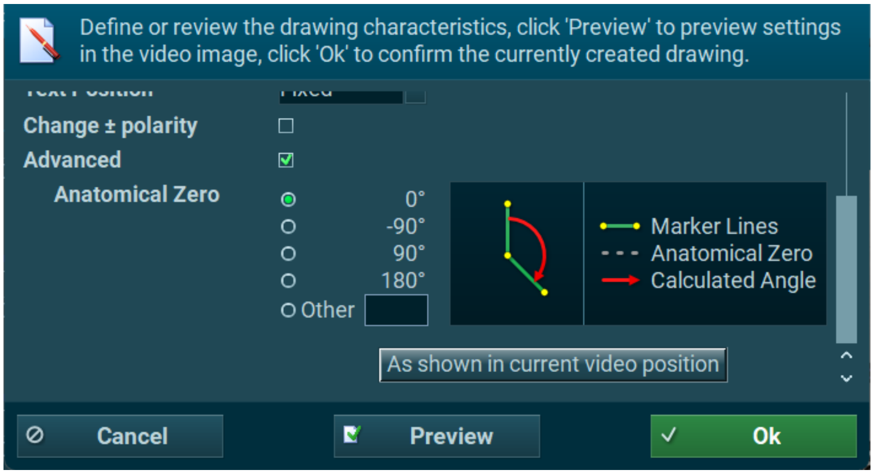
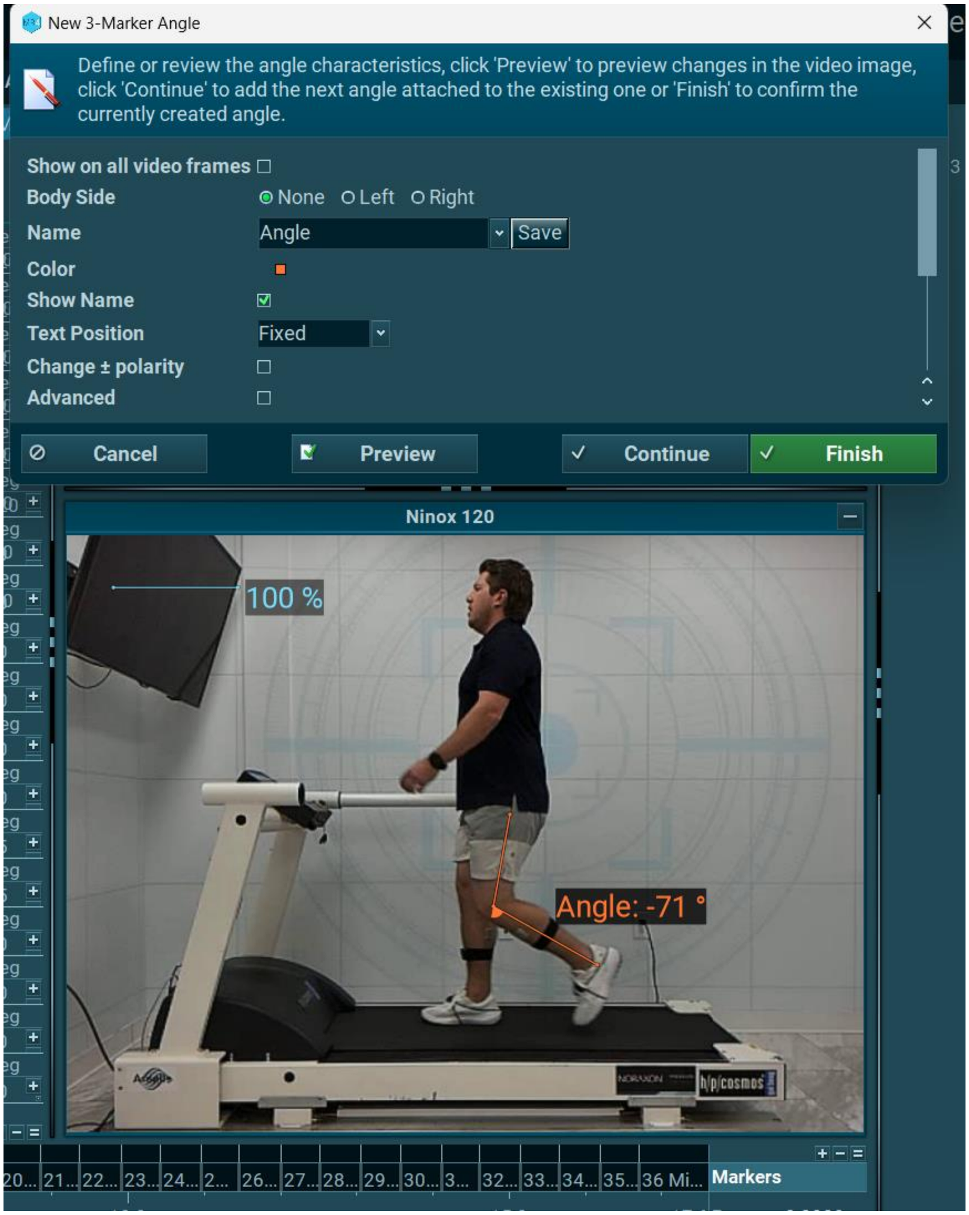


The anatomical zero angle can also be adjusted to the value of the angle/chain in the current video frame by clicking **As shown in current video position**. This function is useful if the subject begins or ends the recording at anatomical zero. Pause the recording when the given angle is at a position of zero degrees, click on the 3-marker angle, open the **Properties** menu, and click this button.

Any setting definitions created in the standard or advanced section can be stored under an angle name by clicking **Save** to the right of the **Name** entry.

3-Marker Angle/Chain

This analysis is based on 3 points, typically placed around a joint. Click on the tool and place 3 markers on the video frame using the left mouse button. To better view a marker or point, use the mouse wheel to zoom into the video picture. Holding down the **Ctrl** key while hovering the mouse pointer over the video will temporarily zoom in on the mouse pointer.



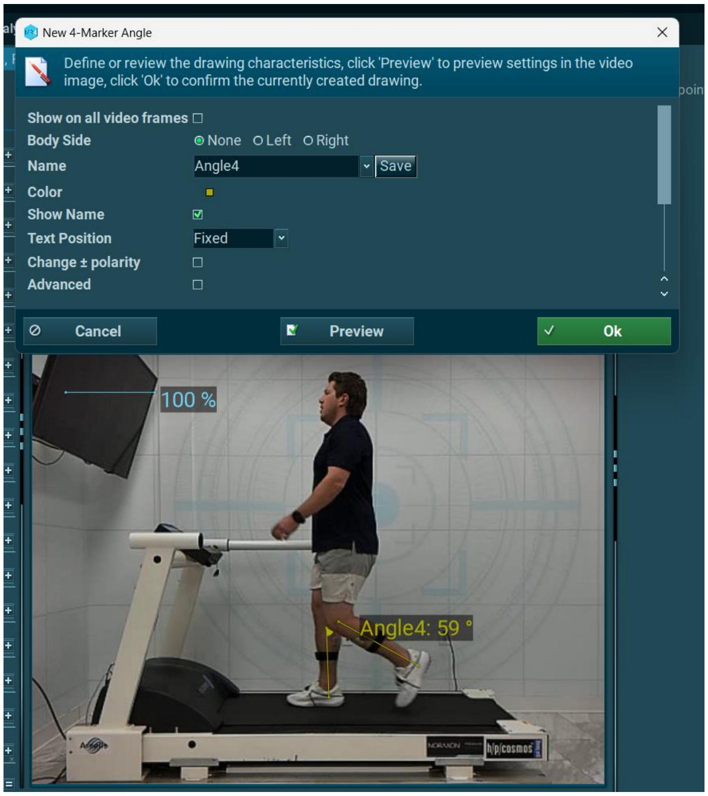
After selecting three points, an angle property menu will pop up and offer a set of helpful options and functions. This menu is identical to the 2-marker angle setup menu; for a more in-depth explanation on each entry and their functions, please refer to the previous section on the **2-Marker Angle** drawing tool.

All drawings added to the given video record are listed in the **Drawing list** on the right toolbar:

4 Marker Angle

The 4-marker angle tool calculates the angular orientation between two lines drawn between four markers. Start by left-clicking on the first two markers to create the first line, then continue left-clicking on the two markers to create the second line.

When done, an angle properties screen will appear. Here exist helpful entries to define the body side, angle name, color, polarity, and anatomical offsets, as explained previously in the **2-Marker Angle** and **3-Marker Angle** tools.



If placed on reflective markers, the 4-marker angle tool can auto-track dynamic changes. This feature is explained in the **Auto-Tracking Markers** section.

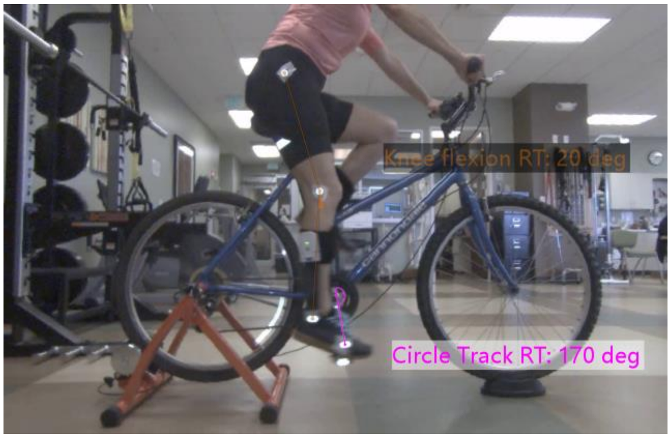
Crosshair Tool

The crosshair tool is purely a visualization tool and can be used to visualize symmetries, side ratios, and distances against vertical and horizontal orientation lines.

If placed on a reflective marker, the crosshair tool can be tracked by the auto-tracking tool by clicking the green Track button in the toolbar in the **Video Analysis** menu.

Circle Tool

The circle tool was designed for bike fitting analysis. It can track a reflective marker on a bike pedal that measures the angular rotation of the crank cycle related to a reference point:



The reference point can be defined or adjusted in the **Properties** menu of this drawing; alongside other helpful functions that have been discussed in the previous sections. The angular rotation of the marker is calculated based on the configured **Anatomical Zero** in these settings.

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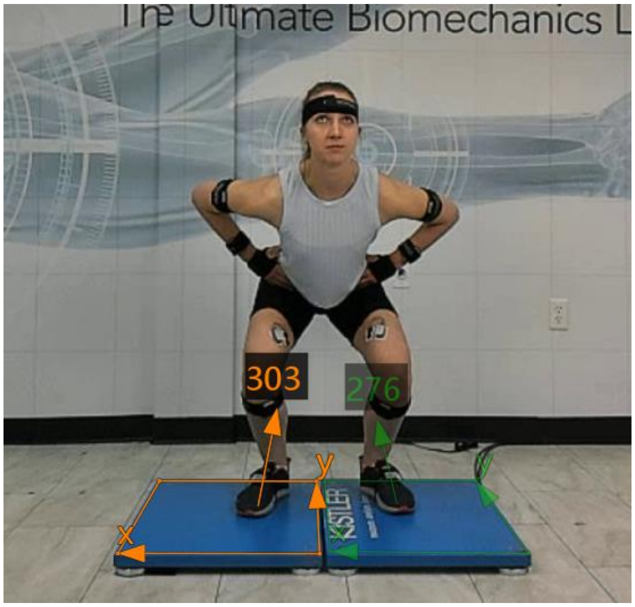
Attention: when changing camera positions, the reference line needs to be redefined. Make sure to place the reference line as close to the video projection plane as possible to avoid perspective errors.

Force Plate (Force Vector Overlay Analysis)

The myoVIDEO recording can be combined with 3D force plates to show the force vector overlay of a subject when in contact with the force plate:



A force plate can be integrated to myoRESEARCH recordings via a connection to a Noraxon AIS analog input board or digitally via a firmware driver of the given manufacturer. Please refer to the Force Plate Systems section in the Measure section of the help documentation or the device-specific manuals for your force plate for more information.

To apply force vector overlay, click the video window, select **Video Analysis** in the right tool bar and choose **Force Plate**. A prompt will appear to set up an **X-Y** coordinate frame by clicking each of the four corners of the force plate, beginning with the origin. Be aware of the orientation of the force plate coordinate system.

*Note: This calibration coordinate frame is only shown in **Video Analysis** mode, not in **Play** mode.*

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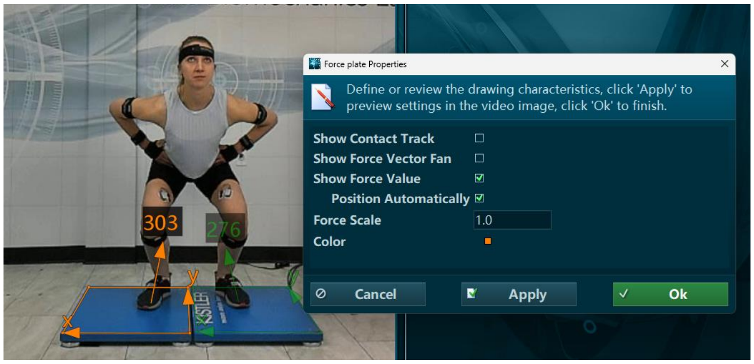

A force plate **Properties** menu will appear to configure the drawing's settings. Each entry within this menu is explained in greater detail in the following sections.

All settings chosen for the force plate setup are automatically stored as a camera setting and are available if the **Force Plate Analysis** option is selected for the new record. The setting will stay active as long as it is overwritten by these new settings.

Show Contact Track	enables visualization of the force plate vector traces for each plate in the recording.
Show Force Vector Fan	visualizes the sequence of vectors over time.
Show Force Value	displays the magnitude value of the force vector in the video.
Force Scale	allows upscale or downscale of the vector magnitude if the force vector exceeds the video window or is too small to see.

Note: *these settings need to be repeated whenever the selected camera position is changed, like a sync light position.*

This FVO process will be automatically repeated if there are multiple force plates involved in the recording. Enter **Play** mode to see the force vector during the replay of the video. To test the accuracy of the force vector visualization, refer to the video tutorial on Force Vector Overlay on Noraxon's website.

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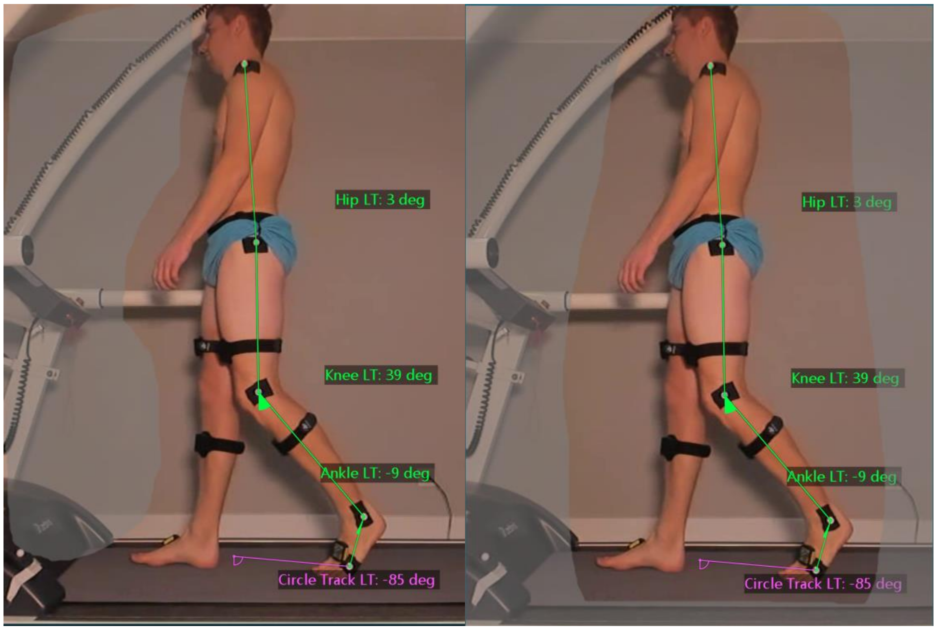
Tracking Mask

The **Tracking Mask** tool is designed to optimize marker auto-tracking and exclude unwanted reflective areas from the tracking algorithm. The marked areas are shown as a gray shaded form.



The **Tracking Mask** tool can be used in two modes:

Include all	This is the default mode. With the left mouse button, areas can be drawn and excluded.
Exclude all	If clicked, the whole screen will be shaded. Pressing the **Shift** key and selecting an area with the left mouse button will be included to be used for the tracking algorithm.

2D Markerless Tracking

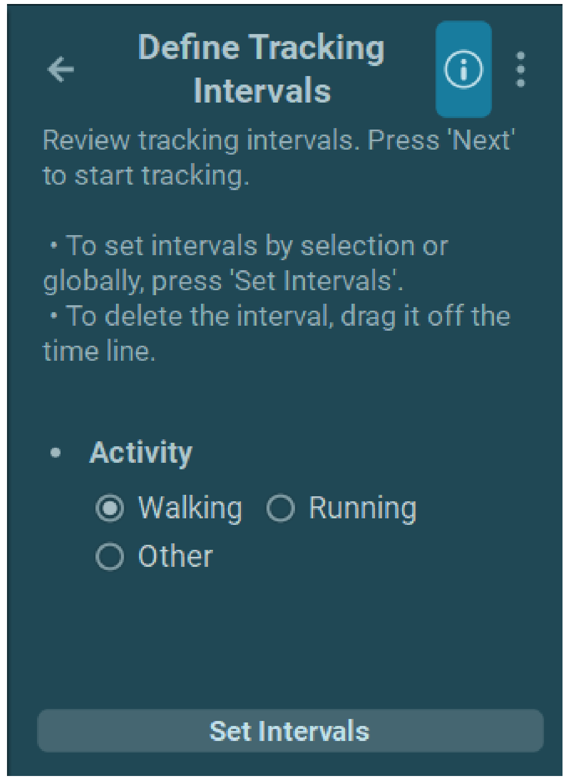
Markerless Tracking is used to measure approximate joint angles in a video recording without the use of physical markers placed on the subject body. Instead, an image processing algorithm calculates joint angles during an activity, such as walking or running.

This feature requires at least one camera positioned to view either the **sagittal** or **frontal** plane of motion as the subject moves but works best with **two cameras** capturing both planes of motion.

Clicking the option reveals this set of options to configure the video tracking behavior:

The software first instructs the user to select the interval to track – highlight the timeline and click set intervals to identify what section of the recording to process.

Activity – Select one of these options to identify which activity the subject is performing in the recording.



After hitting the Next button, myoRESEARCH will analyze the video footage, and will create an overlay over the subject's body, with joint angles being displayed as the footage plays.

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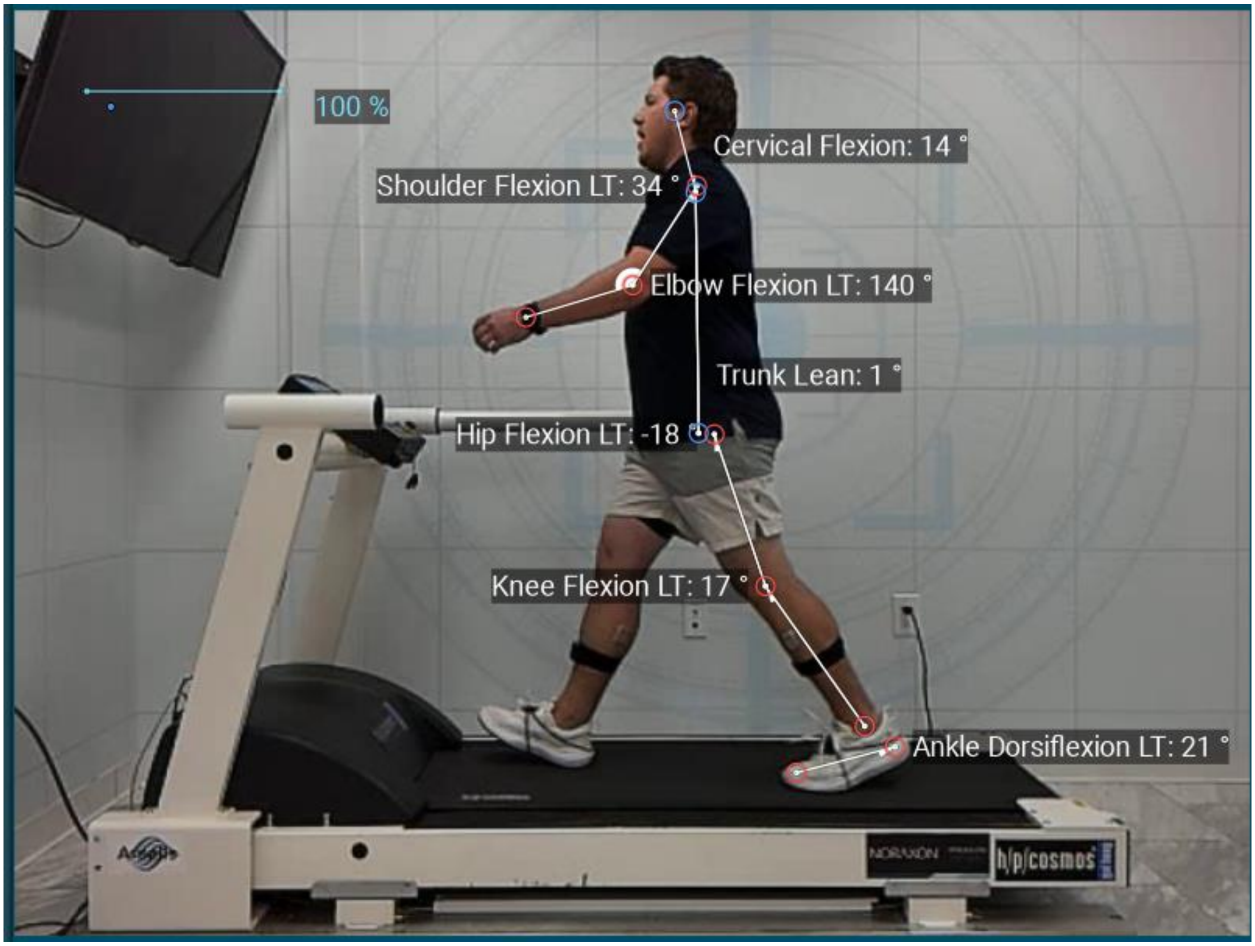

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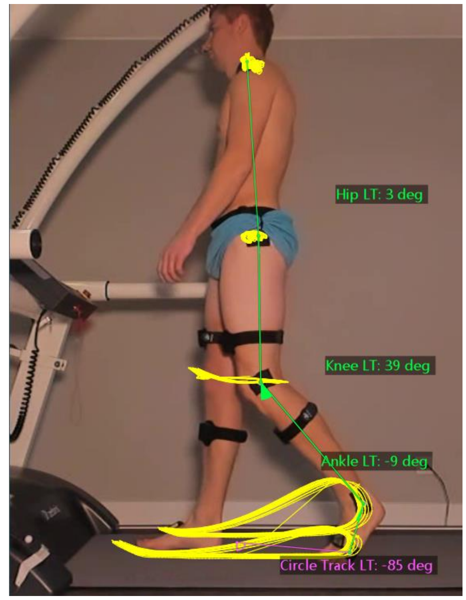
Trajectory

This tool is based on the results of auto-tracked dots or angles and visualizes their two-dimensional path along the video projection plane:

This tool allows the visualization of motion patterns, quality, and constancy.



Auto-Tracking Markers

Marker auto-tracking is a powerful tool that allows for automatic marker detection and tracking over the course of all pictures within a video clip. myoVIDEO's auto-tracking algorithm works on the principle of contrast and object recognition (circular dots). Successful auto-tracking requires well-defined video recording conditions, which are described below.

Equipment

30Hz HD camera(s)

- *Recommended: Logitech c920*

Noraxon Power LED light

Reflective markers (1.9 cm)

125/300c Hz Ninox color HS camera

Sync to myoSYNC

Reflective markers (1.4 – 1.9cm)

General Remarks

It is highly recommended to use equipment provided by Noraxon, as all components have been tested for their ability to work in conjunction with each other and will produce the most accurate automatic angle tracking.

The 30Hz cameras, like the recommended Logitech c920 model, are meant to be used as a scene and documentation camera. Due to the low frame rate, the 30 Hz camera cannot be used for fast motions like running and jumping. It is recommended to use the supported Ninox 120 ~~25 or~~ 300c Hz color cameras for marker auto-tracking tasks. The higher frame rate is an essential component of the tracking algorithm stability, especially in faster motion.

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The tracking algorithm supports compressed and uncompressed videos. It is highly recommended to compress videos.

Best Practices

The quality and stability of auto-marker tracking depends on the light and background conditions as well as optimal camera setup settings. The most important strategy is to keep shutter time as low as possible; otherwise, marker blurring can become an issue. It is also crucial that marker sets stay in a perpendicular plane to the camera For Markerless tracking, it is crucial that the subject's movement is perpendicular to the camera.

For a comprehensive list of best practices, reference the Tips for Markerless Tracking Guide.

Camera Settings in myoVIDEO

Motion JPEG at 800x600 resolution is a good compromise between picture quality and data file size. Certain computers with high-end graphics cards and processors can also use full HD files. However, higher resolution does not necessarily improve tracking stability.

If the camera(s) are rotated to the recommended portrait position, the rotate setting should be clockwise or counterclockwise, depending on the mounting direction.

Below is a list of camera settings that should be adjusted to get the most stable and reliable auto-tracking results:

Exposure	This is the most important setting for stable auto-tracking. It reduces the lens shutter time, which is important to avoid marker blurriness. Select the lowest available setting; for the standard Noraxon cameras, this value is -1.
Gain	The gain factor depends on the overall light condition. The gain factor can be used to brighten up the video, which is automatically darkened by the low exposure value. Find a good balance between visibility and contrast between the reflective markers and body. Typical gain is around 125.
Focus	Enabling manual focus improves tracking stability. The chosen value should provide the most contrast between the markers and the background.

Marker Blurriness

It is important to avoid any marker or object distortion. The most important action here is to use the lowest exposure time available at a given light/gain condition.

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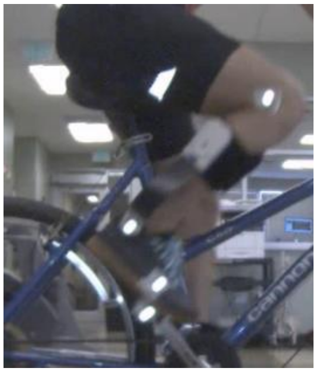
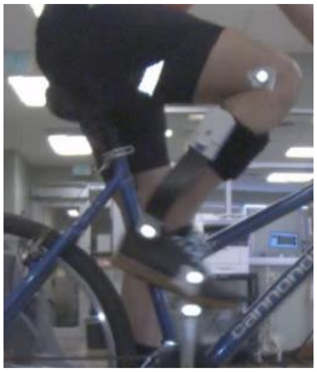
Poor exposure: *Good exposure:*





Tracking

Dots, crosshair, and marker sequence angles can all be tracked in myoRESEARCH. Typically, 3 markers are used to define a joint angle.

First, a tracking object must be determined or created, such as an angle or a chain of angles. Once an angle is selected from the list of drawing objects on the right toolbar, press the green **Track** button to start the tracking process.

The system also tracks multiple angles acting in a chain. To create multi-joint chains, start with the definition of the first angle; label it under the **Name** entry. Press **Continue** and define the second angle, and so on for each angle. When done with the angle chain, press **Finish**:

The algorithm can also track black dots on bright skins or surfaces, although it is strongly recommended to use the white reflective markers included in the package from Noraxon for tracking. For the black dots, low shutter time is required, as well as ensuring that other black objects do not distract the tracking algorithm.

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Auto-Tracking of Trajectory

When single markers are tracked, their two-dimensional pathway can be visualized by the **Trajectory** tool introduced in an earlier section of the chapter. If access to the 2D coordinate data is necessary, use the record **Export** function in the **Database** menu; it will export the timeline data and the **X-Y** coordinates of each picture position.

To the right is an example of 2D coordinate data exported to an Excel spreadsheet.

Name	Gait - Treadmill + Marker Tracking (POI Gait Report LT)			
Frequenc	100			
Date	30-10-2018			
Time,s	Markers	Marker Na	Activities	Activity Names
-1.2	1	Video	1	Activity
-1.19	0		0	
-1.18	0		0	
-1.17	0		0	
-1.16	0		0	
-1.15	0		0	
-1.14	0		0	
-1.13	0		0	
-1.12	0		0	
-1.11	0		0	
-1.1	0		0	
-1.09	0		0	
-1.08	0		0	
-1.07	0		0	
-1.06	0		0	
-1.05	0		0	
-1.04	0		0	
-1.03	0		0	
-1.02	0		0	
-1.01	0		0	
-1	0		0	
-0.99	0		0	
-0.98	0		0	

When single markers are tracked, their two-dimensional pathway can be visualized by the **Trajectory** tool introduced in an earlier section of the chapter. If access to the 2D coordinate data is necessary, use the record **Export** function in the **Database** menu; it will export the timeline data and the **X-Y** coordinates of each picture position.

Below is an example of 2D coordinate data exported to an Excel spreadsheet:

Two Video Camera Recording

The myoVIDEO module can support as many cameras as necessary. By default, myoVIDEO ships with two HS video cameras. A second camera can be added to any configuration anytime by entering the **Measurement Configuration** for a given protocol.

All video analysis functions can be used as normal. The selected video clip window is highlighted a light blue color, limiting all drawings to this video window only.

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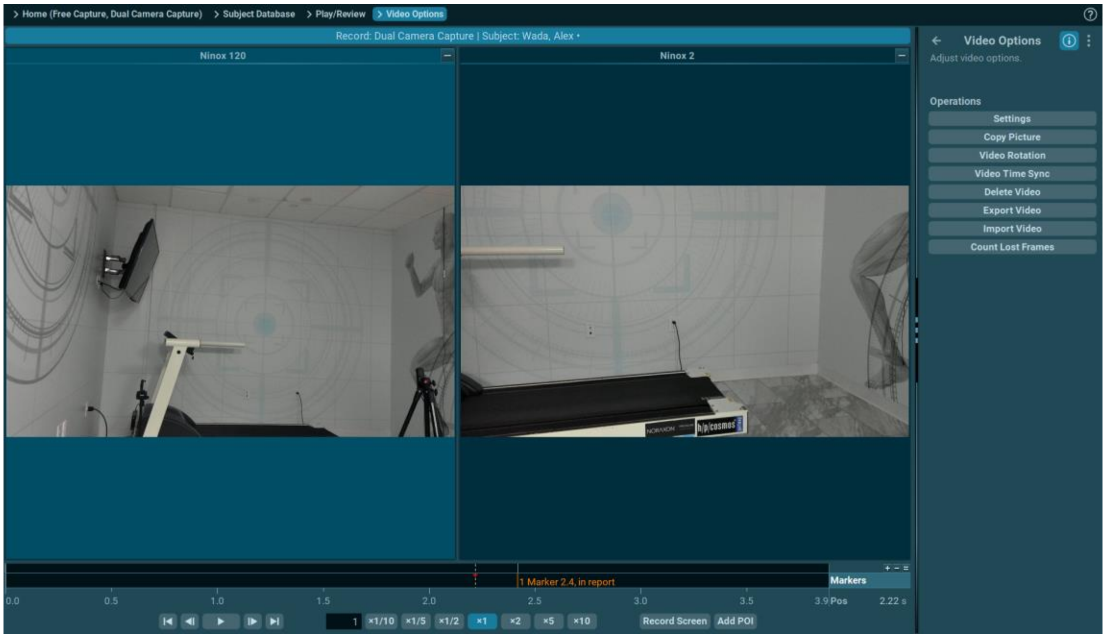
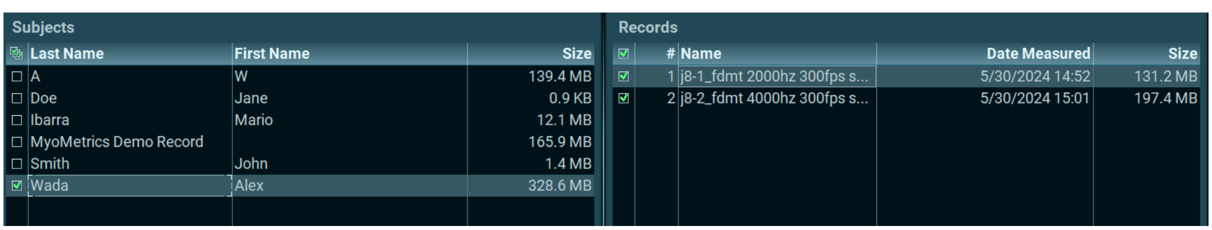


Both videos are automatically included at any point of interest analysis– use the **Add POI** Button to define POIs.

Two Video Record Comparison

The myoVIDEO module supports a direct comparison of two video records in the record viewer. To load 2 records into the **Viewer**, select a record from the **Database** menu, hold the **CTRL** key and select a second record. Then click the green **View** action button or **Viewer** on the top navigation bar:



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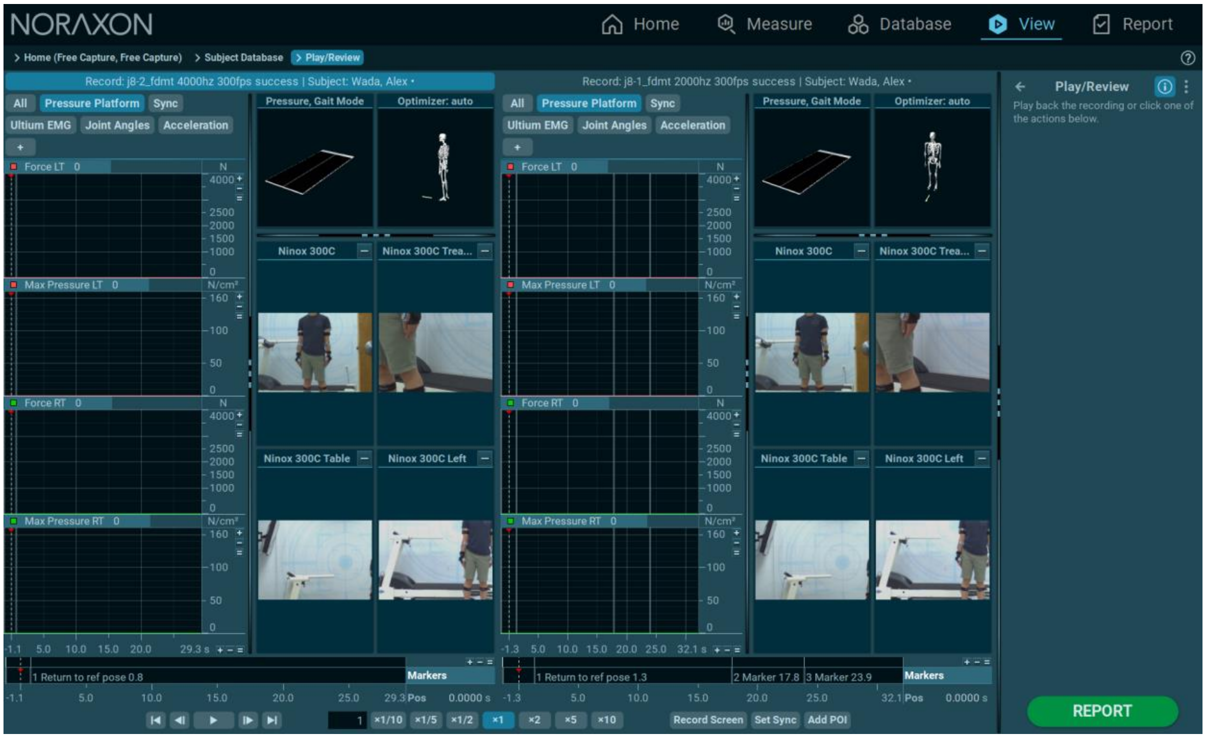
Both records will be shown in a direct comparison. The **Play** button on the video control bar will play both videos simultaneously:



Click the **Set Sync** button on the video control bar, beside the **Add to Report** button, to start the video at a defined position and repeat the same procedure with the other video record. The **Play** button will now play both videos from the sync points.

Video Options

The second video-related menu under the toolbar on the right side of the **Viewer** tab is the Video Options menu. **Video Options** consists of a setup for **Count Lost Frames**, **Delete Video**, **Export Video**, **Import Video**, **Video Time Sync**, **Video Rotation**, **Copy Picture** and **View Settings**.

View Settings

This button opens a setup screen that can rename the camera, adds a grid that enters the number of vertical/horizontal lines that are wanted, a reference line that is used in conjunction with the distance tool, and the ability to adjust the font size.

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Copy Picture

This button takes a snapshot of the **POI** (point of interest) in the video. It copies it to the clipboard and can be pasted into any other Windows based document or an email.

Video Rotation

This function rotates the orientation of the video as needed, which may be useful if the cameras were configured incorrectly prior to recording the video.

Video Time Sync

This function syncs videos with other peripheral devices in a recording, such as EMG or another video that may have been imported from an external location. It can also be applied to the current video, or all videos associated with the record. This function assumes there is a sync event both in the given video and in the associated signal recording.

1) Locate the **sync event on Video**.
2) Locate the **sync event on reference Signal**.
3) Verify and adjust **Video Offset** if necessary.

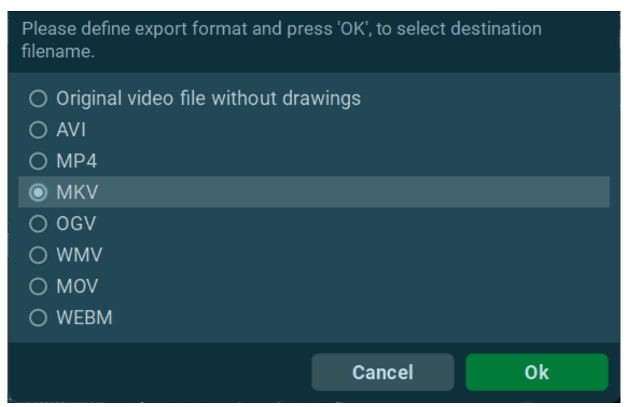
This procedure is identical to importing a video from an external source, as explained in a previous **Import Video** section, and it is identical to importing a video from the **Edit** menu.

Delete Video

This function deletes the selected video from the recording. The video cannot be recovered once the changes to the record have been saved and the **Viewer** menu has been exited.

Export Video

This menu option exports the original raw video or a video that includes all video analysis drawings. The default format is *.mkv. You may also export in any other supported filetypes such as .avi and .mp4, listed on the right. Select a path for the stored video after clicking **OK**.



Import Video

See how to **Import Video** under the **Edit Menu** section of this help document.

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Count Lost Frames

If data loss is expected in a recording, myoRESEARCH can process and calculate the percentage of dropped frames in a recording to assist with troubleshooting.

Marker Menu

The marker menu manages the marker labeling in the recording. The main purpose of the markers is to define important time periods in the recording. Markers can be placed by three major actions:

- in real time, by pressing the Enter button while recording, or double clicking a marker category under the New Marker section on the Operations Menu.
- in the Record Viewer by double clicking a signal portion with the left mouse button.
- by using the auto-marker algorithms presented in this menu.

The markers and period system is needed for tasks like selecting signal portions, TTL triggered periods, timing related onsets, or sub phases in gait analysis. myoRESEARCH offers several marker methods to place markers in a record:

- Manual Placement
    - Mouse
    - Enter Key
    - Marker Menu
- Automated
    - Steps
    - Min/Max by trigger channel
    - Rise/Fall by trigger channel

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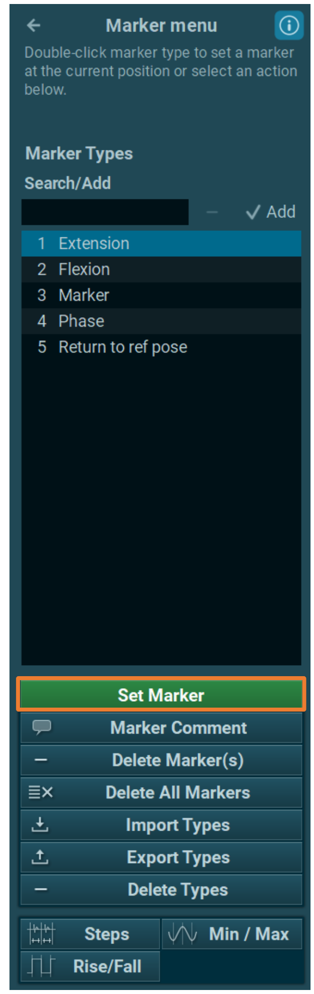
Manual Markers

By default, a double click with the left mouse or pressing the **Enter** button on the keyboard on any signal portion in the channel screens will place a marker on the record. A comprehensive marker placement menu is available in the **Operations menu** of the **Record Viewer**, which supports several helpful functions and marker definition modes.

By default, this menu is in the **manual marker placement** mode. To create a new marker type, click on the entry line of **Search/Add** and click again on the ✓ **Add** button.

To search for an existing marker type, enter the name in the entry line to automatically select it in the marker type list. To use a marker type from the list, double click on the name or press the green **Set markers button** under the toolbar to place this marker at the cursor position in the signal viewer:



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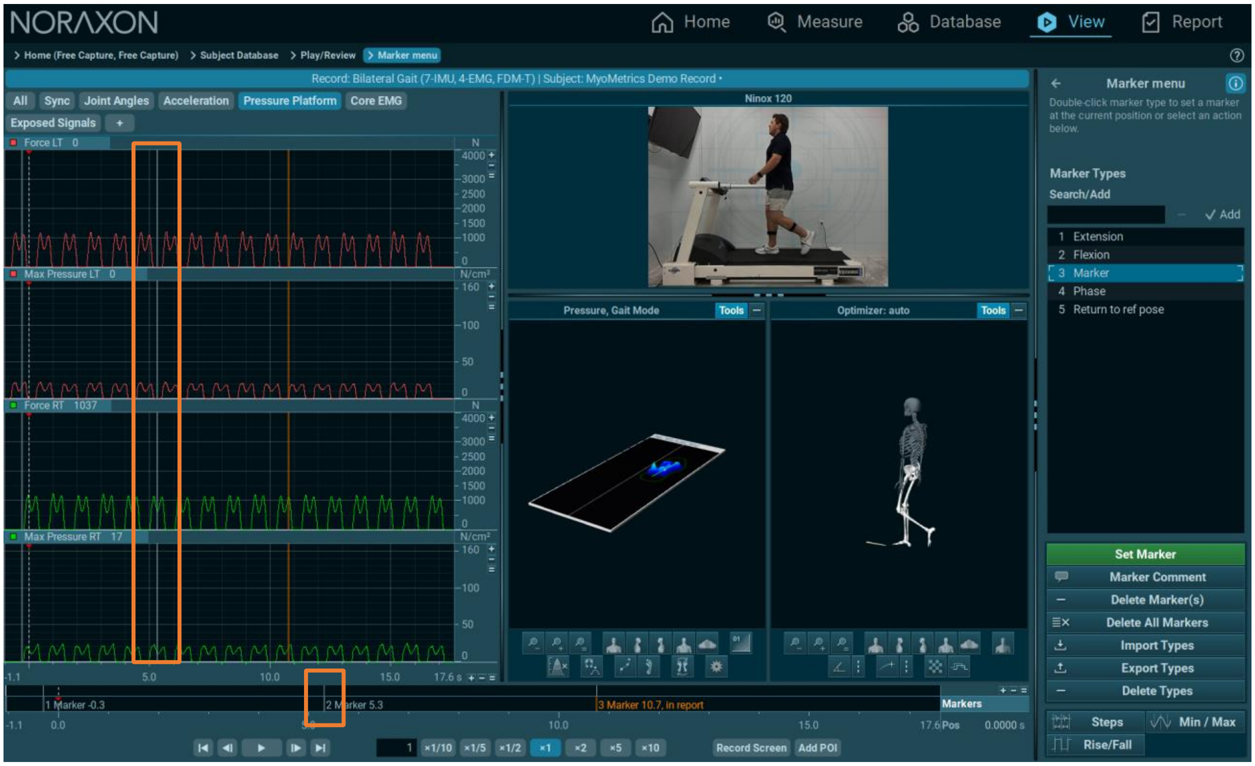

Markers on the timeline can be seen on all channels at the corresponding timestamp.

To delete markers from the signal screen, select one with the mouse, or click and drag to highlight multiple markers, and click **Delete marker(s)**. **Delete all markers** clears all markers within the record.

A description of all marker functions is shown below:

Set marker	marker is placed at cursor position.
Marker Comment	will add a text comment to this marker which can later be seen in the POI report.
Delete Marker(s)	deletes a selected marker or all markers in a mouse-marked area.
Delete All Markers	deletes all placed markers in a record.
Export Types	exports the list of marker labels.
Import Types	imports a list of marker labels

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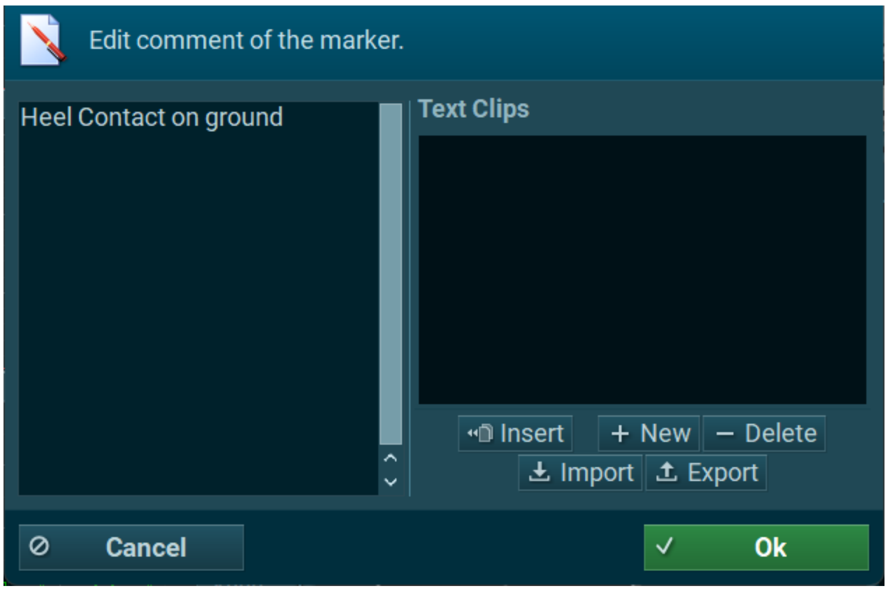
Marker Comments

The **Marker comment** feature allows user-defined comments, explanations or pre-configured text clips to a given marker. These comments can be shown later in the report alongside the POI (Point of Interest) video picture and other POI related analysis. Any labeled marker can become a POI marker for the POI report by clicking on it and pressing **Add to Report** or right click and select **Add to Report.**

To add a Comment to an existing marker, select it on the signal screen, then right click and select **Edit Comment**. Alternatively, click on **Comment** on the right toolbar of the marker menu. The Comment window will open:



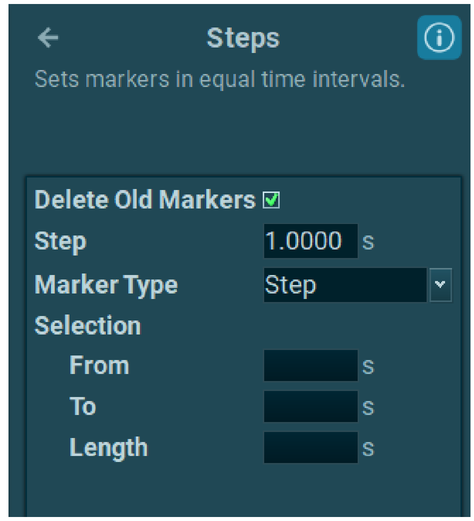
Click on the left text entry box to type in the comment's text or press **CTRL+V** to insert text from the clipboard. The **Text Clip Menu** can be useful in situations where multiple, repeated texts are used to comment on markers – it allows the user to create comment templates that can be inserted quickly.

New	when text is highlighted in the left text window, will create a text clip .
Insert	will load a text from a selected text clip to the left text window.
Delete	will delete a text clip.
Import & Export	imports or exports text clips.

Step Markers

This method is used for continuous records or signal portions where the time domain changes. Any curve parameter is analyzed in a sequence of fixed intervals.

First, it is necessary to define a signal portion in the record that would like to be marked and analyzed by the **Step** feature. By default, the whole record is selected. Areas highlighted with the mouse are also recognized, and the mouse mark can be used to define the signal portion of interest.



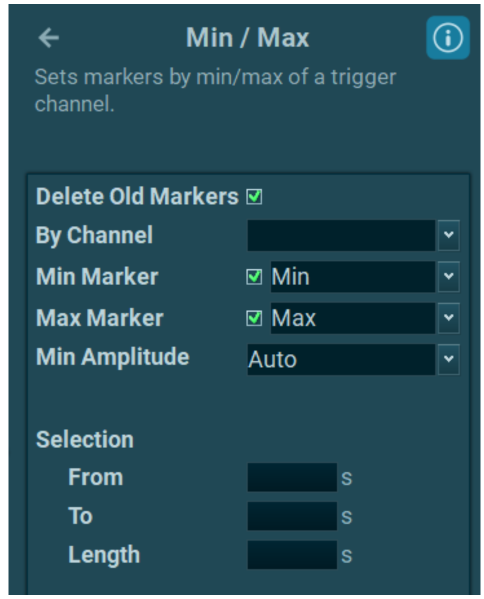
The options to customize the step markers are explained below:

Delete Old Markers	deletes all previously placed markers.
Step	defines the step size in seconds.
Marker Type	selects a marker category for the step markers
Selection	enters a precise start (**From**) to end point (**To**). **Length** indicates the duration of the selected interval in which the auto-marker routine is operated.

Min/Max Markers

This method automatically detects the local minimums and maximums on the channel's curve for records including an angle, goniometer, or inclinometer signal. Typically, the local events represent a start, point of return, and end position within a movement sequence.

This process is similar to Steps Markers, however, a channel to use to identify a min/max must be selected.



Delete Old Markers	removes the existing markers.
By Channel	defines the channel which should be used for the automarker routine.
Min / Max Marker	selects a marker category for the markers, labels them with value
Min Amplitude	Auto, the default setting, automatically finds this by analyzing the highest range found within the record and accepting local ranges higher than 50% of this maximum range. For manual calculation, a threshold value for the minimum range between Min and Max can be entered and must be exceeded to identify a local Min/Max value.
Selection	enters a precise start point (From) and an endpoint (To) in seconds. Length indicates the duration in seconds of the selected interval in which the auto-marker routine is operated.

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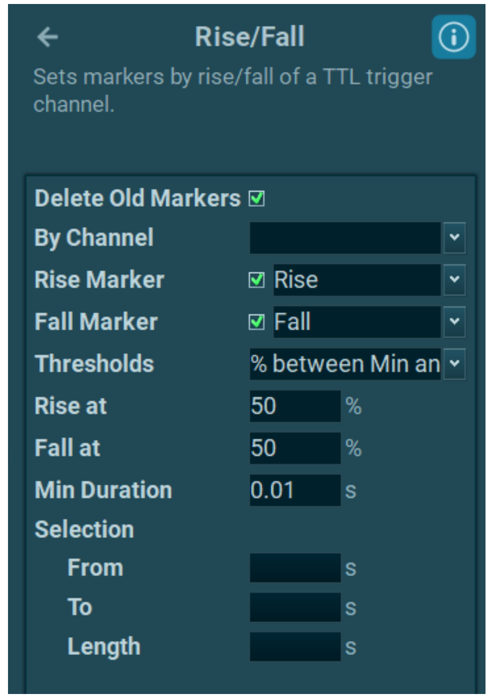
Rise/Fall Markers

For records including a TTL based trigger signal, such as foot switches, this method can identify the trigger up and down events. There are numerous applications that use this trigger function. In gait analysis, the trigger represents the heel strike and toe-off events in the gait cycle. A trigger up can be a synchronization impulse between two measurement devices to achieve time synchronization.



Similar to Min/Max, a channel must be selected to analyze, as well as additional options to tweak markers.

Delete Old Markers	removes existing markers.
By Channel	defines the channel which should be used for the auto-marker routine.
Rise/Fall marker	selects a marker category for the markers, labels them with value

Thresholds has two modes:

% between Min & Max	the TTL range and places markers at a user-defined percentage position between the rise and fall level. This location can be customized by the controls Rise at and Fall at a % between min and max TTL level.
Absolute	works in the same mechanism but uses the given channel dimension for the desired Rise and Fall event level instead of %.

Minimum Duration	defines the minimum duration a TTL event needs to last below or above the specified threshold level to be recognized by the

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algorithm. This switch can help avoid incorrect marking by artifact spikes which are spikes with a short duration.
Note: For very short trigger events like jump ground contacts, this "anti-rattle" duration must be adjusted to lower values!

Selection — enters a precise start point (From) and endpoint (To). Length indicates the duration of the selected interval in which the auto-marker routine is operated.

Edit Menu

Unlike Signal Processing, all Edit operations physically change the record data set and the original recording cannot be restored. Therefore, it is generally recommended to first make a backup or a copy of the data. The latter can be done by entering the Database Menu, selecting a record, and clicking the Duplicate button on the right toolbar. Alternatively, save any changes in Edit mode with the Save Record as option, which creates a new record file while keeping the source record in its original status.

Depending on the given screen resolution, it may be needed to resize the right toolbar to make all functions and entry lines visible in the Edit Menu.

New Channel(s)

This function allows for creating new channel lines at any location within the record. All entries are defined as shown below:

Place After — defines a certain location for the insertion in the channel order.

Name — edits the name of the channels.

Color — loads a color palette to define a trace color for this channel.

Channel type — offers a pull-down list of all available channel types to be selected for the new channel.

Raw Signal — organizes the amplitude scaling settings for the new channel.

- Units — defines the physical channel unit.
- # of digits — is the decimal value when showing the actual amplitude value.
- Bottom/Top value — is the lowest (Bottom) and highest (Top) amplitude scaling value for the Y-axis.

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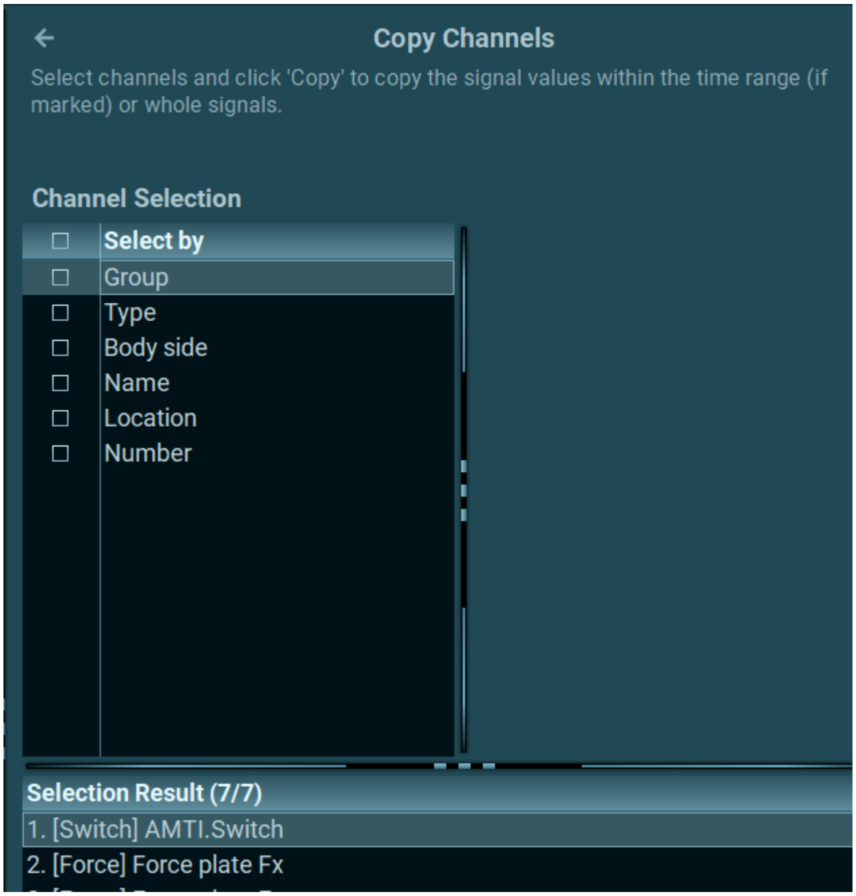
Delete, Copy and Paste Channels

All three options have the same functionality. The selection for the given operation is done via the **Channel Selection menu** as similarly explained in the Signal Processing menu in the Record Viewer.

The section **Select by** will offer a set of filters to exclude and include channels. Each filter has subcategories to apply sub selection criteria. The filters work in cascading mode, and their order can be changed by mouse dragging a filter to another position. To alter the selection of a channel for the given **Edit** operation, use any of the following selection modes:



Group	selects channels by registering tab groups.
Channel type	selects channels by type.
By body side	selects left or right side labeled channels.
Name	selects via muscle or segment name.
Channel number	selects by the channel list number.
	Note: *All 3 modes can be combined with each other.*

A description of each important channel operation that can be performed is listed below.

Delete Channels — physically erases a channel.

Attention: This channel cannot be restored after confirming and leaving the Record Viewer menu!

Copy channels — copy the selected channel data to the clipboard.

Paste values — fill channel data coming from the clipboard to the selected target channel.

Paste channels — paste channels copied to the clipboard below the last channel.

Fill Values

The **Fill Values** function is an editor for amplitude values. Based on the highlighted signal portion, any amplitude value entered will overwrite the existing values. For example, this function can be used to manually eliminate artifact spikes or other invalid data portions within a channel.

> *Attention: This operation will physically change the data of the source signal. It is imperative to back up files to prevent the loss of raw data.*

Scale Values

This function multiplies the selected amplitude data of a given channel by any factor, which can be used to mathematically amplify the signal up or scale it down to lower values.

Offset Values

This edit function is designed for manual correction of the baseline to any desired value.

Recalibrate Channels

This function corrects EMG baseline shifts by determining the mean amplitude value of a highlighted signal portion. Because raw EMG is a bipolar signal with equal value distribution across the x-axis, its mean is typically zero.

Revert Changes

Any edit operation done on the currently loaded record can be undone if the record changes are not confirmed when leaving the **Edit/Viewer** menu:

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Save Record As

This function saves an edited record under a new name and keeps the original record as it is. This function is highly recommended for any edited record since editing operations cannot be restored to the original version. This operation is also another method to duplicate a given record.

Import GFX File

This function imports time stamped GPS data from external devices.

Import Video (only available in myoVIDEO)

This operation imports externally recorded videos to an existing myoRESEARCH record. To merge external video, it is recommended to use a synchronization event – an event that can be clearly measured by all devices in the recording. Examples of a synchronization event could include using a Noraxon led syncLIGHT – which illuminates when connected to a myoSYNC, or a physical event such as tapping an EMG or IMU sensor with an object. The goal is to create an event that can be used to link an imported video with a sensor.

The external video and myoRESEARCH -based recording can run independently, and in the merging dialog of the import routine, there is a multi-step scheme that aligns the external video with the internal myoRESEARCH recording:

1) Rotate the video to the appropriate orientation.
2) Locate the **sync event on Video**.
3) Locate the **sync event on reference signal**.
4) Correct **Video offset** if necessary.

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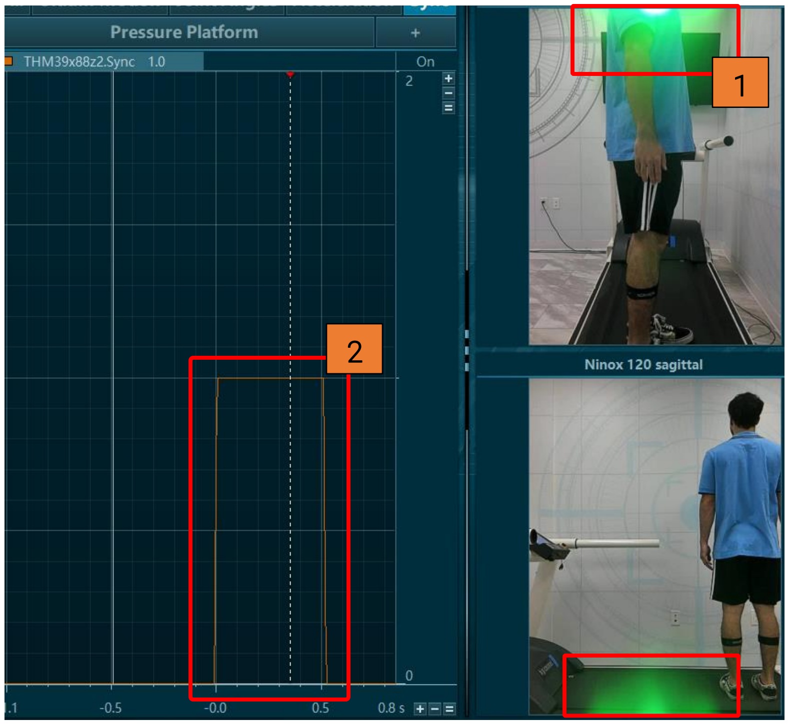
Watch the preview window of the imported video and play it to the sync light illumination position. Click **Next**, then move the cursor to the **sync event position** on the sync channel.



The offset time should be automatically calculated from the sync events in the reference signal and the video. If this shared sync scheme is not available, skip the first two dialog screens by clicking **Next** and entering an offset value if known. When done, click **Next** to check the result in the record viewer.

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Signal Bars

This function is helpful when overviewing the amplitude data of a given activity at a given cursor position:

Show channels with units	selects what type of signal/channel will be displayed as a bar graph.
Smoothing	defines a smoothing interval based on the mean value of this selected interval. This function is useful when reviewing EMG data.

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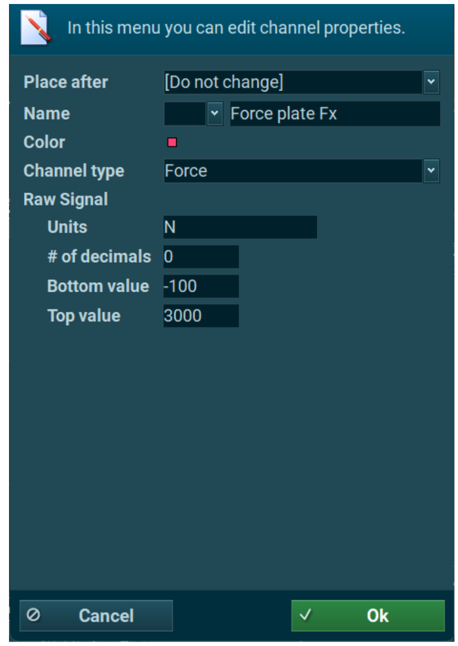
Edit Channel Properties

This menu can be opened by right-clicking on a **channel name** on the signal viewer screen.

The following **Edit** functions are available for use:



Place after	moves a selected channel around. Use this function to reorder channels.
Name	changes the current channel name and side indicator.
Color	loads a color palette to define trace colors for this channel.
Channel type	offers a pull-down list of all available channel types that can be selected for the new channel.
Raw Signal / Processed Signal	organizes the amplitude scaling settings for the selected channel.
Units	define the physical channel unit.
# of decimals	displays the decimal value when showing the actual amplitude value.
Bottom value / Top value	defines the lowest and highest amplitude scaling value for the Y-axis.

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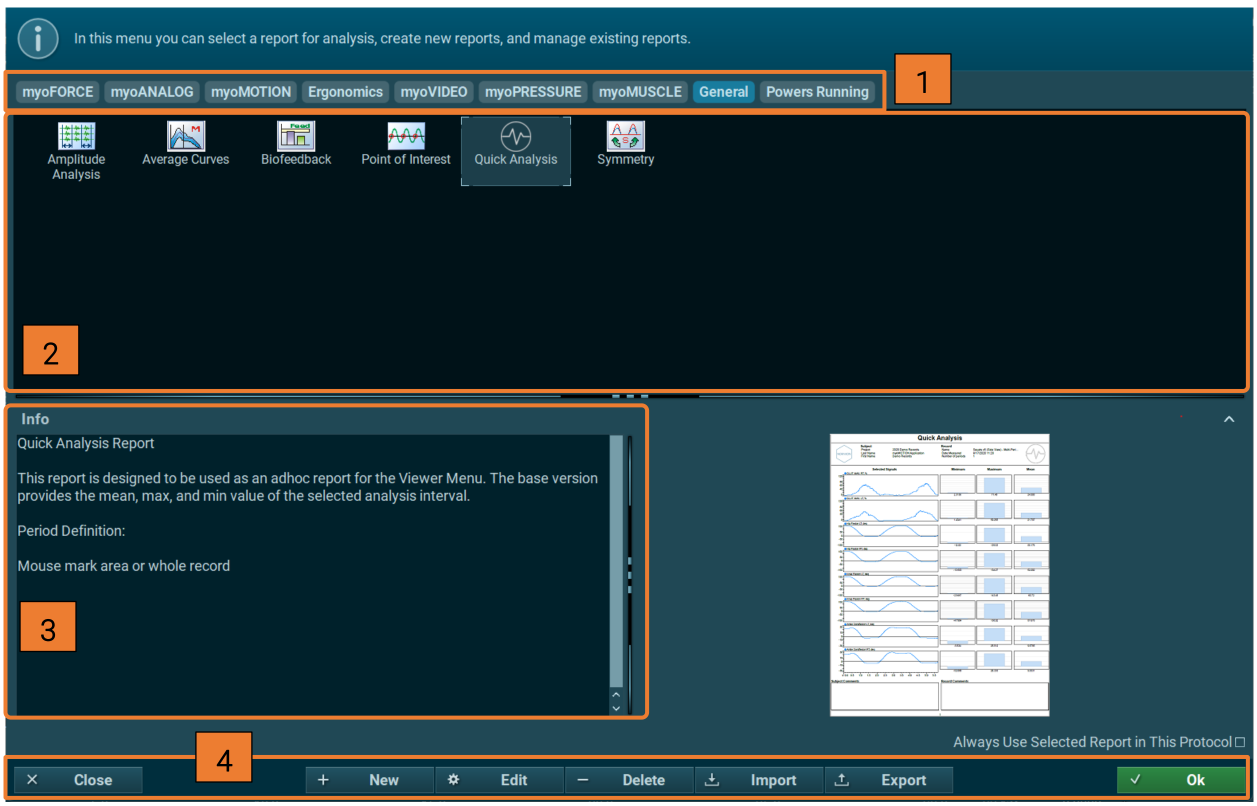
Report

The **Report** menu is responsible for analyzing relevant data from existing records and generating organized reports for a comprehensive review.

***Note**: This section only covers the report selection process within the **Select Report** menu. For a comprehensive guide on all the built-in functions and analysis tools myoRESEARCH offers in the **Report** menu, please refer to the separate Report Elements help documentation.*

Report Selection Menu

The **Select Report** menu opens after clicking the green **Report** button at the bottom of the right toolbar or clicking the large **Report** icon within the **Navigation bar**. Depending on the number of installed modules, a selection of available analysis reports is displayed.



[1] Report selection organized by modules(myoFORCE, myoMOTION, etc.)
[2] Selection of a report within an application group
[3] Report info section with report preview
[4] Report Options

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1. Report Selection

Reports are sorted into nine major tab sections. The accessibility of these tabs is dependent upon the equipment and modules installed.

- myoFORCE – for 3D force plate jump analysis
- myoANALOG – for isokinetic machines and analog signal analysis
- myoMOTION – for all myoMOTION based 3D kinematic records
- Ergonomics - for ergonomics-related assessments
- myoVIDEO – for myoVIDEO 1 and 2 video camera records
- myoPRESSURE – for all pressure related records
- myoMUSCLE (Essential, Clinical, Master) – for all myoMUSCLE EMG and sensor records
- General – Generalized reports that can be used to process and analyze multiple types of data
- Powers Running – Dr. Chris Powers' running assessment for performance optimization

The list of available default reports may grow with new program updates.

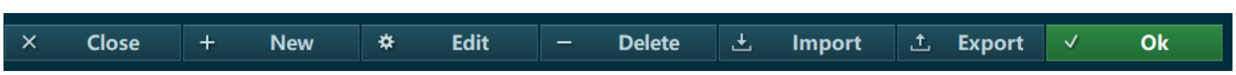
2. Select Report

From the available templates in each section, click on a report type to view the relevant details concerning this report. From this point, clicking the green OK button at the bottom of the window will proceed with the next step of selecting signals for further processing in the Viewer menu.

3. Report Info with Report Preview

A short description of the selected report addresses its main purpose, the way analysis periods are defined (Period Definition), analysis elements, and typical signal processing steps. To the right is a static picture that acts as a preview of the typical contents of the selected report.

4. Report Options



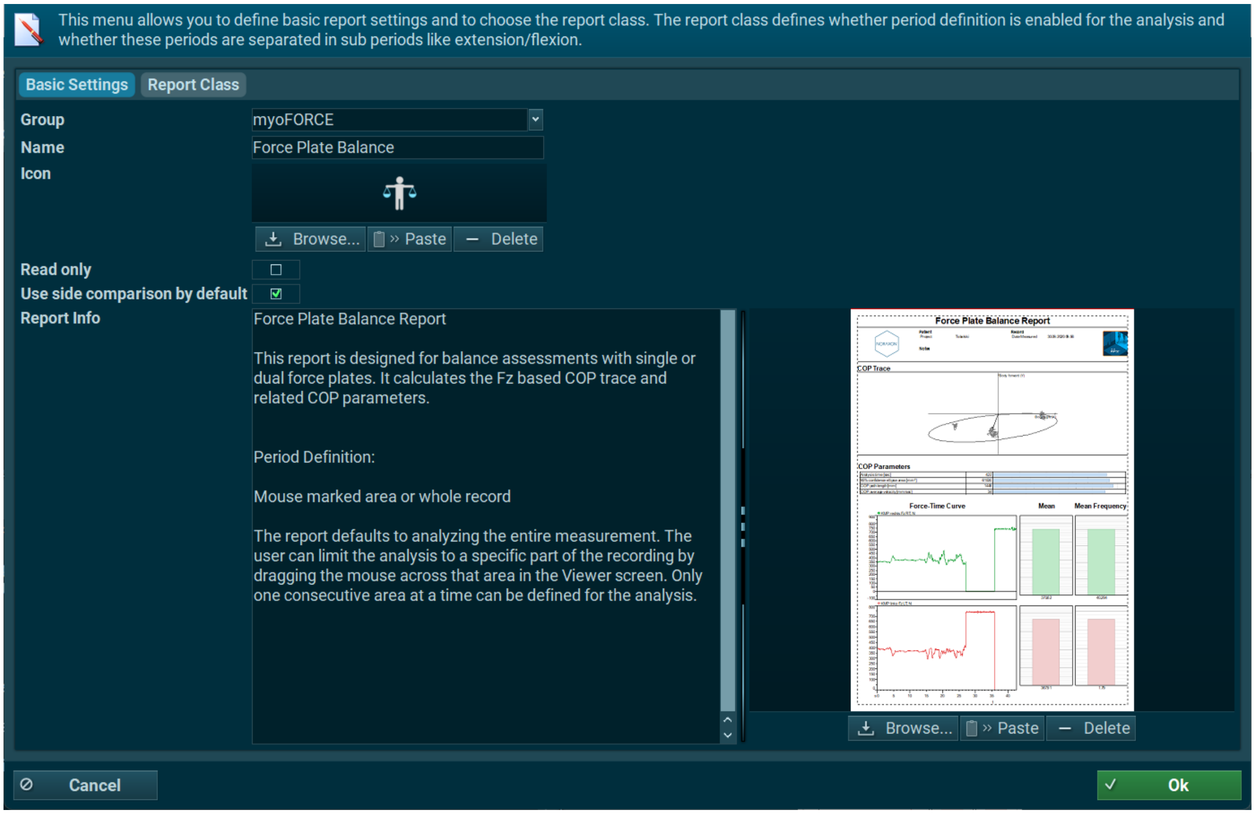
New

Specialists can create their own reports by selecting a report class and manually choosing analysis, layout, and info elements. A description of this Report Generation module is given in an additional document.

Edit

Basic Settings

A report's **title**, **tab**, **sections**, **description text**, and **preview window** can be changed with this edit menu:



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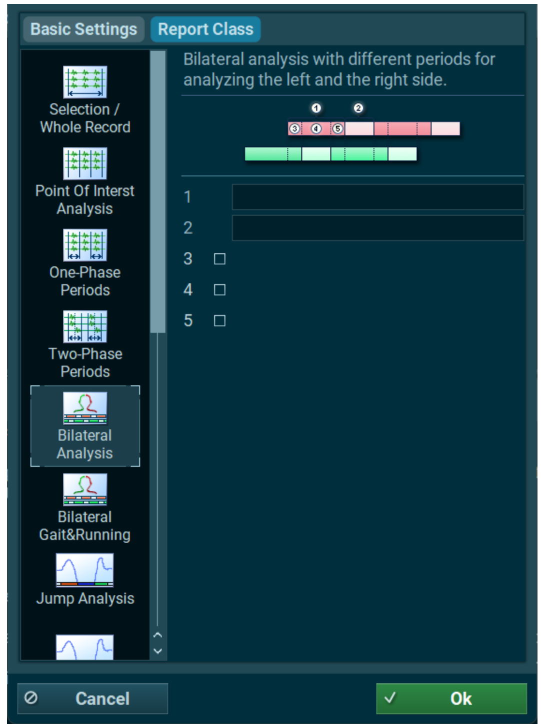
Report Class

The report class editor is meant to be used by experienced report designers only. Reports can vary in complexity in terms of periods and phase definition. For example, some reports do not require any period or phase definition because the whole record or only the highlighted area is analyzed. Other reports, like **myoMOTION Gait** in the myoMOTION Application Reports, require a complete lower body set of IMUs to generate foot contacts. These contacts are used to generate the period phases stance and swing to include in the report. The current version of myoRESEARCH offers six levels of complexity to manage these differing tasks:



Selection / Whole record	The whole record or the highlighted area will be analyzed.
Point of Interest	Single points of interest, marked with **Add to Report** in the **Viewer menu**, will be included in the report. This is designed for video analysis.
One-phase periods	A sequence of markers or events based on periods without additional sub phases will be analyzed. They are defined by trigger channels.
Two-phase periods	A sequence of periods with two sub phases will be analyzed such as extension/flexion or the stance/swing phase. The first and second phases can be freely named. The default names are Pre and Post event.
Bilateral analysis	This class is designed for left to right comparison analysis in bilateral gait measurements and splits the period definition into left and right-side periods.

Bilateral Gait analysis	A subversion of Bilateral analysis is needed for MyoPRESSURE recordings.

Delete

This function will delete a record without the ability of restoring it. Carefully use this option and consider making a backup to an external location using the Export function beforehand.

Export/Import

Reports can be exported or imported from external locations, such as other HDD drives or network directories. An exported report is automatically stored in a container directory called Noraxon MR4 data. To import reports, set a path to the location where this directory is located.

OK

Clicking this green button operates the selected report. Once the correct report has been selected press OK to continue with record signal selection for data analysis.

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